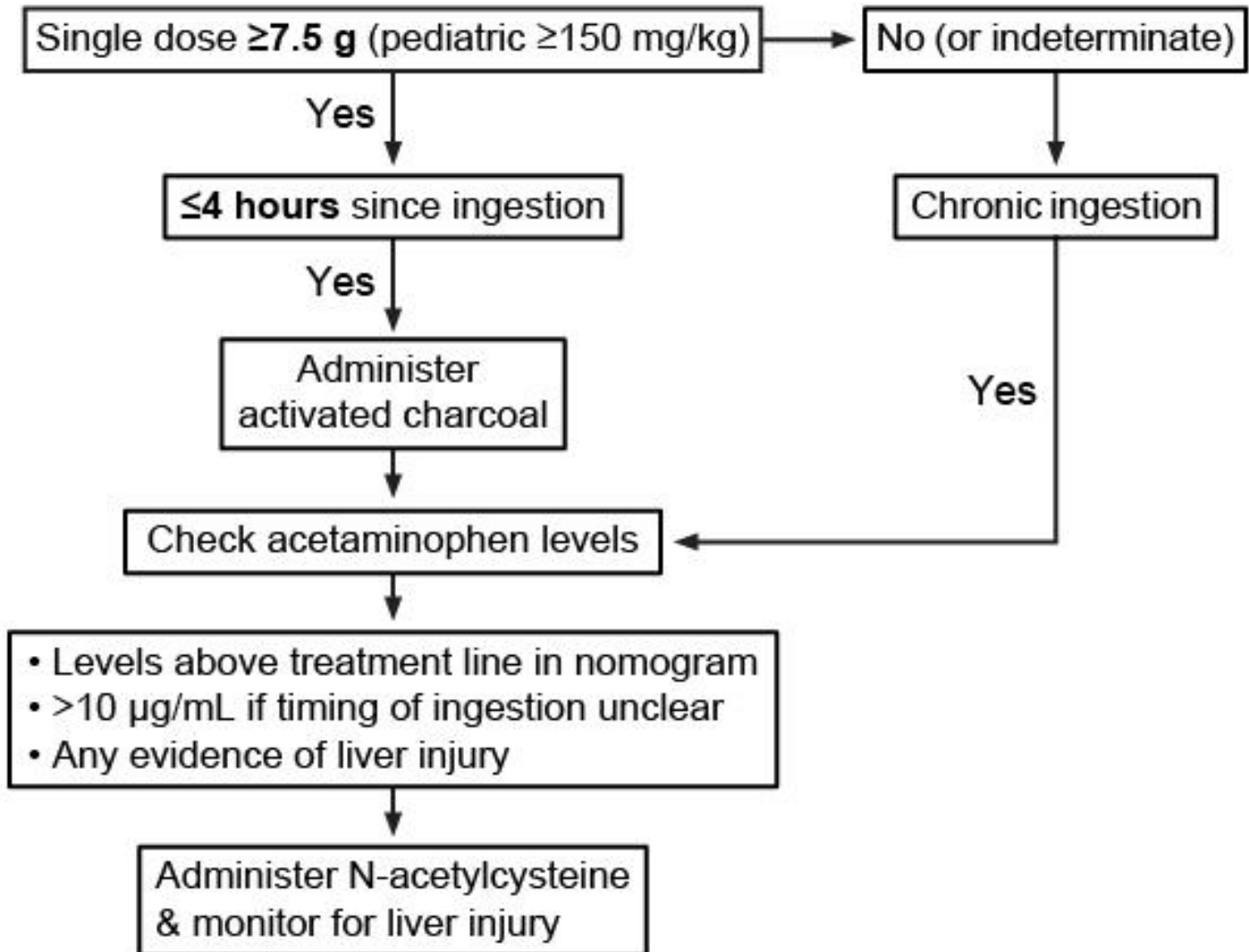
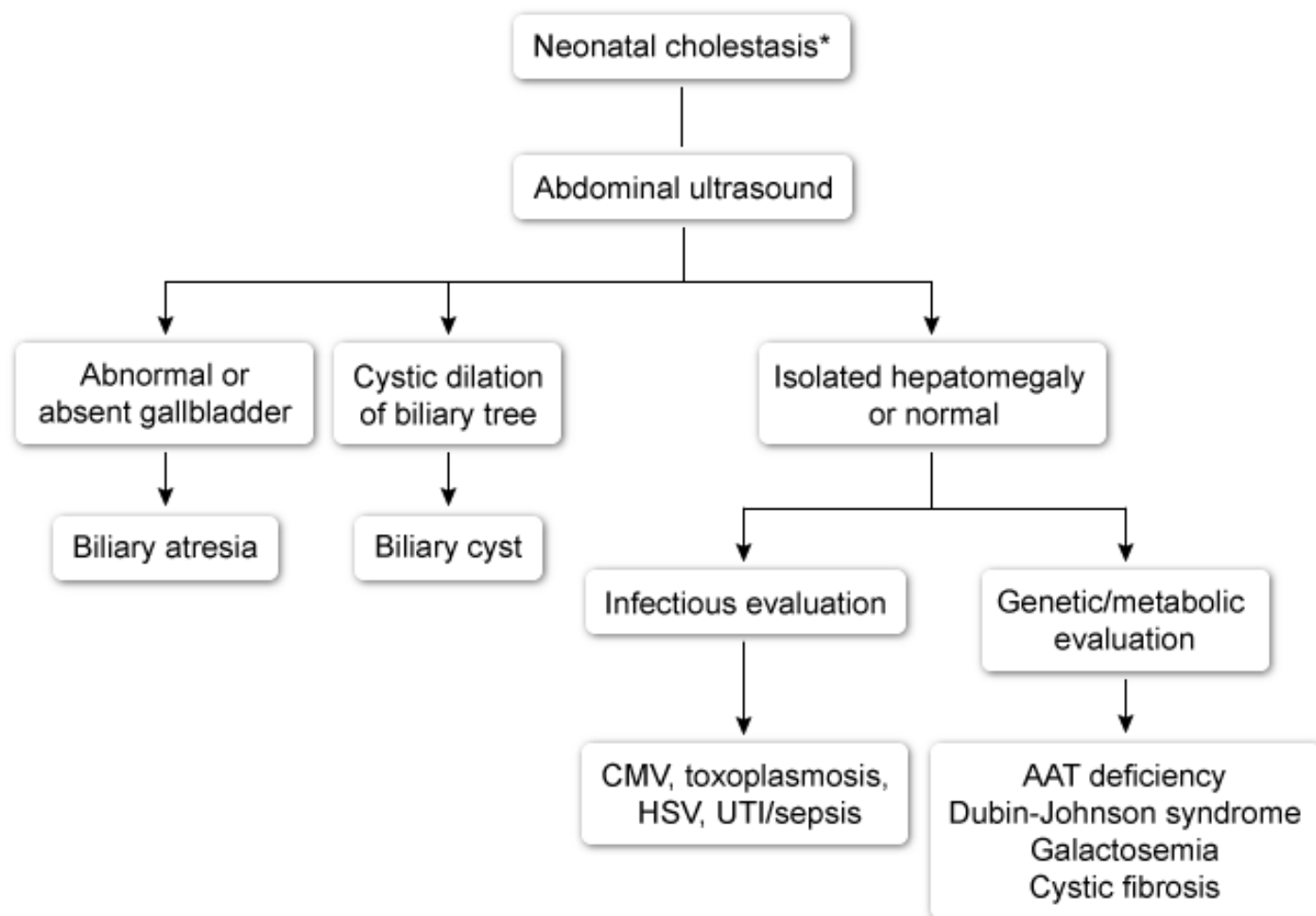


ALGORITHMS

Acetaminophen intoxication



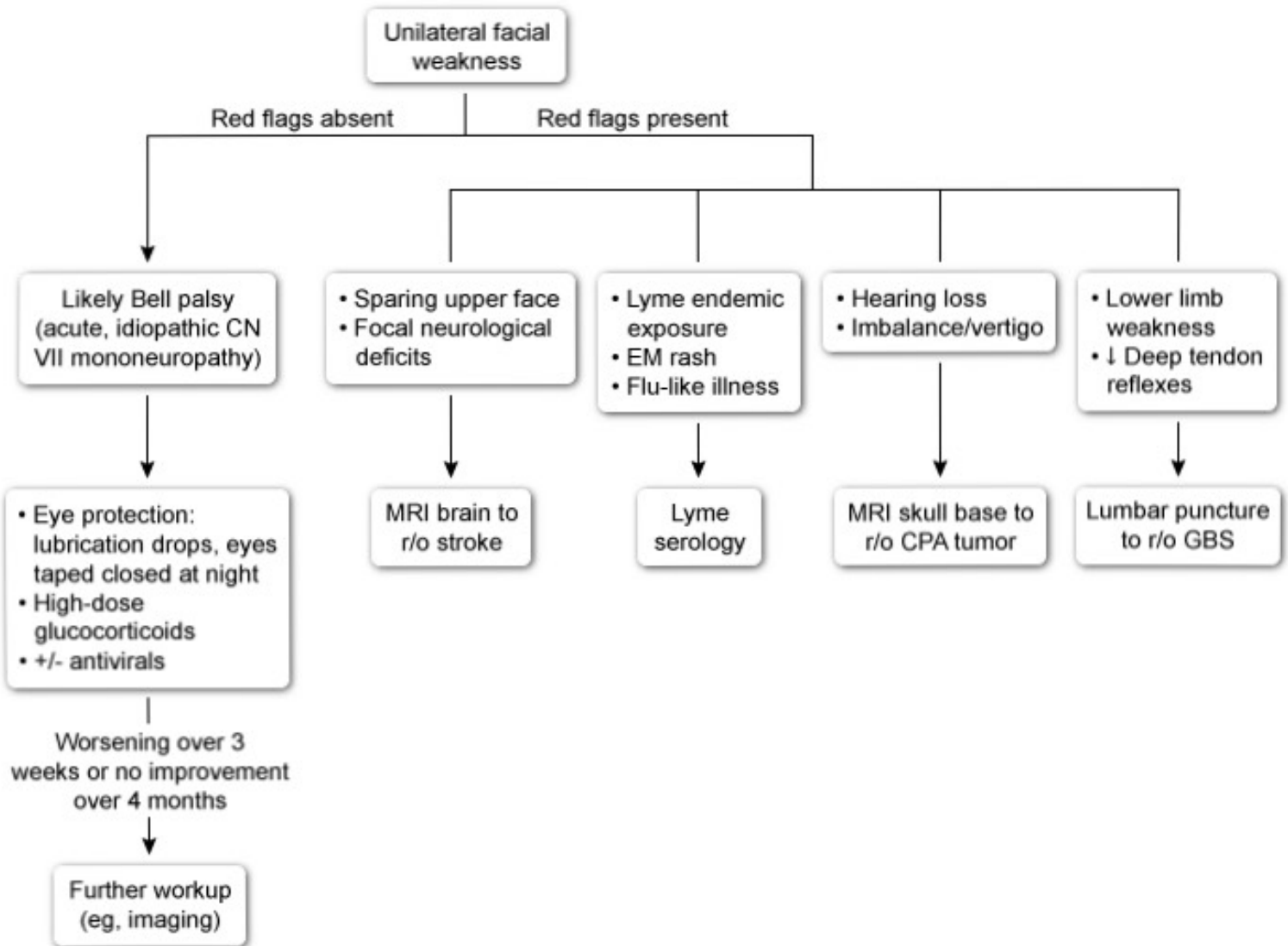
Approach to neonatal cholestasis



*Dark urine, acholic stools, conjugated hyperbilirubinemia.

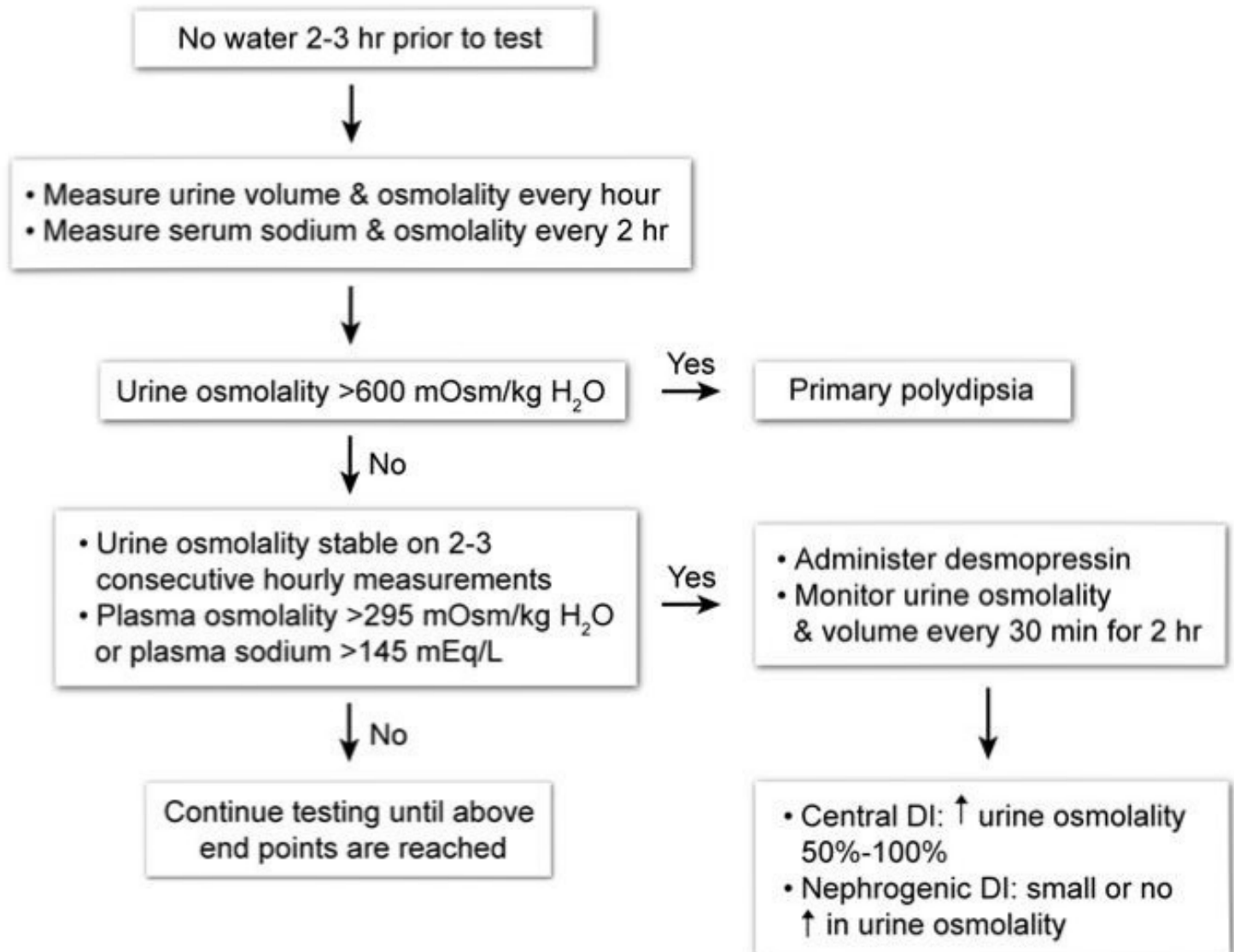
AAT = alpha-1 antitrypsin; **CMV** = cytomegalovirus; **HSV** = herpes simplex virus; **UTI** = urinary tract infection. ©UWorld

Management of unilateral facial weakness

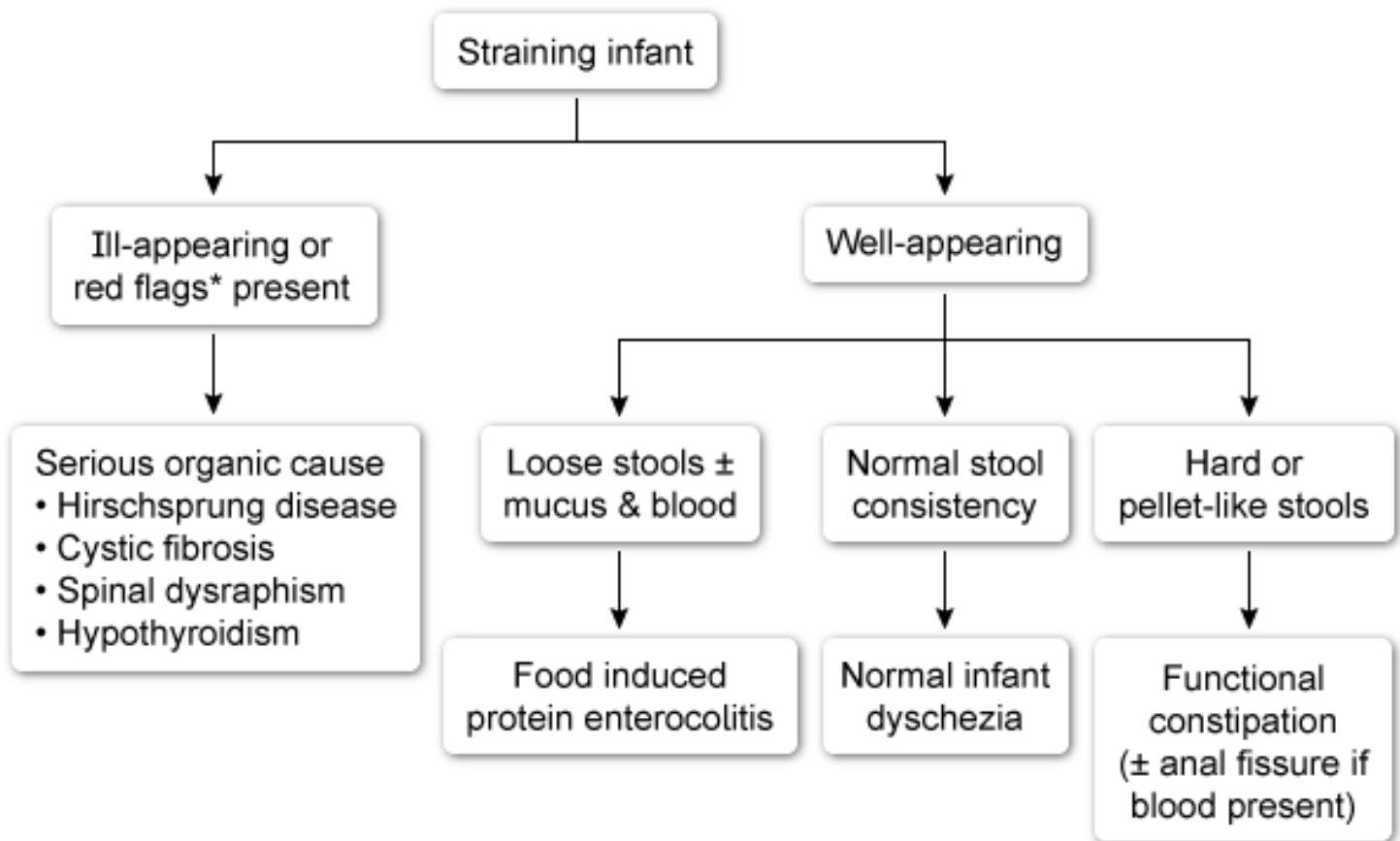


EM = erythema migrans; **CPA** = cerebellopontine angle; **GBS** = Guillain-Barré syndrome.

Water deprivation test



Approach to the straining infant

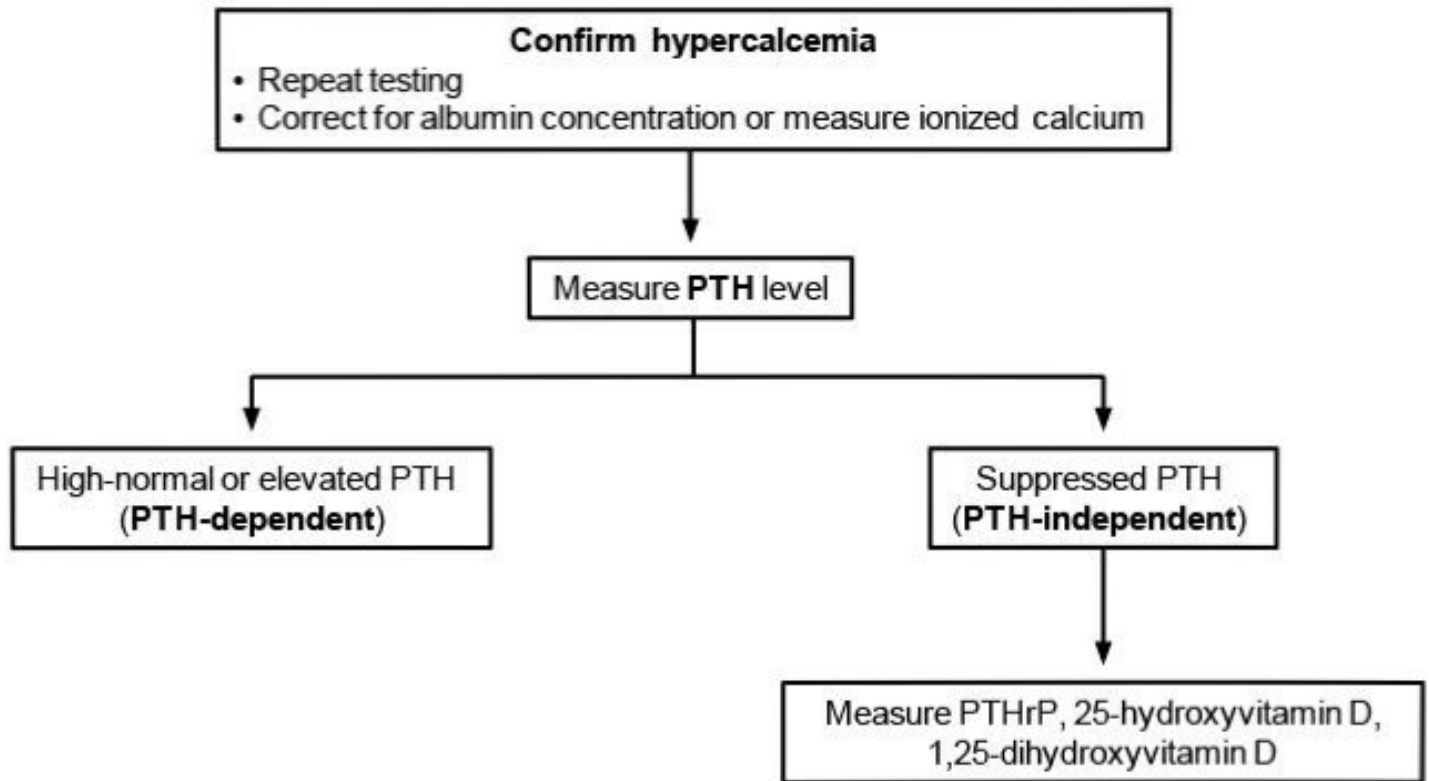


*Severe abdominal distention, abnormal rectal tone or sacral findings, delayed passage of meconium, failure to thrive

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Important to note that anal fissure in a child can be under functional constipation and does not constitute something serious

Diagnosis of hypercalcemia



Causes

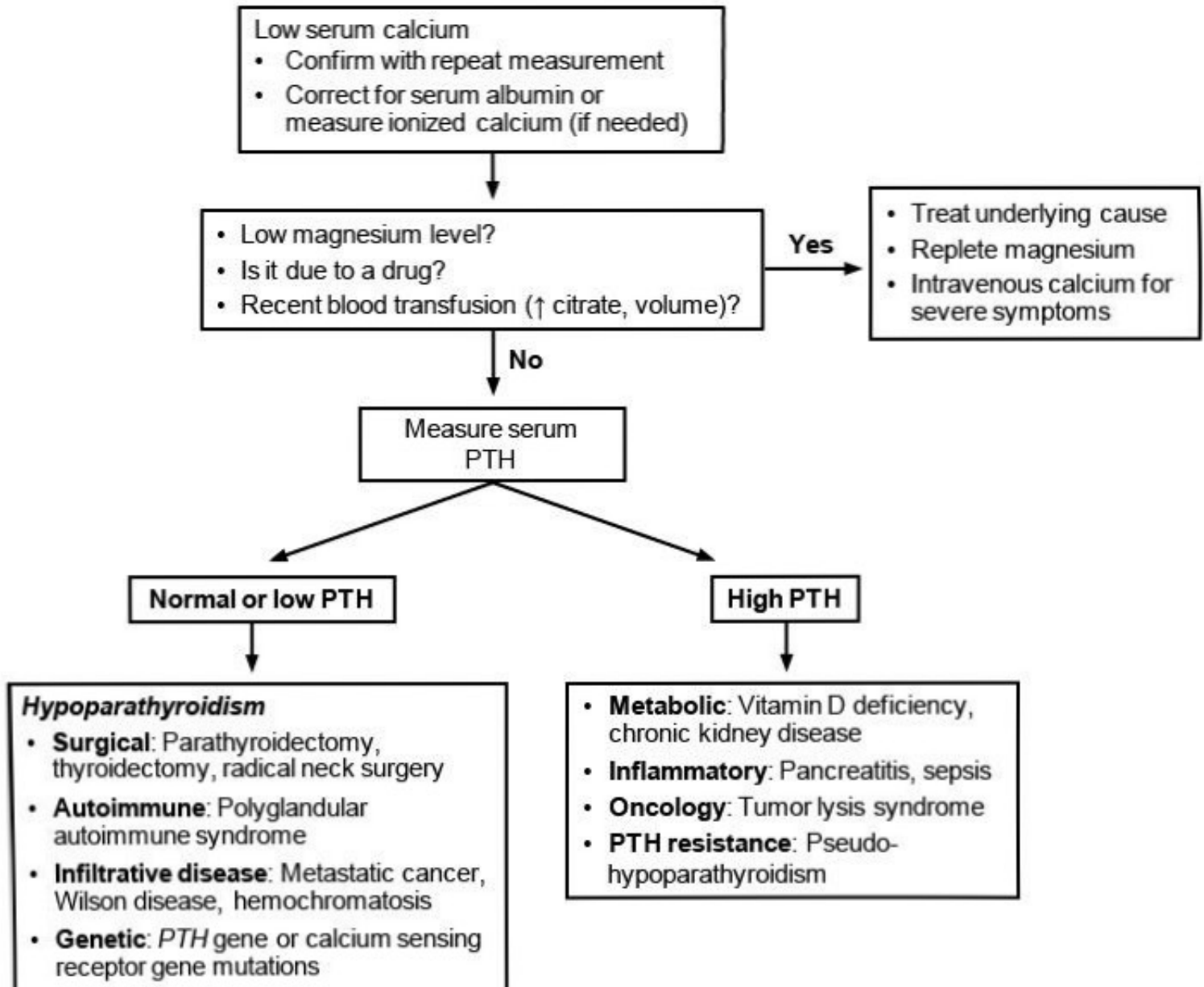
- **Primary** (or tertiary) **hyperparathyroidism**
- Familial hypocalciuric hypercalcemia
- Lithium

Causes

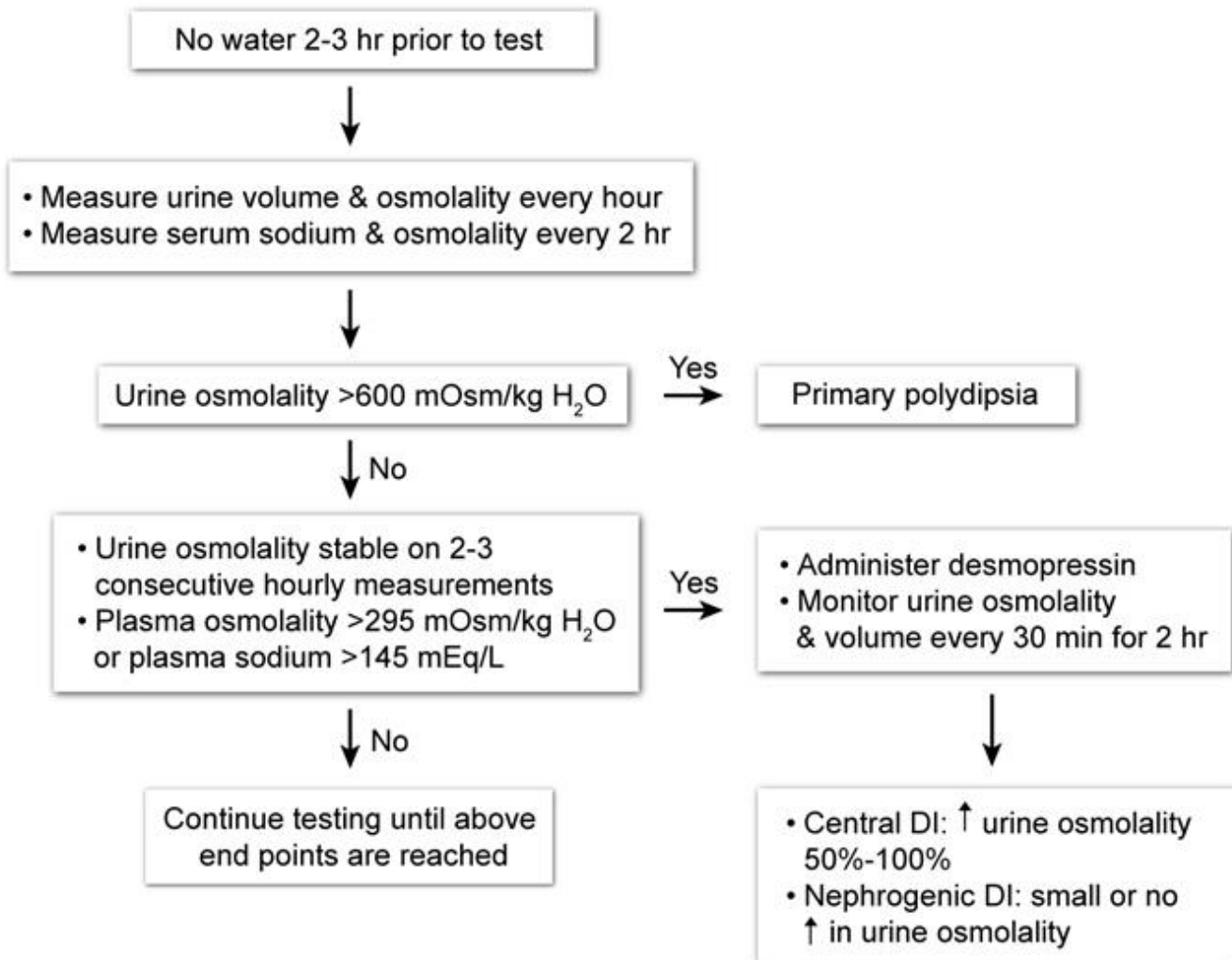
- **Malignancy**
- Vitamin D toxicity
- Granulomatous diseases
- Drug-induced (eg, thiazides)
- Milk-alkali syndrome
- Thyrotoxicosis
- Vitamin A toxicity
- Immobilization

PTH = parathyroid hormone; PTHrP = parathyroid hormone -related protein.

Approach to hypocalcemia



Water deprivation test



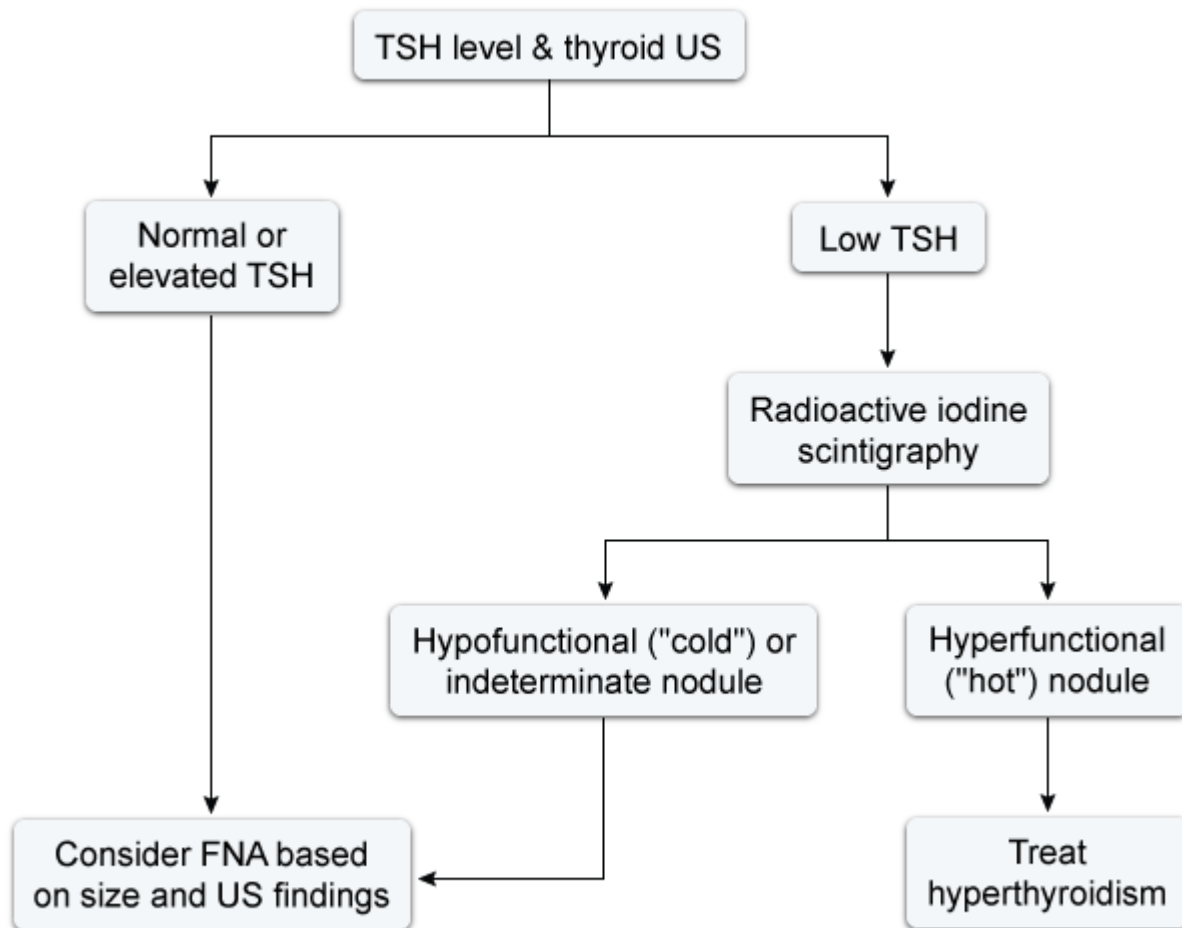
The **inner ear** contains the endolymphatic fluid-filled semicircular canals (which convey movement and position of the head) and the cochlea (which is the sensory organ of hearing). Conditions that cause disruption of endolymph flow can present with vertigo (semicircular canals) and/or sensorineural hearing loss (cochlea).

Perilymphatic fistulas are a rare, but debilitating, complication of head injury or barotrauma. They cause leakage of endolymph from the semicircular canals and cochlea into surrounding tissues, resulting in characteristic clinical features:

- **Progressive sensorineural hearing loss** caused by damage to cochlear hair cells from loss of endolymph.
- **Episodic vertigo with nystagmus** triggered by pressure changes in the inner ear (eg, **Valsalva** maneuver, elevation changes [eg, riding in elevator]) due to acutely increased endolymph leakage. This can be demonstrated clinically by performing a loud clap (ie, pressure change due to sound conduction through the ossicles) near the patient's ear and observing for nystagmus (Tullio phenomenon).

Patients are advised to limit activities that increase inner ear pressure; they also require ENT referral for further management.

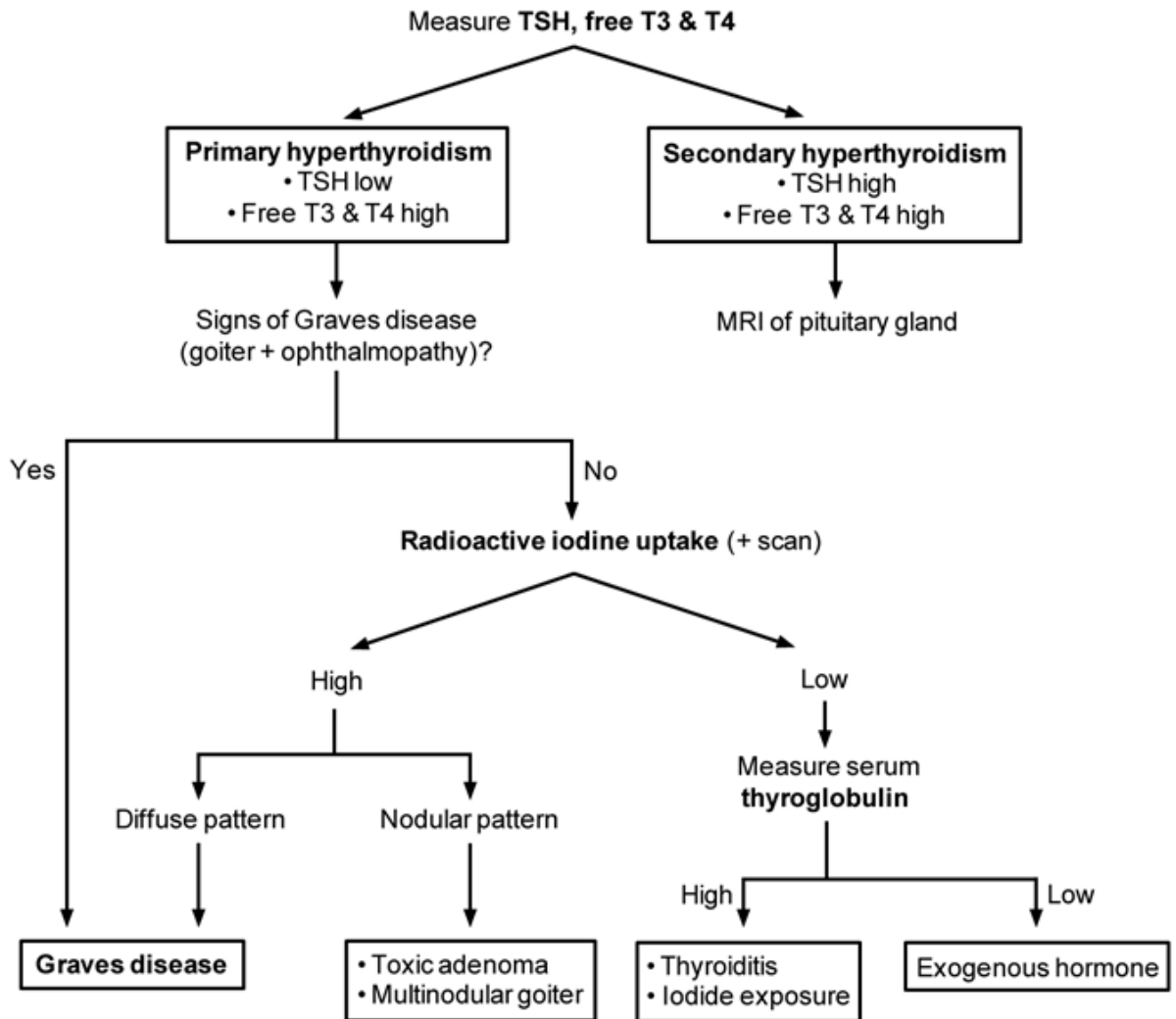
Evaluation of thyroid nodules



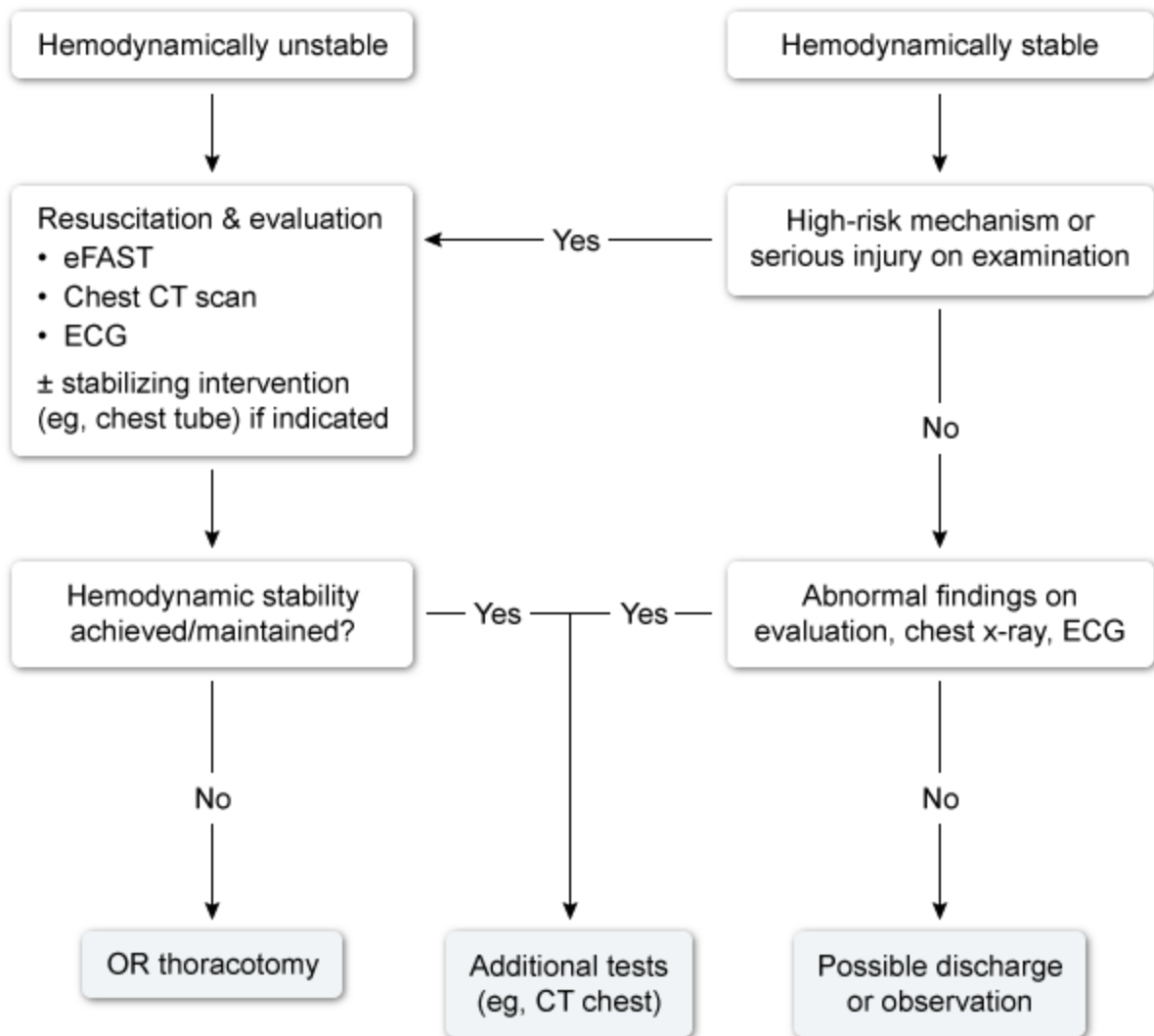
FNA = fine-needle aspiration; US = ultrasound.

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Evaluation of hyperthyroidism



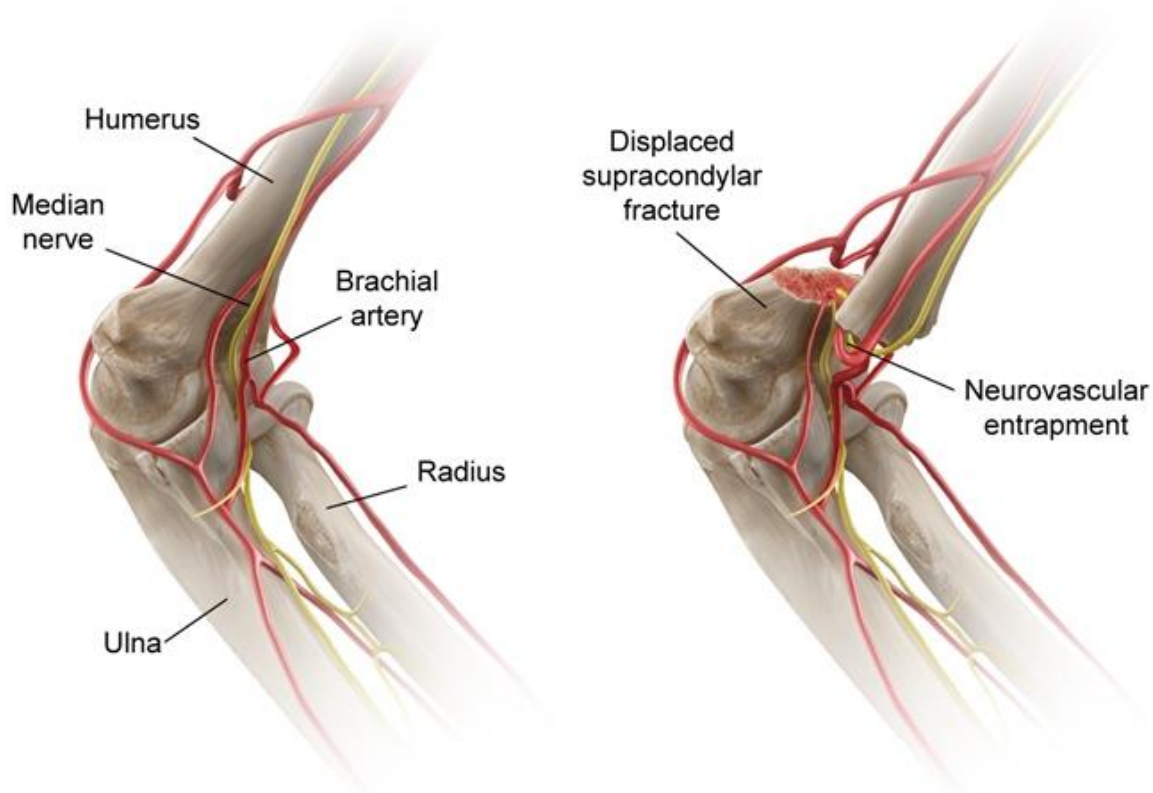
Blunt chest trauma



eFAST = extended focused assessment with sonography for trauma; **OR** = operating room.

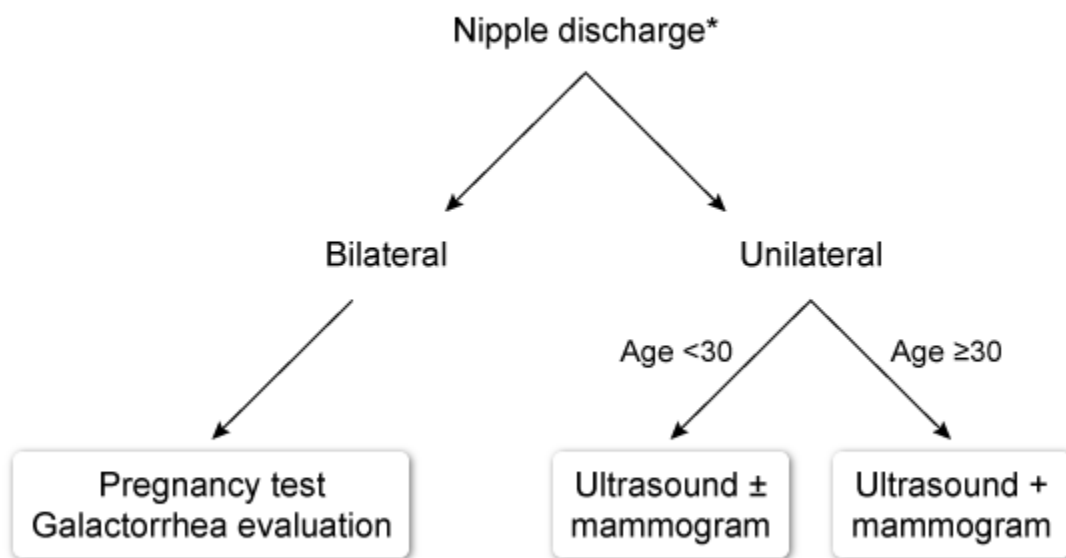
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Complications of supracondylar fracture



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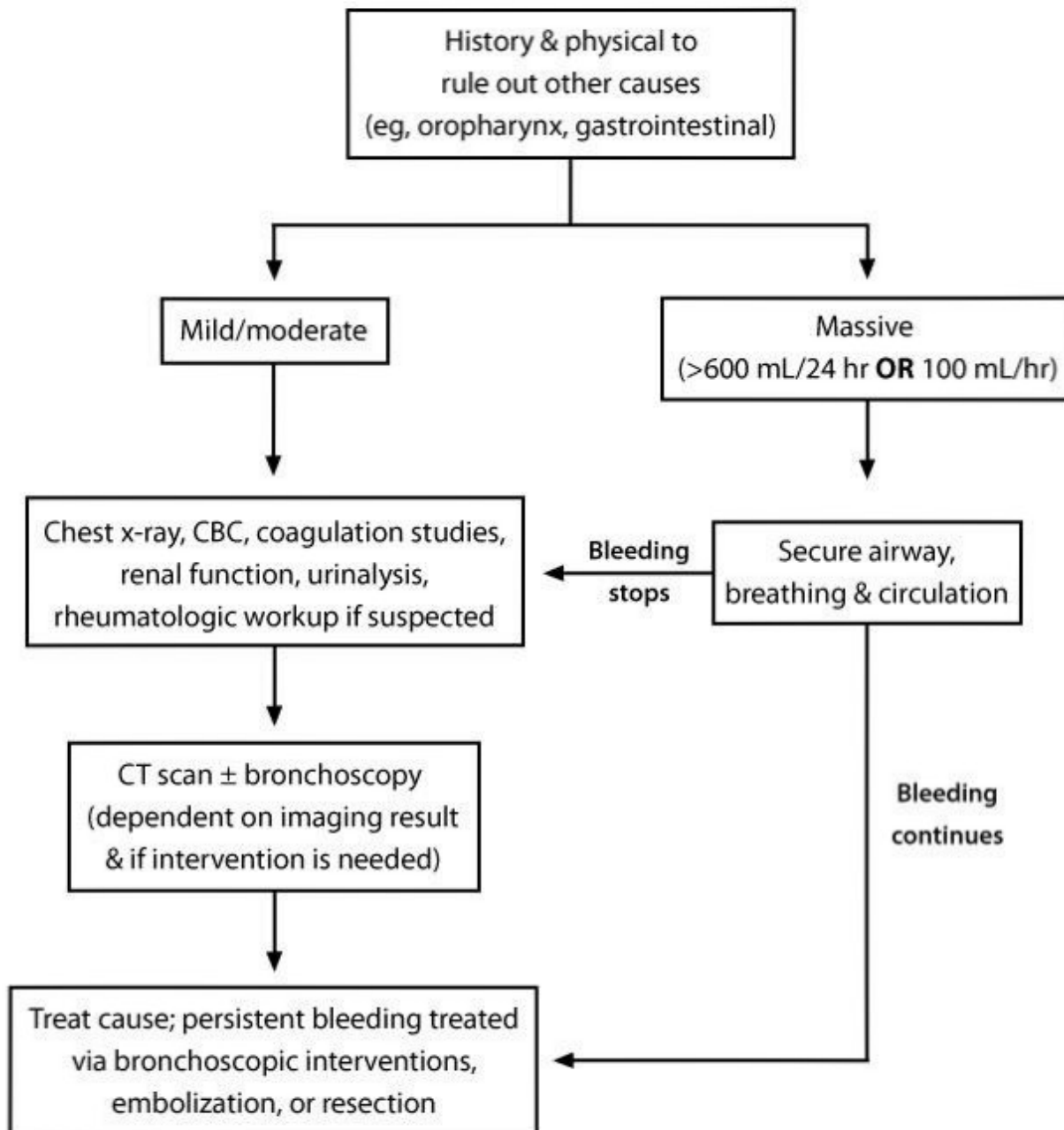
Breast discharge evaluation



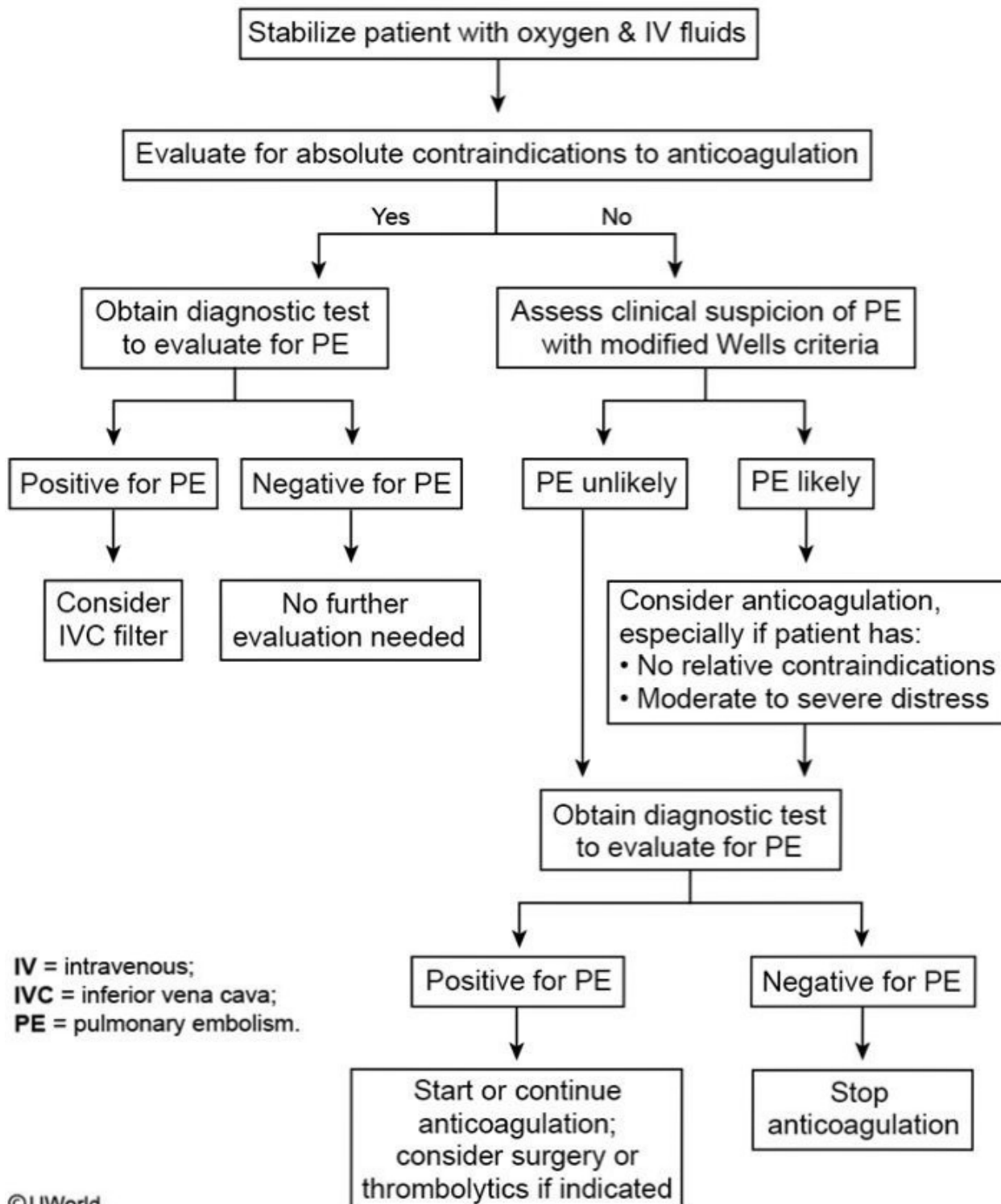
*Nonbloody discharge & normal breast examination (eg, no masses, lymphadenopathy, or skin changes).

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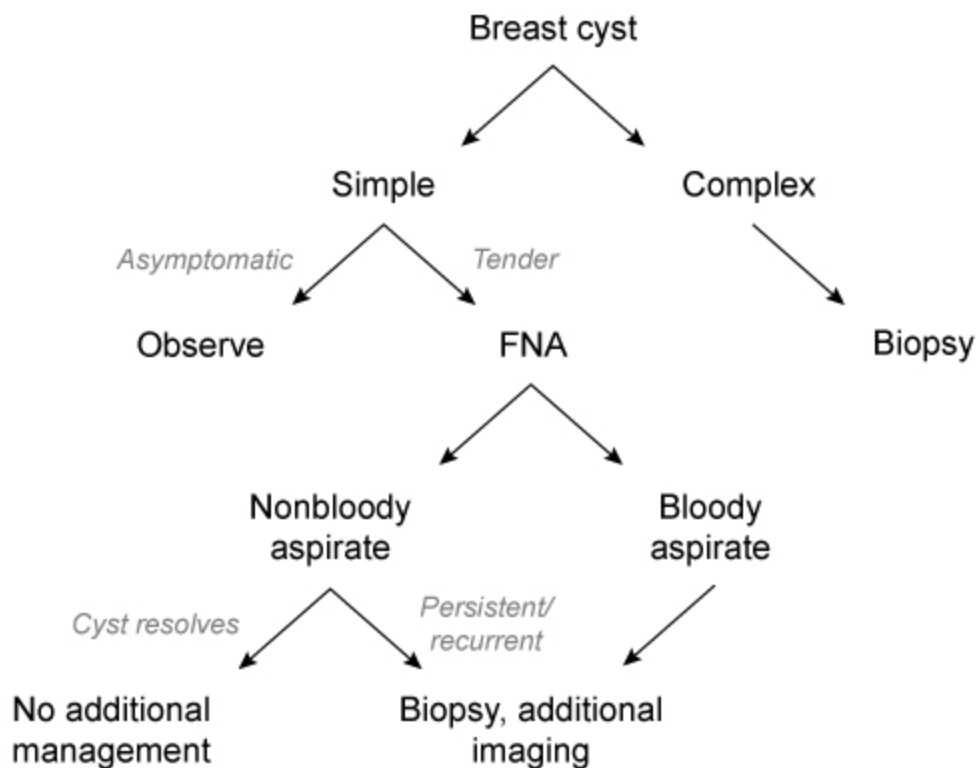
Evaluation of hemoptysis



Approach to patient with suspected pulmonary embolism



Breast cyst management

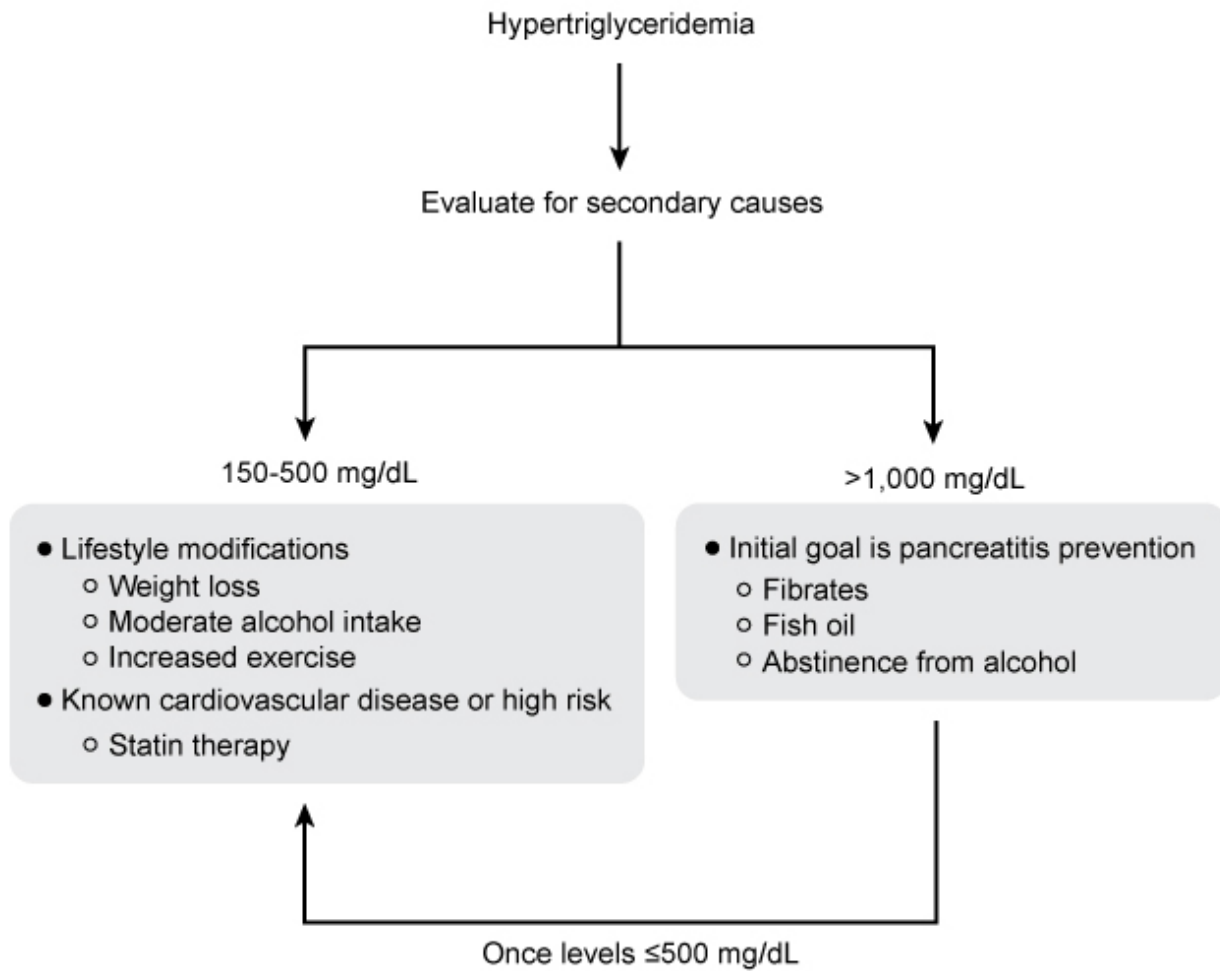


FNA= fine-needle aspiration.

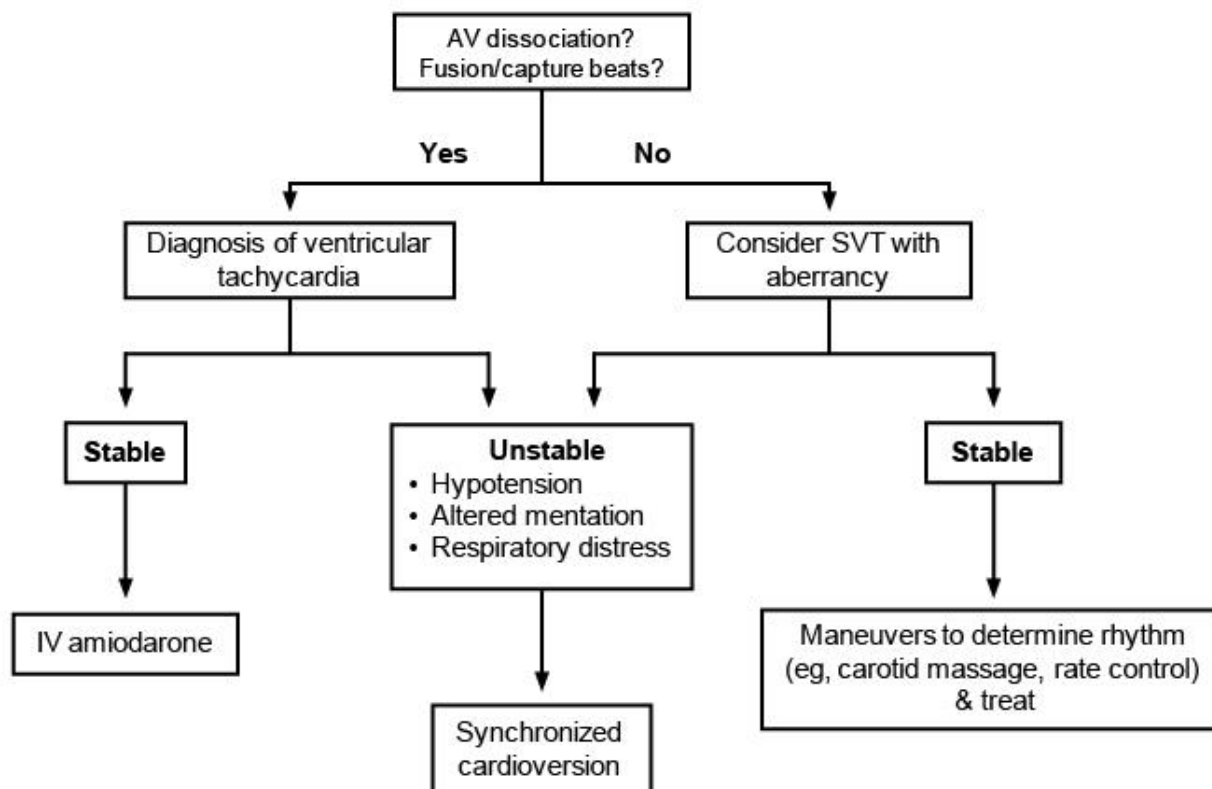
©UWorld

Diagnostic criteria for anaphylaxis	
Anaphylaxis is likely if there is rapid symptom onset & any 1 of the following criteria:	
1	Skin/mucosa involvement (eg, hives, lip/tongue swelling) & either hypotension or respiratory distress
2	Involvement of ≥ 2 organ systems after exposure to a likely allergen • Skin/mucosa (eg, hives, lip/tongue swelling) • Respiratory (eg, wheezing, stridor, dyspnea) • Cardiovascular (eg, hypotension, tachycardia, syncope) • Gastrointestinal (eg, abdominal pain, vomiting, diarrhea)
3	Hypotension after exposure to a known allergen

Treatment of hypertriglyceridemia



Approach to wide-complex tachycardia

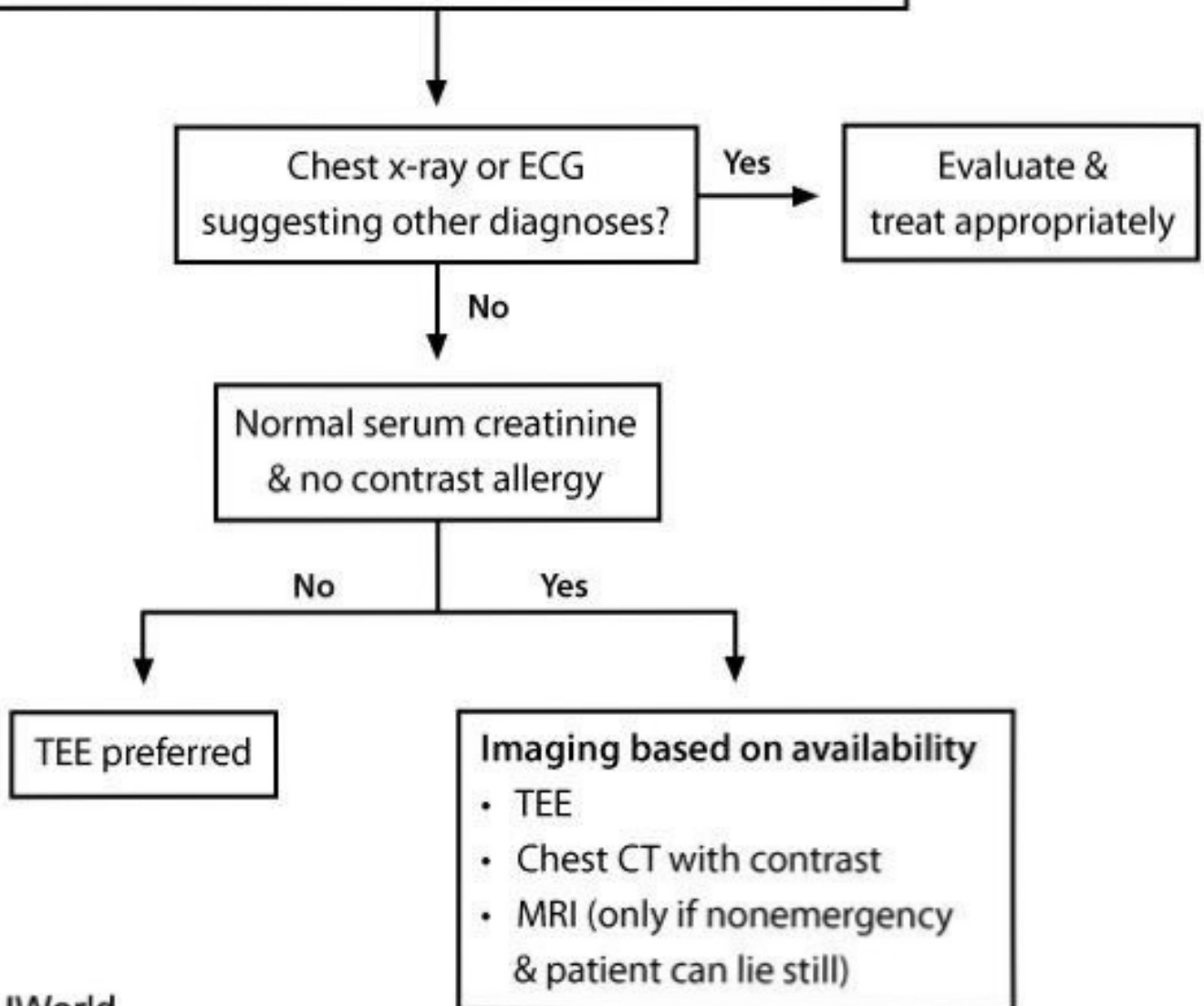


AV = atrioventricular; **IV** = intravenous; **SVT** = supraventricular tachycardia.

Diagnostic approach for suspected aortic dissection

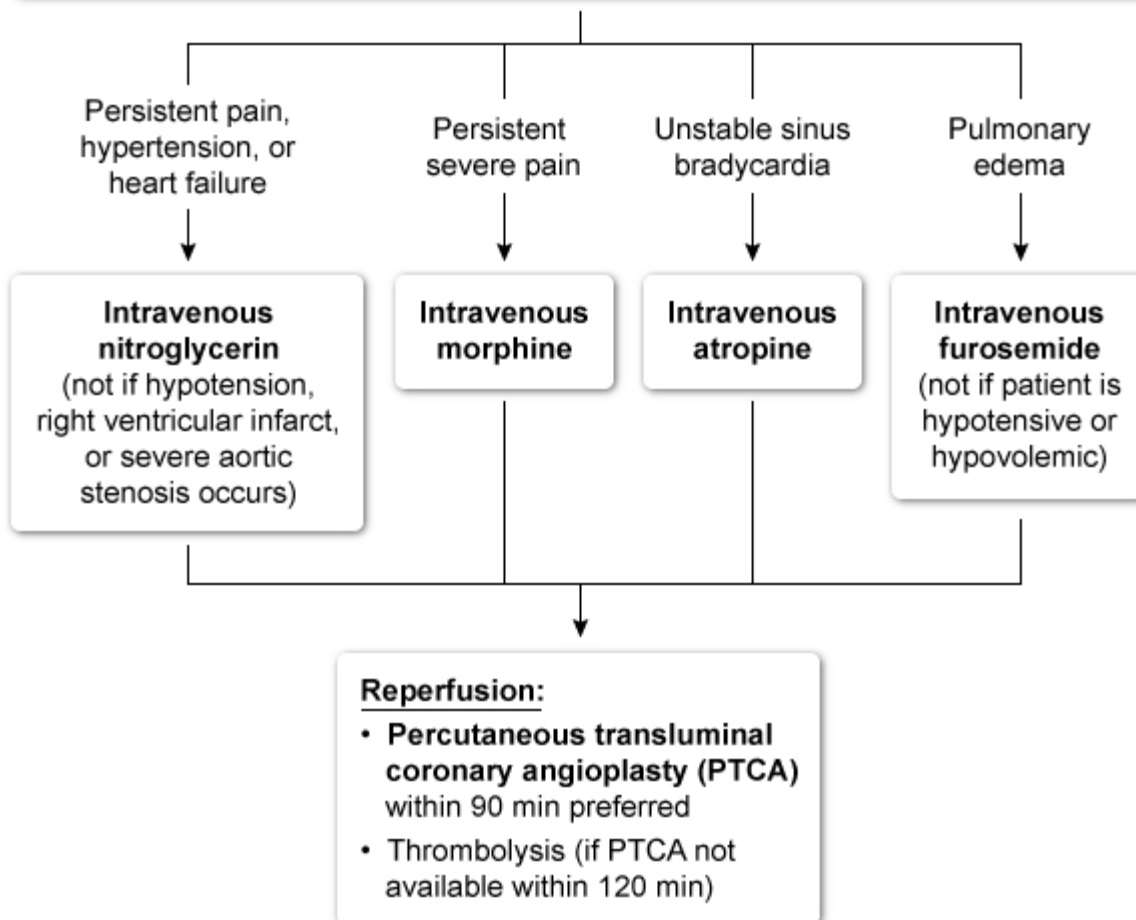
Suspected aortic dissection

- Risk factors (eg, Marfan syndrome, connective tissue disease, hypertension)
- Tearing chest/abdominal pain radiating to the back
- Perfusion deficits (eg, pulse deficit, >20 mm Hg blood pressure difference between right & left arm, aortic regurgitation murmur)

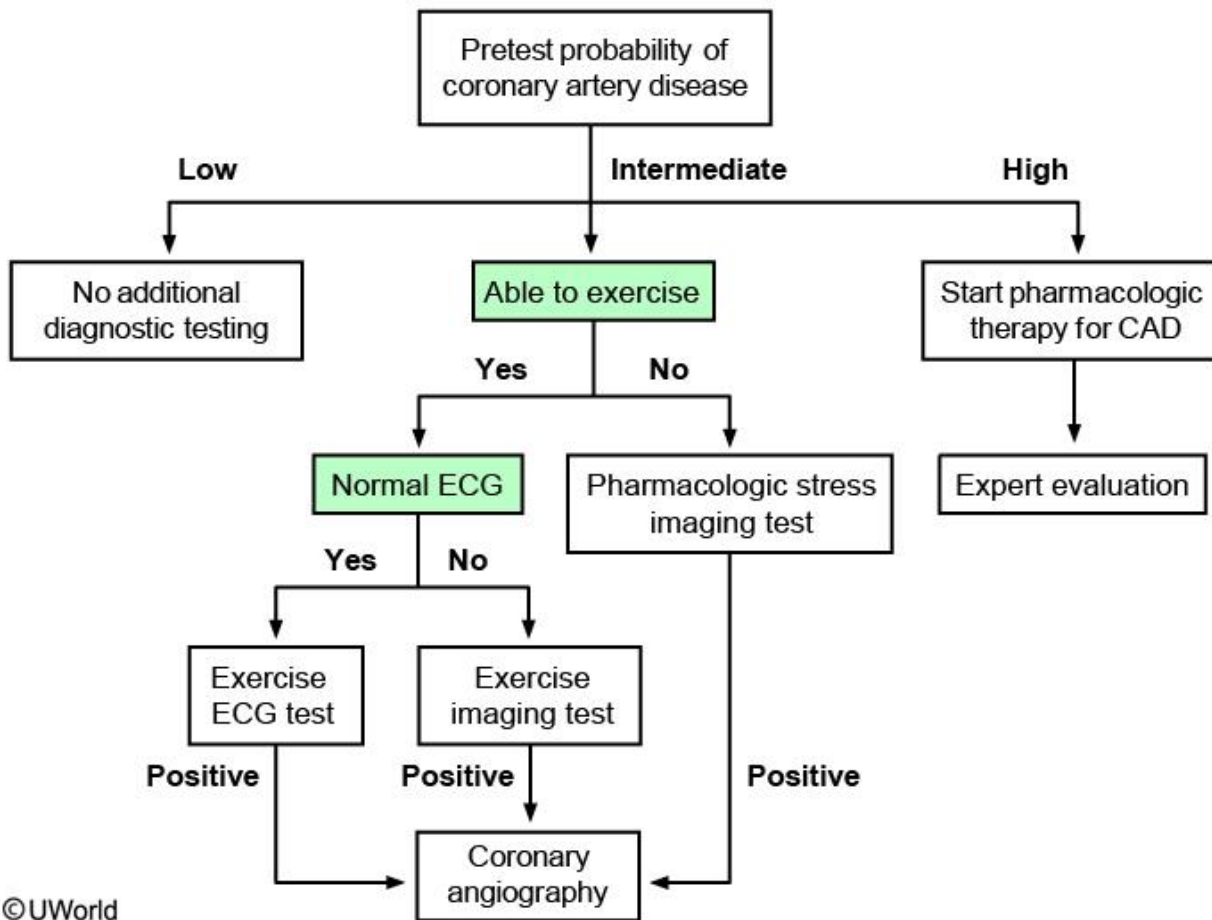


Initial stabilization of acute ST-elevation MI

- Supplemental oxygen (if $\text{SaO}_2 < 90\%$ or dyspnea)
- **Aspirin** 325 mg
- **P2Y12 inhibitor** (eg, clopidogrel)
- Nitrates (sublingual)
- **Beta blocker** (unless hypotension, bradycardia, acute heart failure, heart block)
- High-dose **statin** (eg, atorvastatin 80 mg)
- **Anticoagulation** (drug depends on planned revascularization)

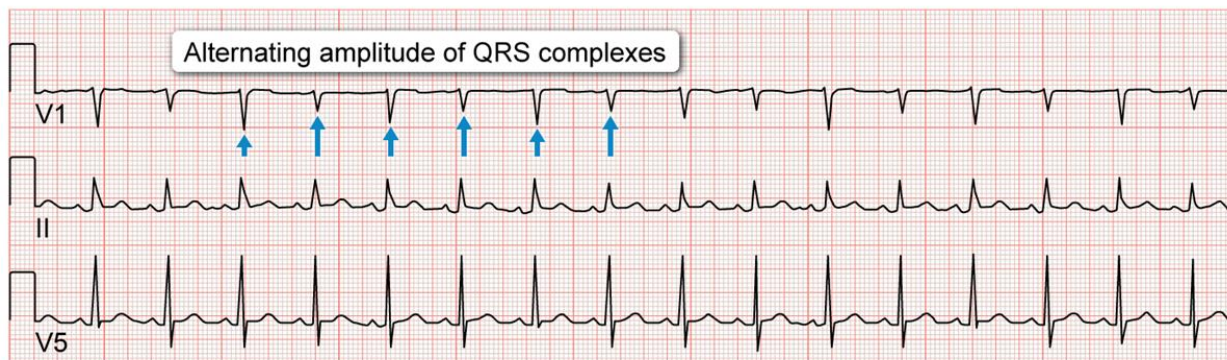


Evaluation of chest pain



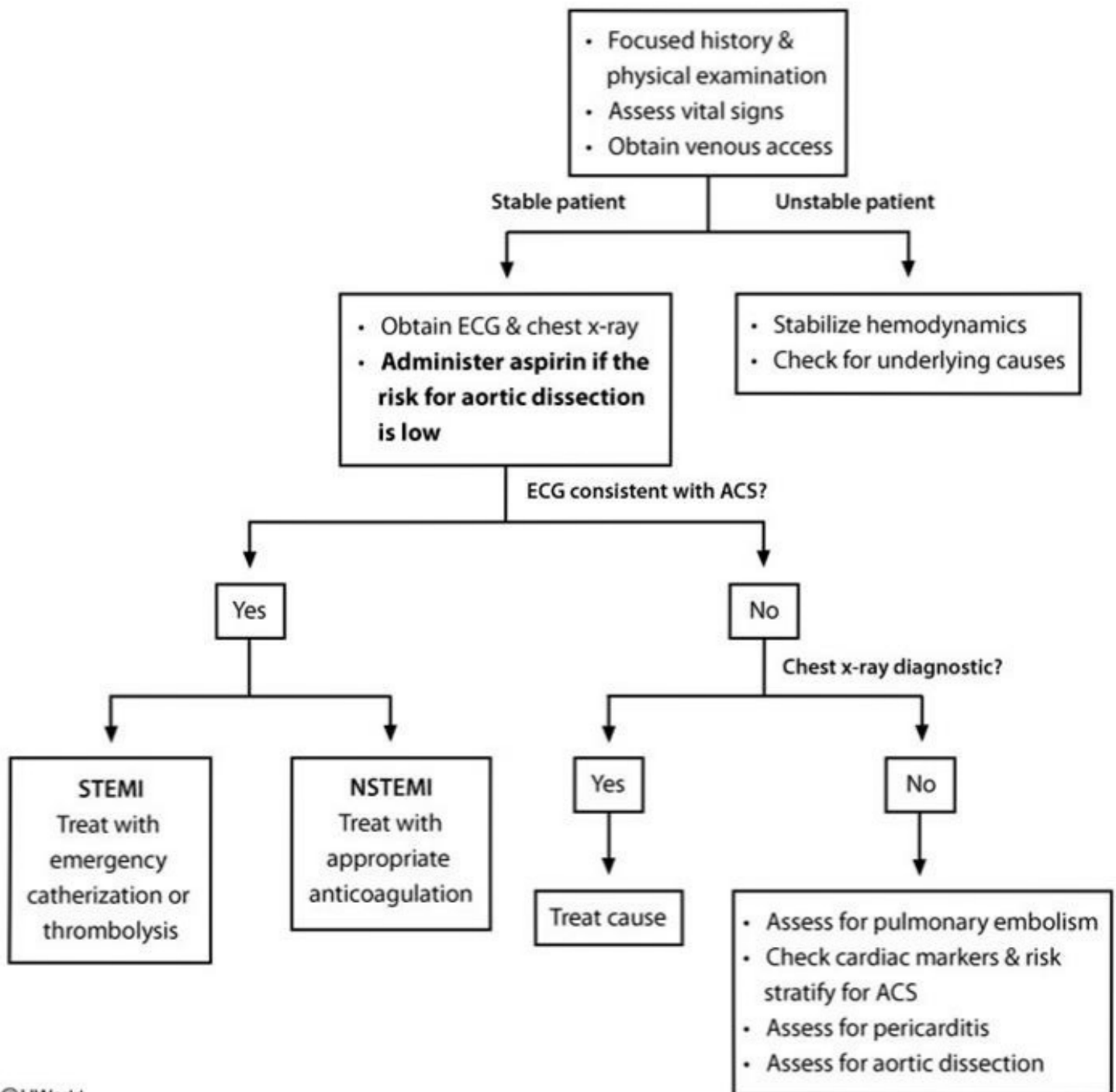
©UWorld

Electrical alternans

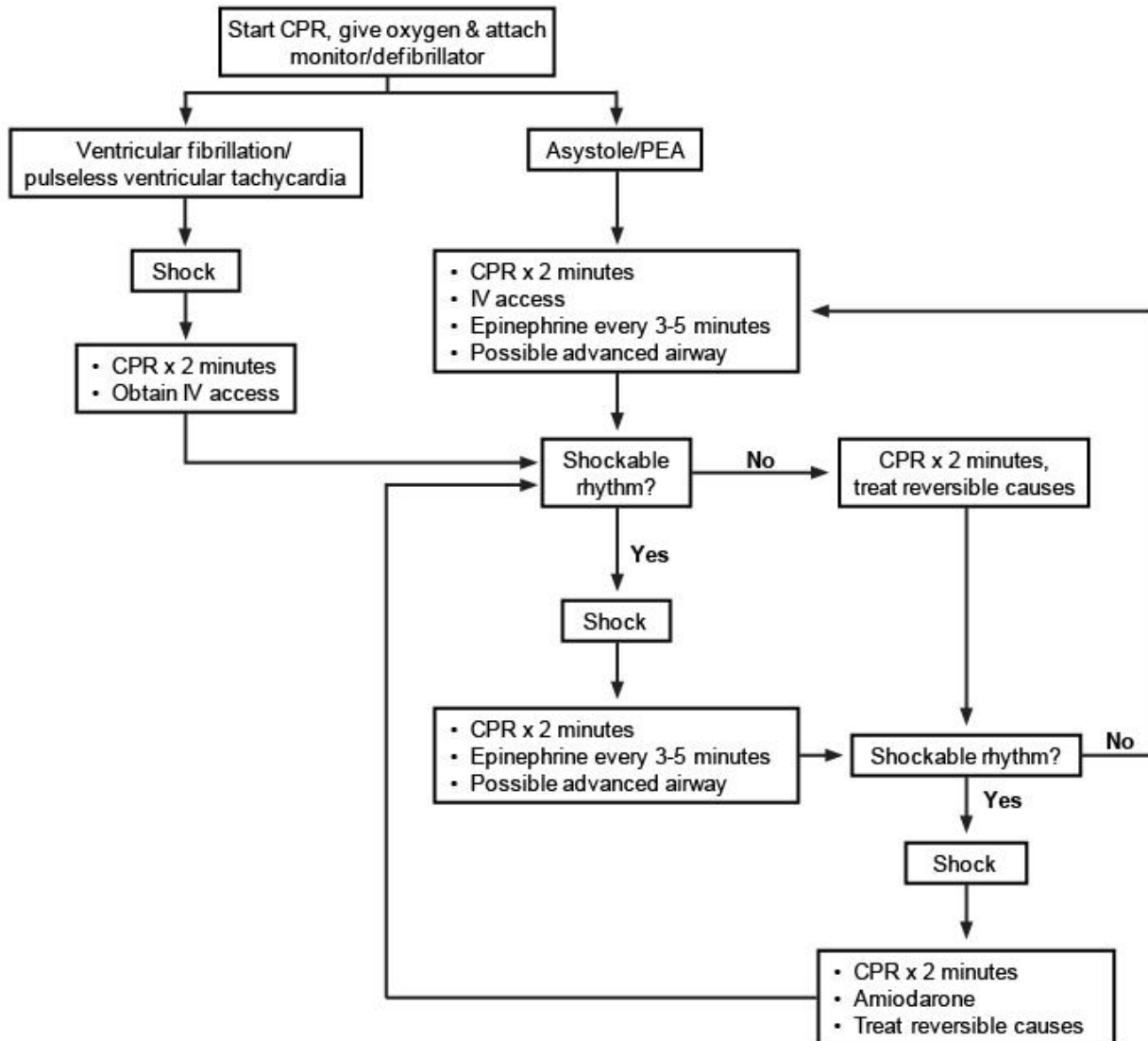


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Evaluation of chest pain in the emergency department

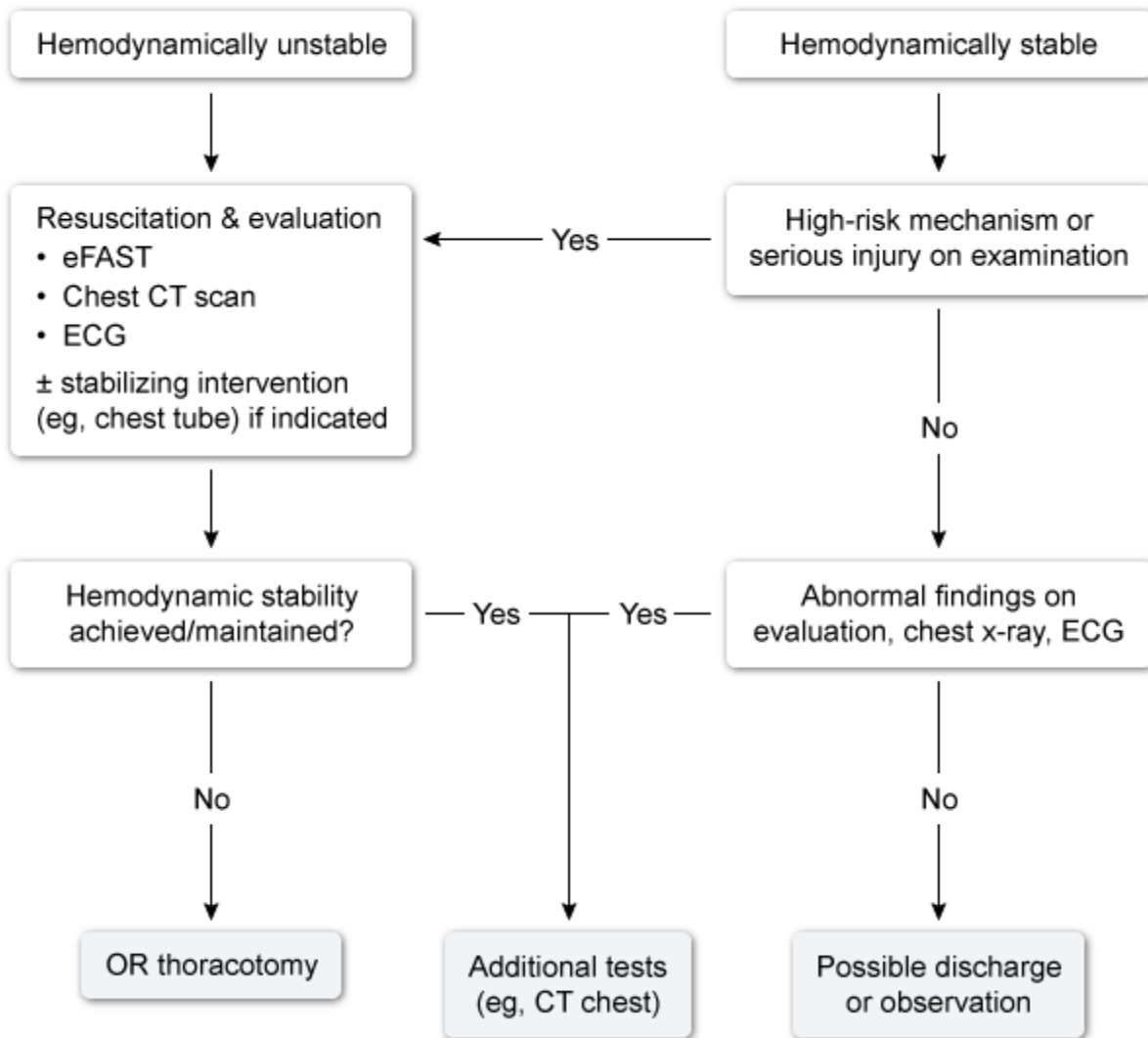


Approach to adult cardiac arrest



CPR = cardiopulmonary resuscitation; PEA = pulseless electrical activity.

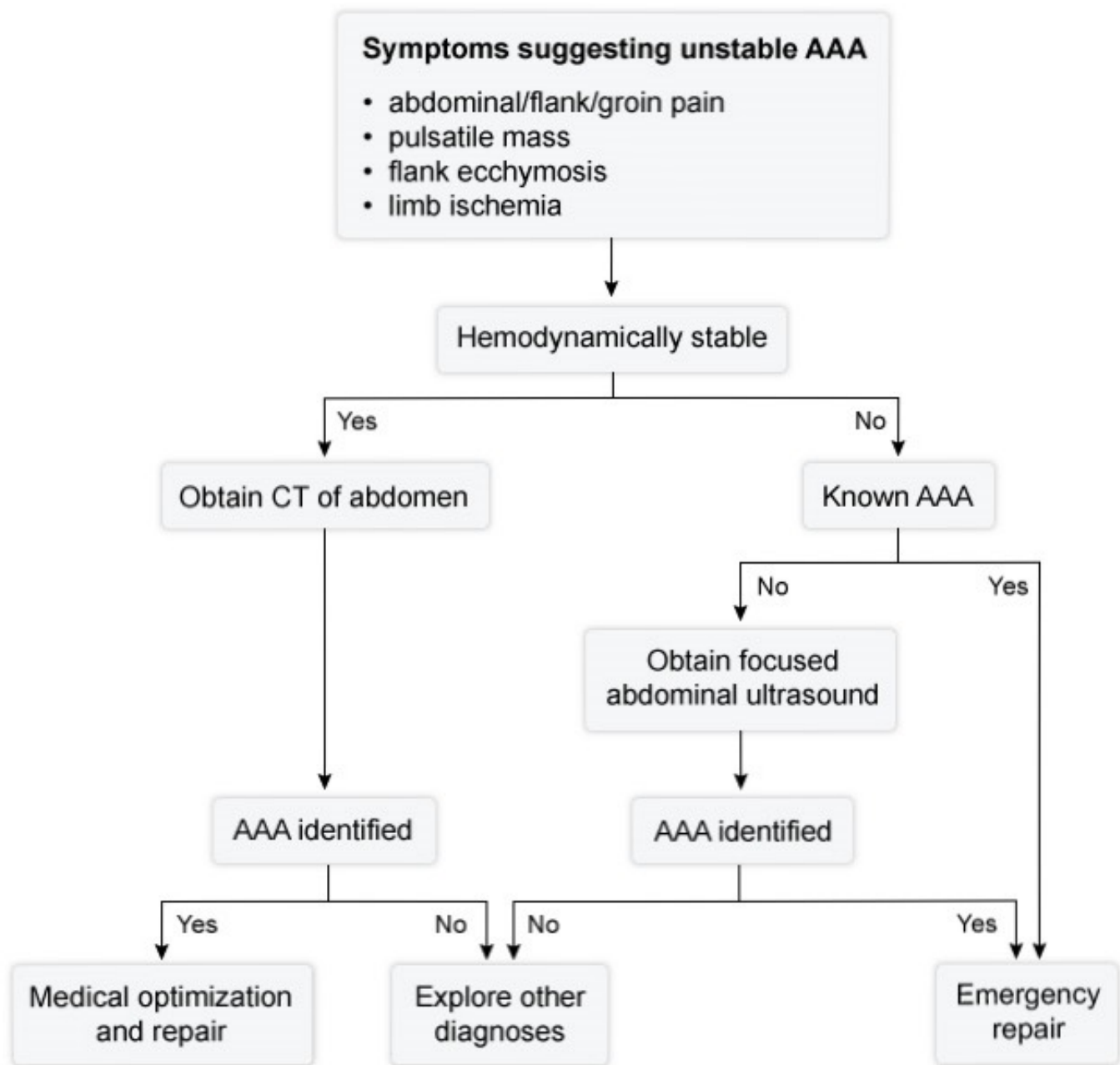
Blunt chest trauma



eFAST = extended focused assessment with sonography for trauma; **OR** = operating room.

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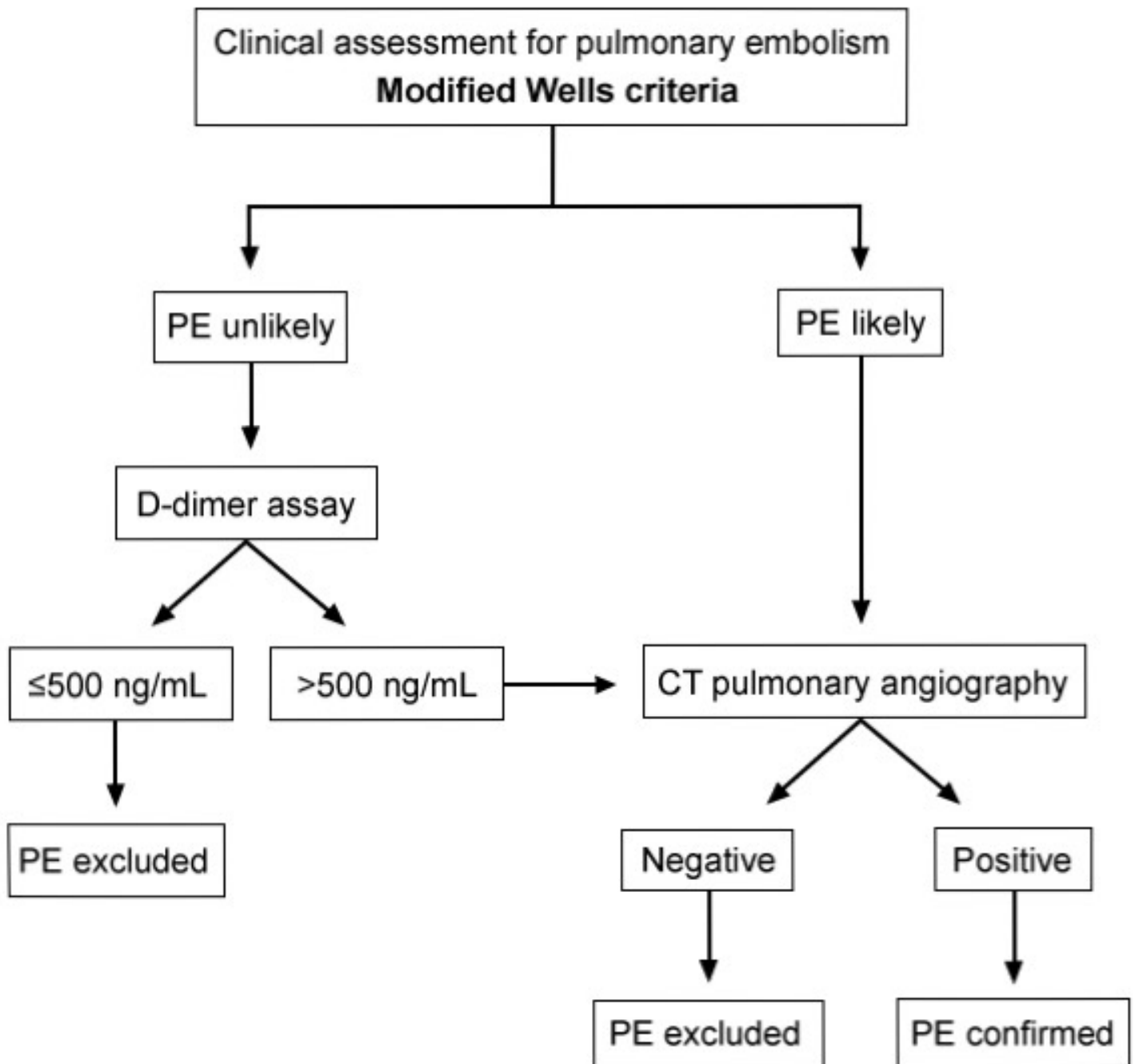
Evaluation of suspected unstable abdominal aortic aneurysm



AAA = abdominal aortic aneurysm.

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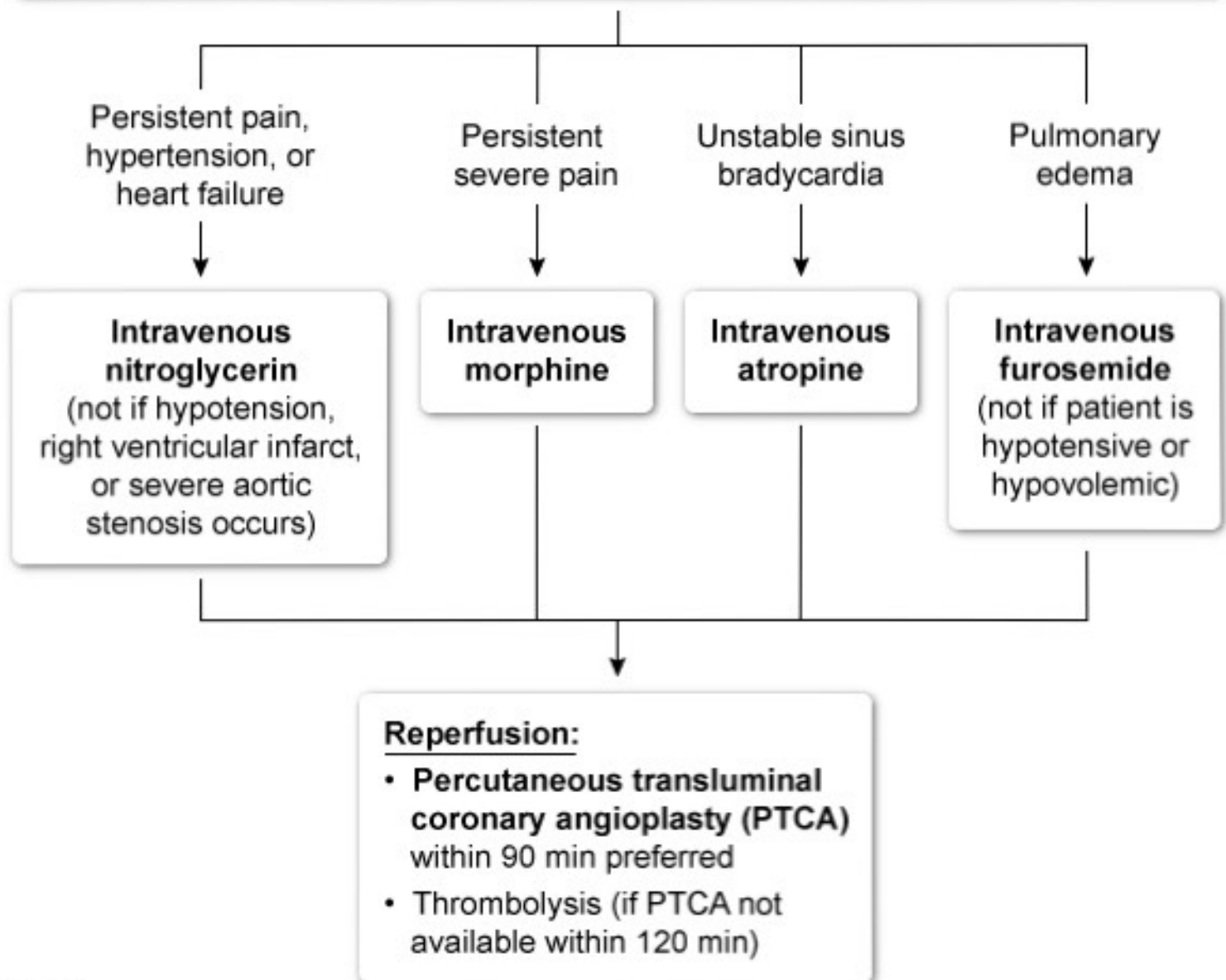
Diagnostic strategy in suspected pulmonary embolism



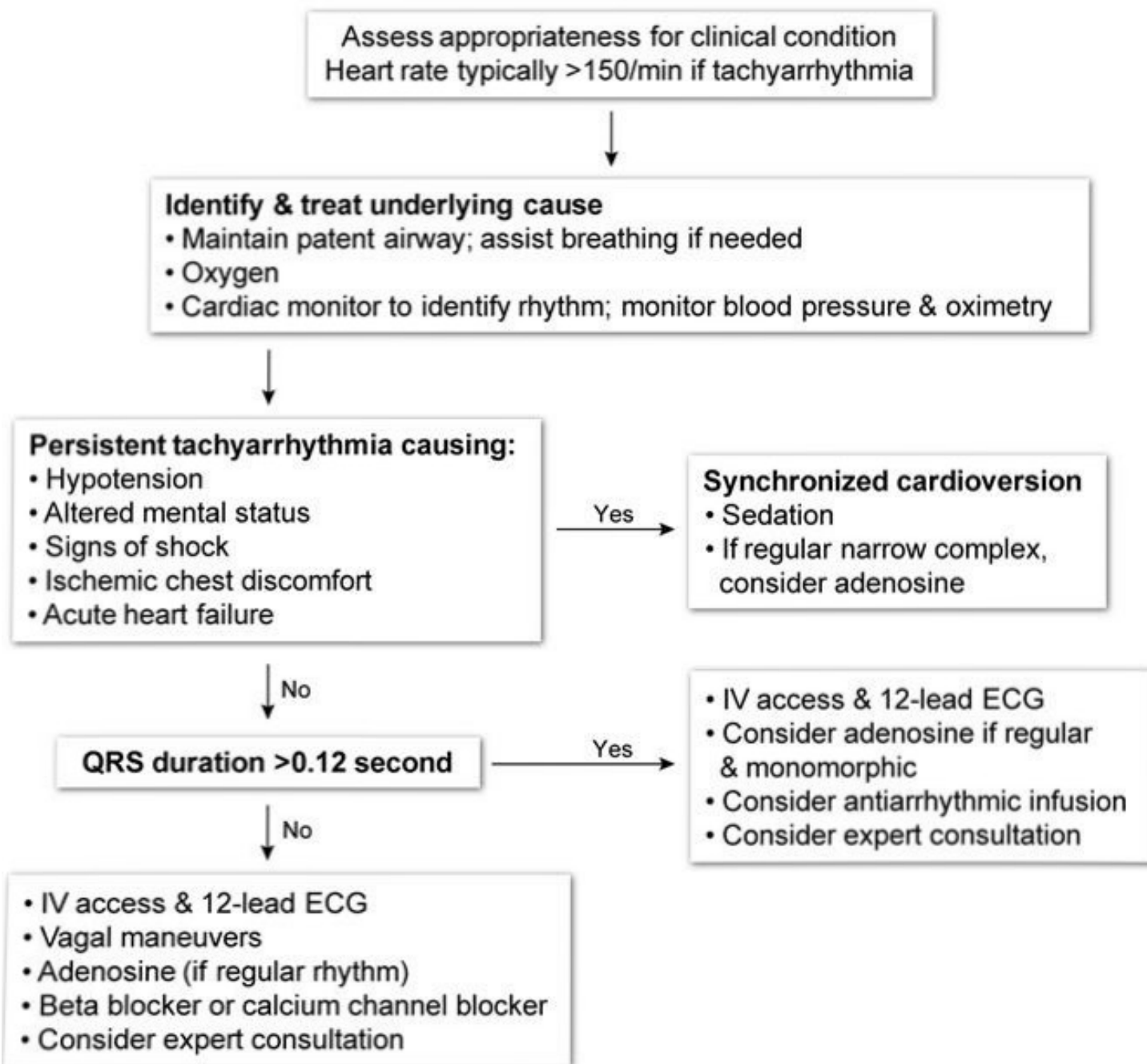
PE = pulmonary embolism.
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Initial stabilization of acute ST-elevation MI

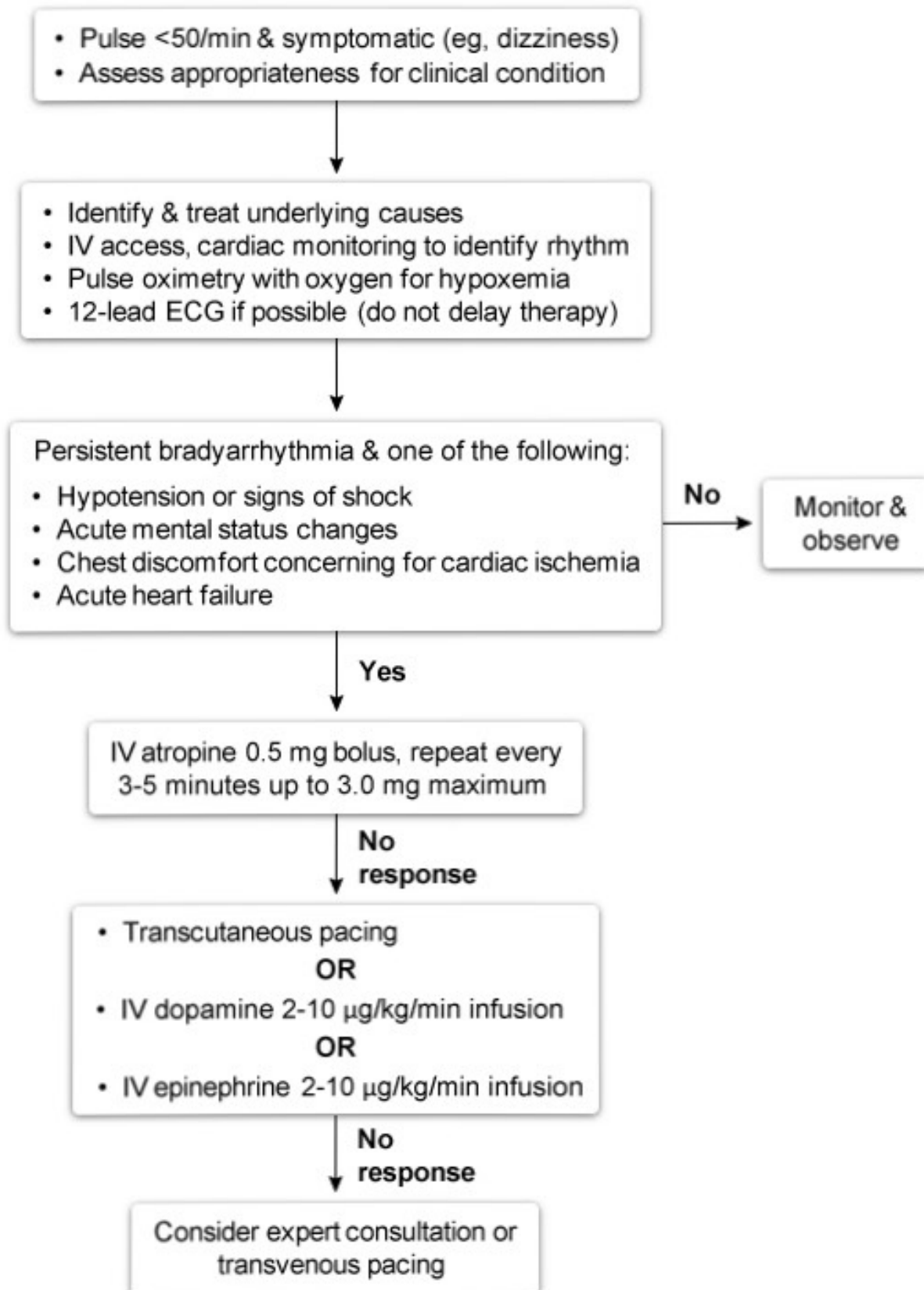
- Supplemental oxygen (if $\text{SaO}_2 < 90\%$ or dyspnea)
- **Aspirin** 325 mg
- **P2Y12 inhibitor** (eg, clopidogrel)
- Nitrates (sublingual)
- **Beta blocker** (unless hypotension, bradycardia, acute heart failure, heart block)
- High-dose **statin** (eg, atorvastatin 80 mg)
- **Anticoagulation** (drug depends on planned revascularization)



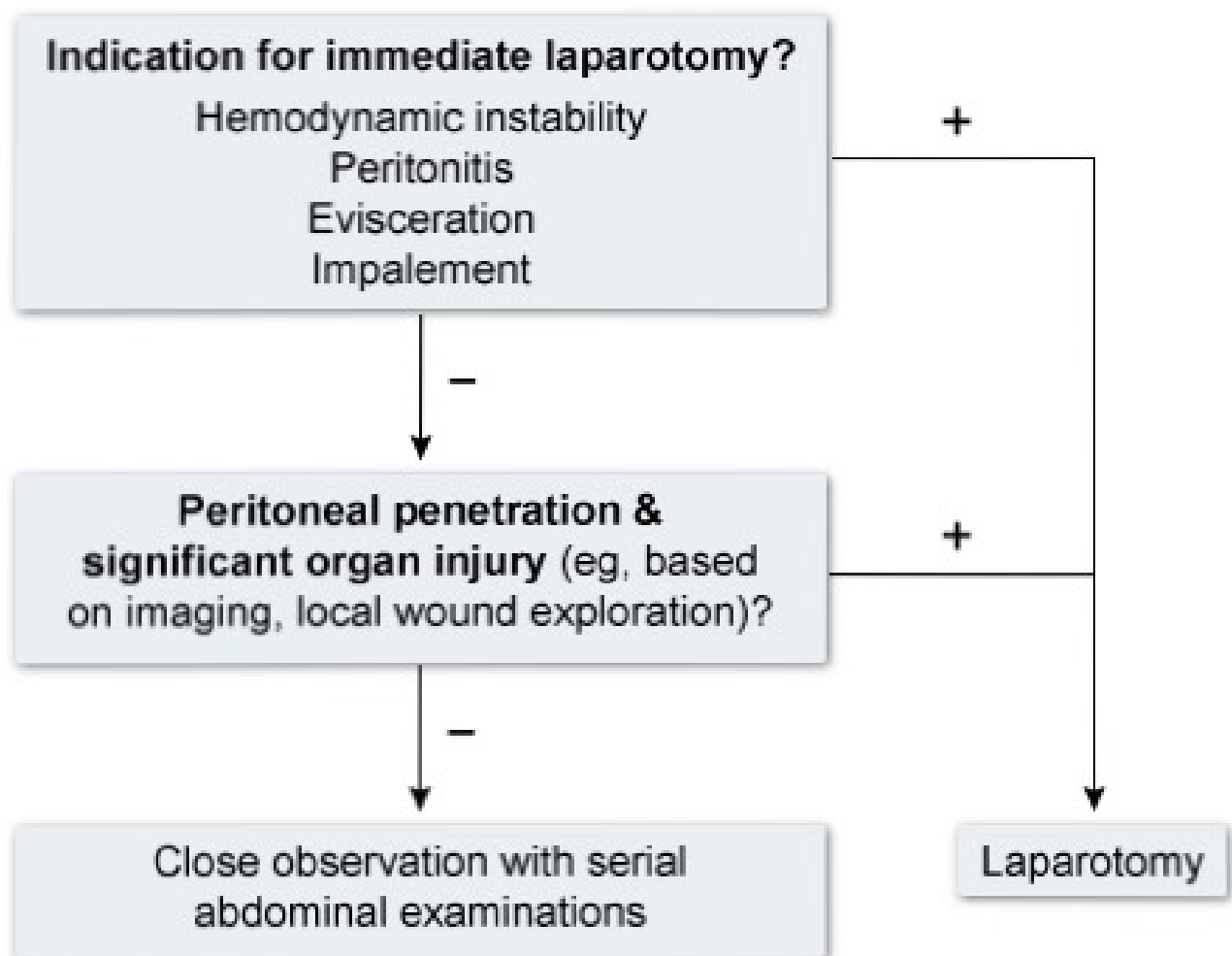
Adult tachycardia algorithm (with pulse) – Advanced Cardiac Life Support guidelines



Approach to sinus bradycardia in adults



Penetrating abdominal trauma

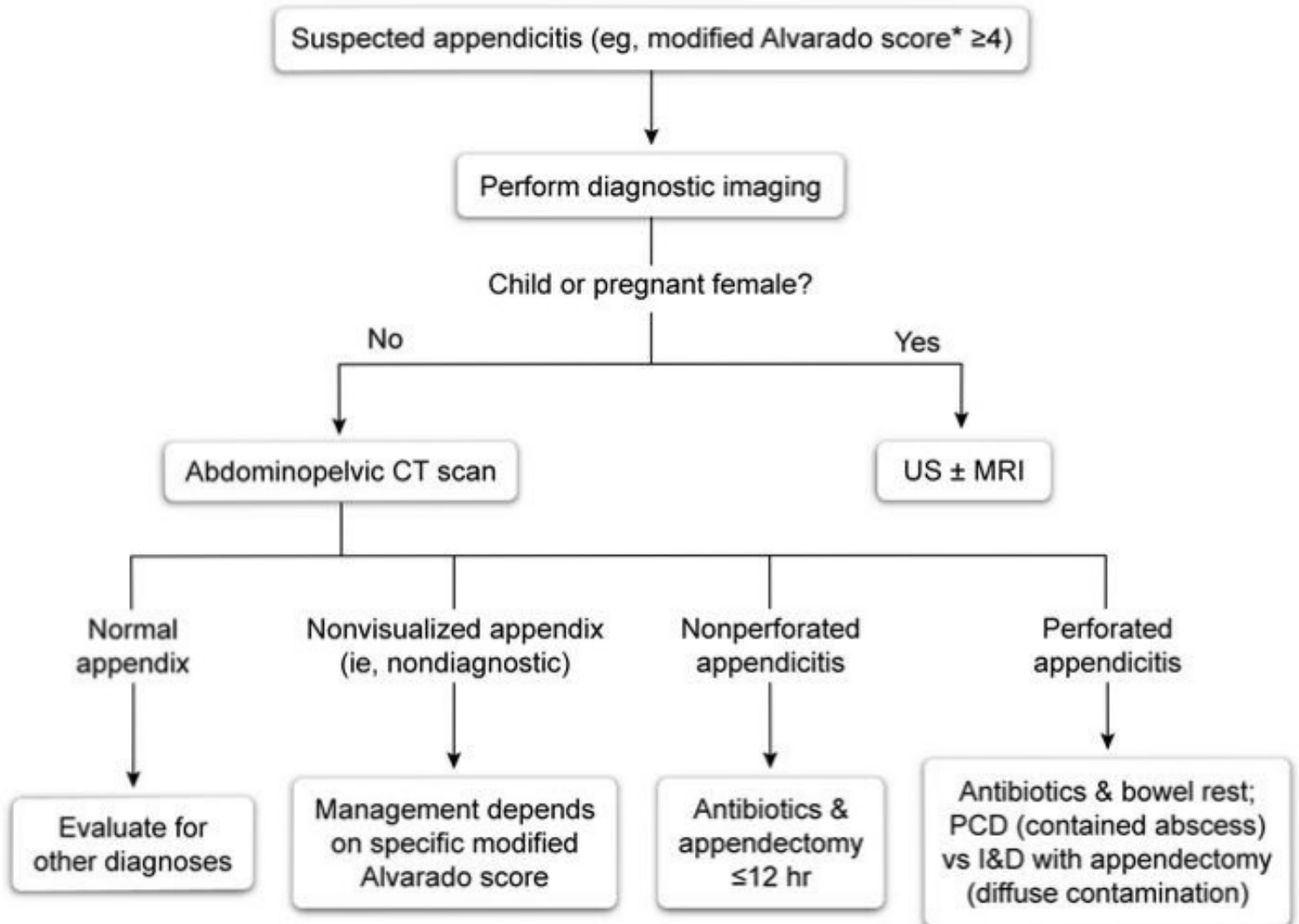




a common inflammatory disease that affects the scalp (dandruff), face (eyebrows, nasolabial folds, and external ear canal/posterior ear), chest, and intertriginous areas. SD occurs in all ages but is most common in the first year of life and again at age 30-60. It is also associated with central nervous system disorders (especially **Parkinson disease**) and **HIV**.

The diagnosis of SD is clinical, with typical findings characterized by pruritic, erythematous plaques with fine, loose, yellow, and **greasy-looking scales**. SD primarily affects areas with numerous sebaceous glands, although sebum production in affected patients is typically normal. *Malassezia* species may play a role in the pathogenesis of SD, and **topical antifungal agents** (eg, ketoconazole, selenium sulfide) are effective in treating this condition

Management of suspected appendicitis



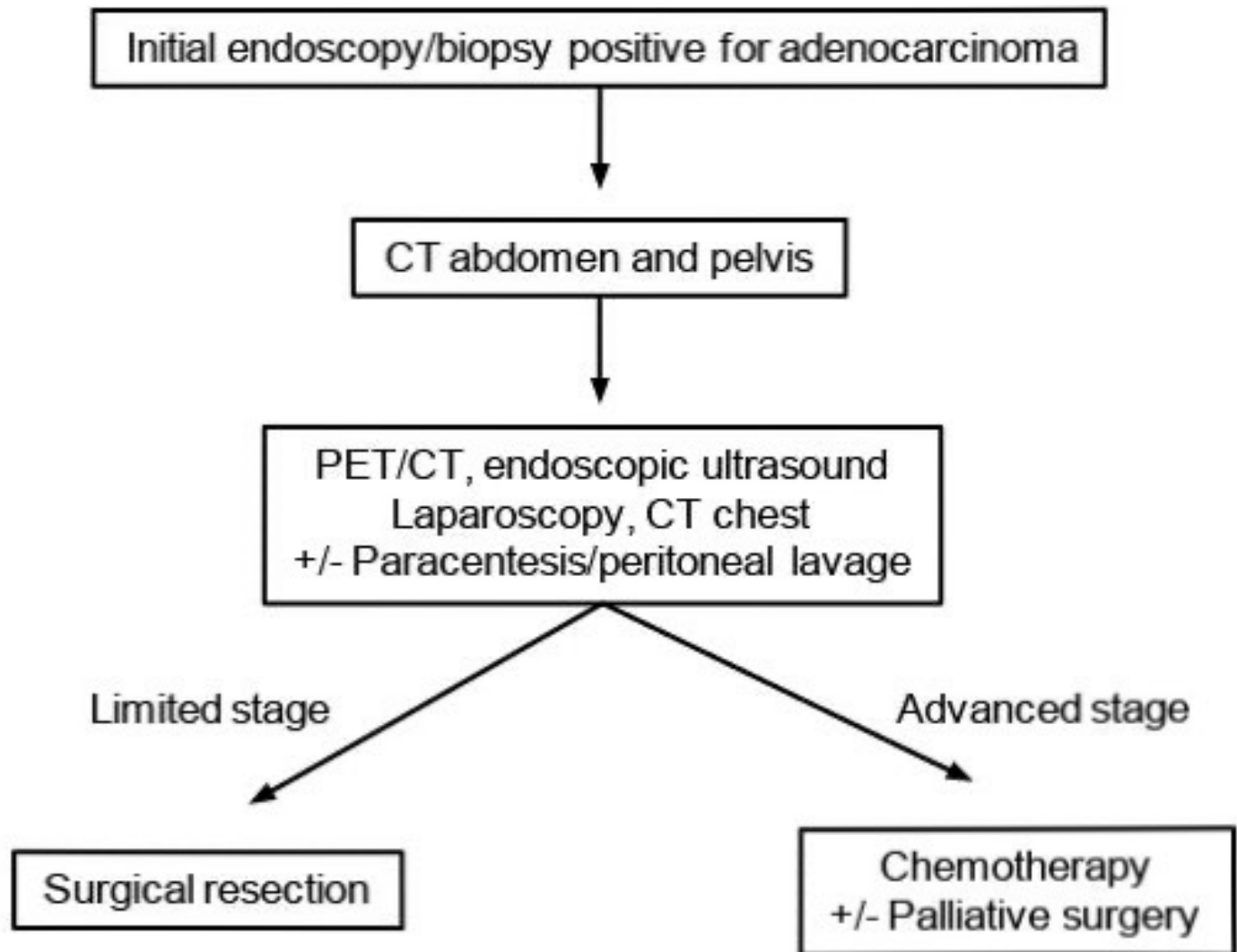
*Modified Alvarado score

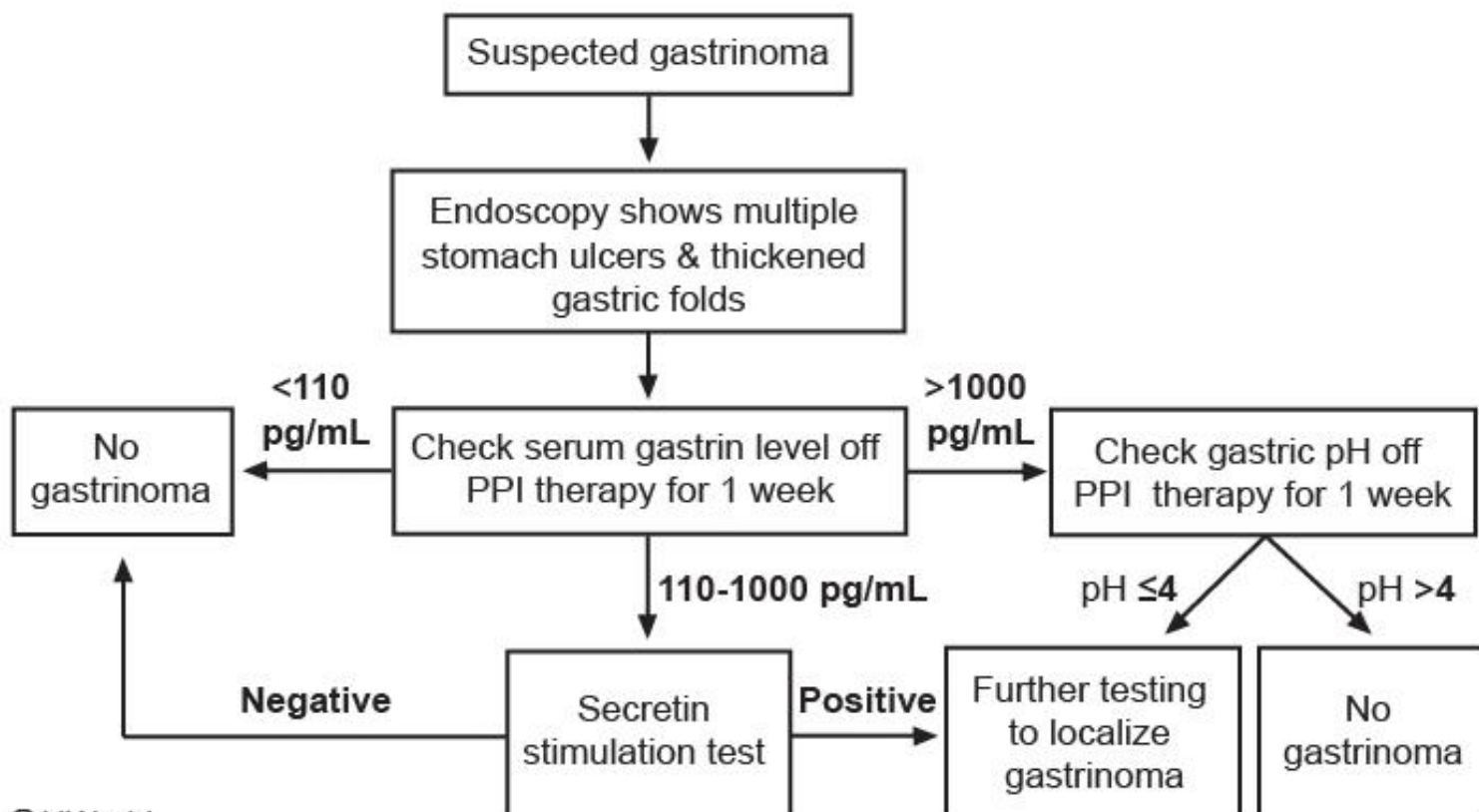
1 point each: migratory RLQ pain, anorexia, nausea or vomiting, RLQ rebound tenderness, fever >37.5 C (99.5 F).

2 points each: RLQ tenderness, leukocytes >10,000/mm³.

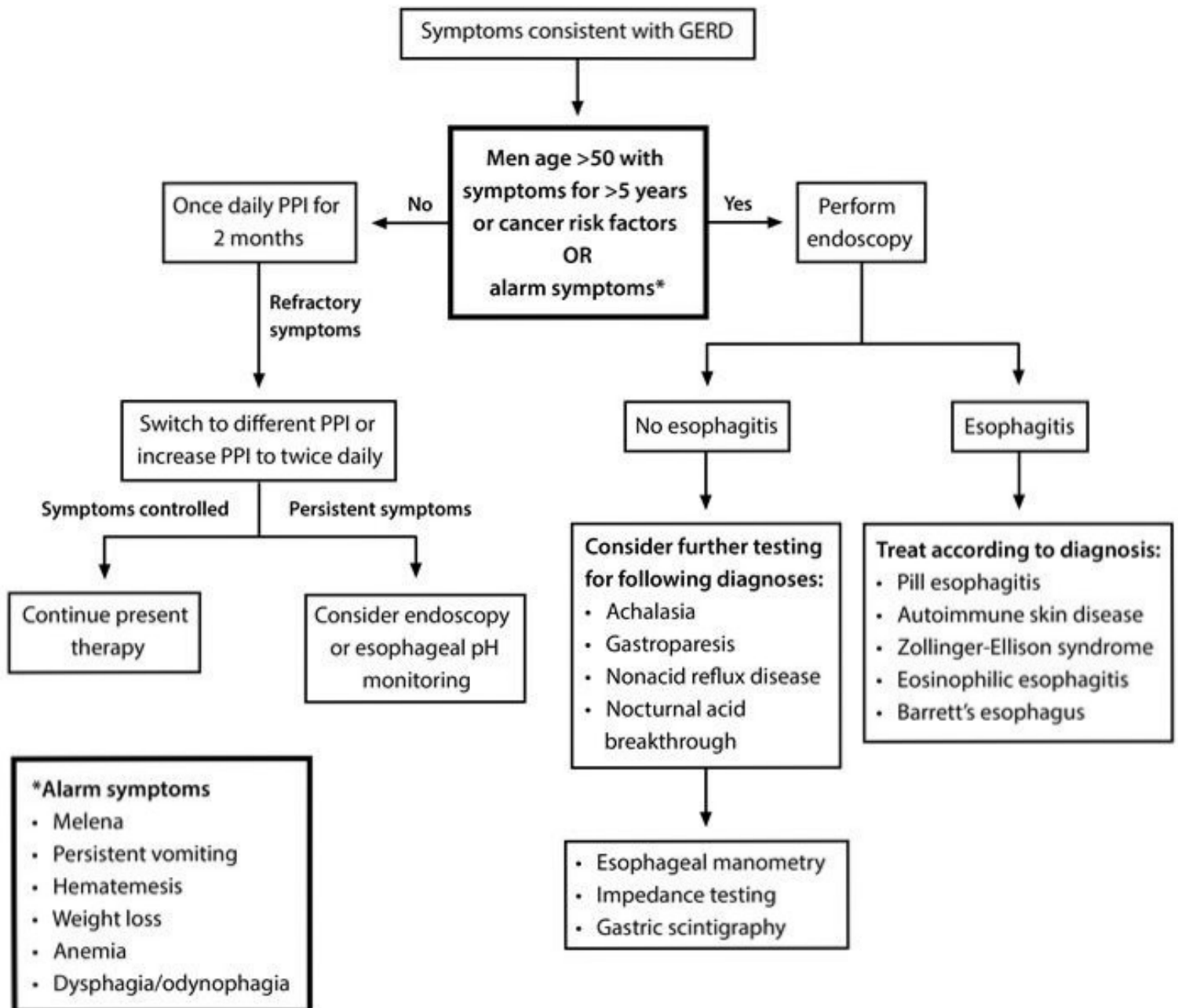
I&D = irrigation & drainage; PCD = percutaneous drainage; RLQ = right lower quadrant; US = ultrasound.

Staging of gastric adenocarcinoma



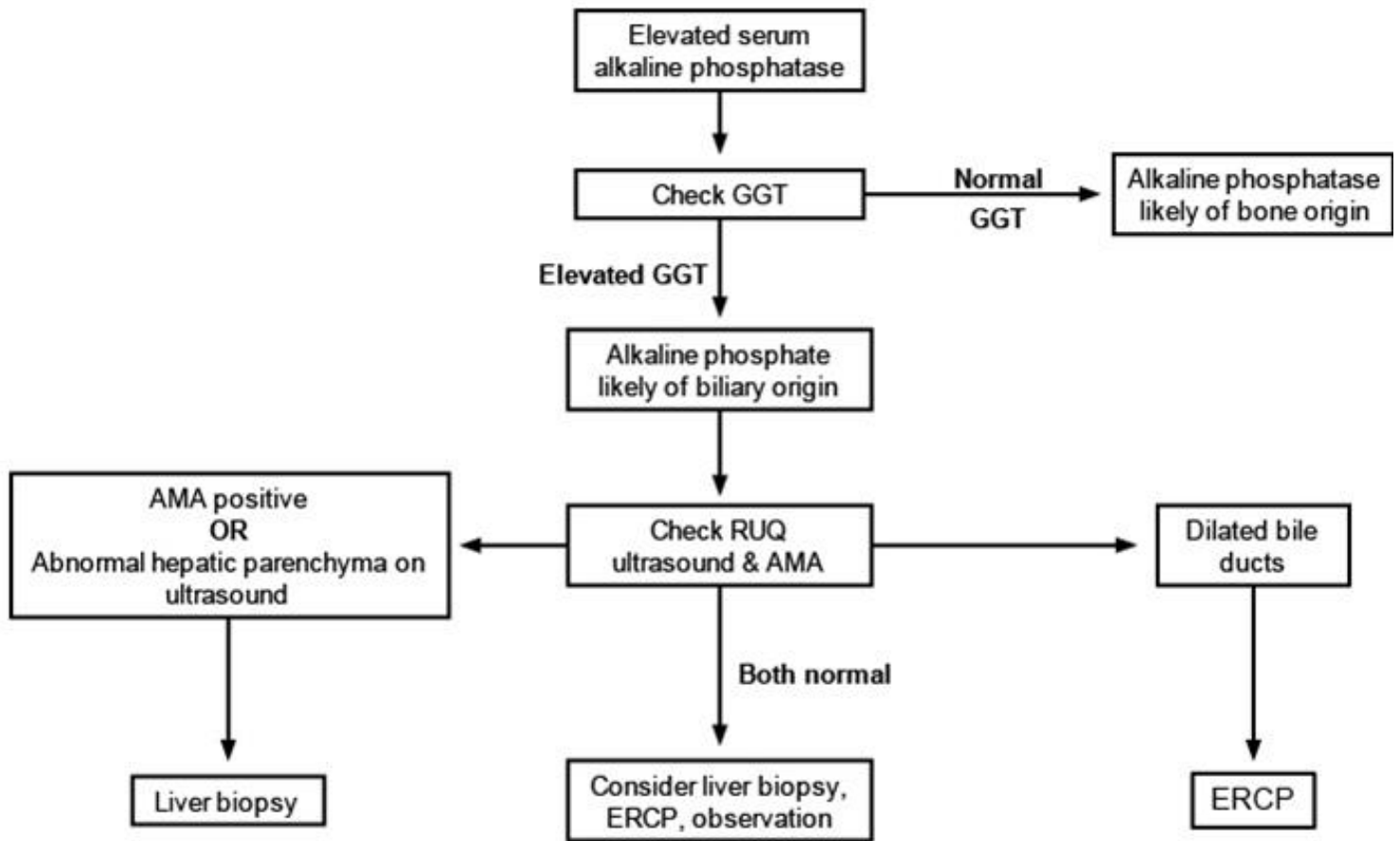


Management of gastric esophageal reflux disease



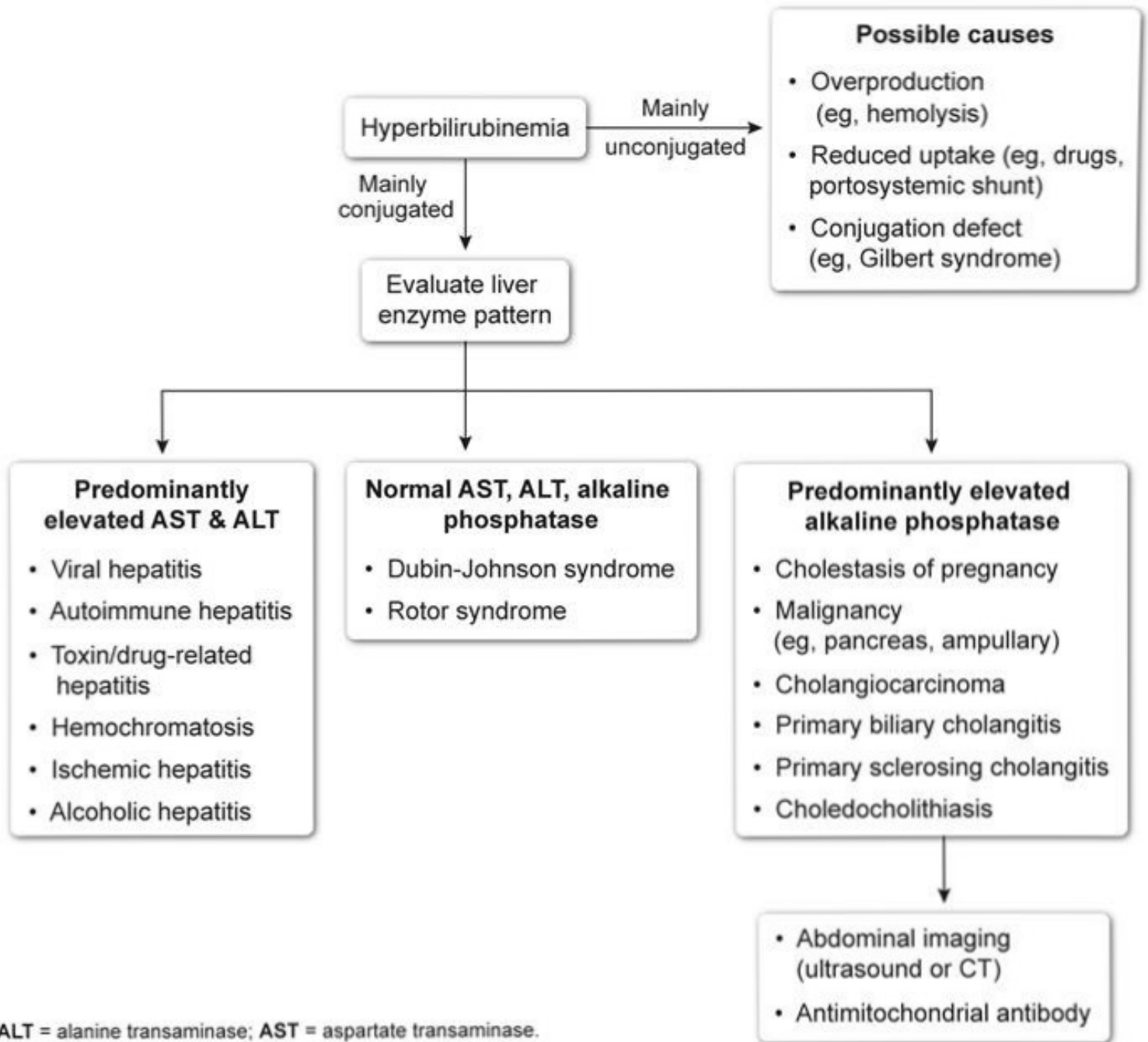
GERD = gastric esophageal reflux disease; PPI = proton pump inhibitor.

Evaluation of elevated alkaline phosphatase



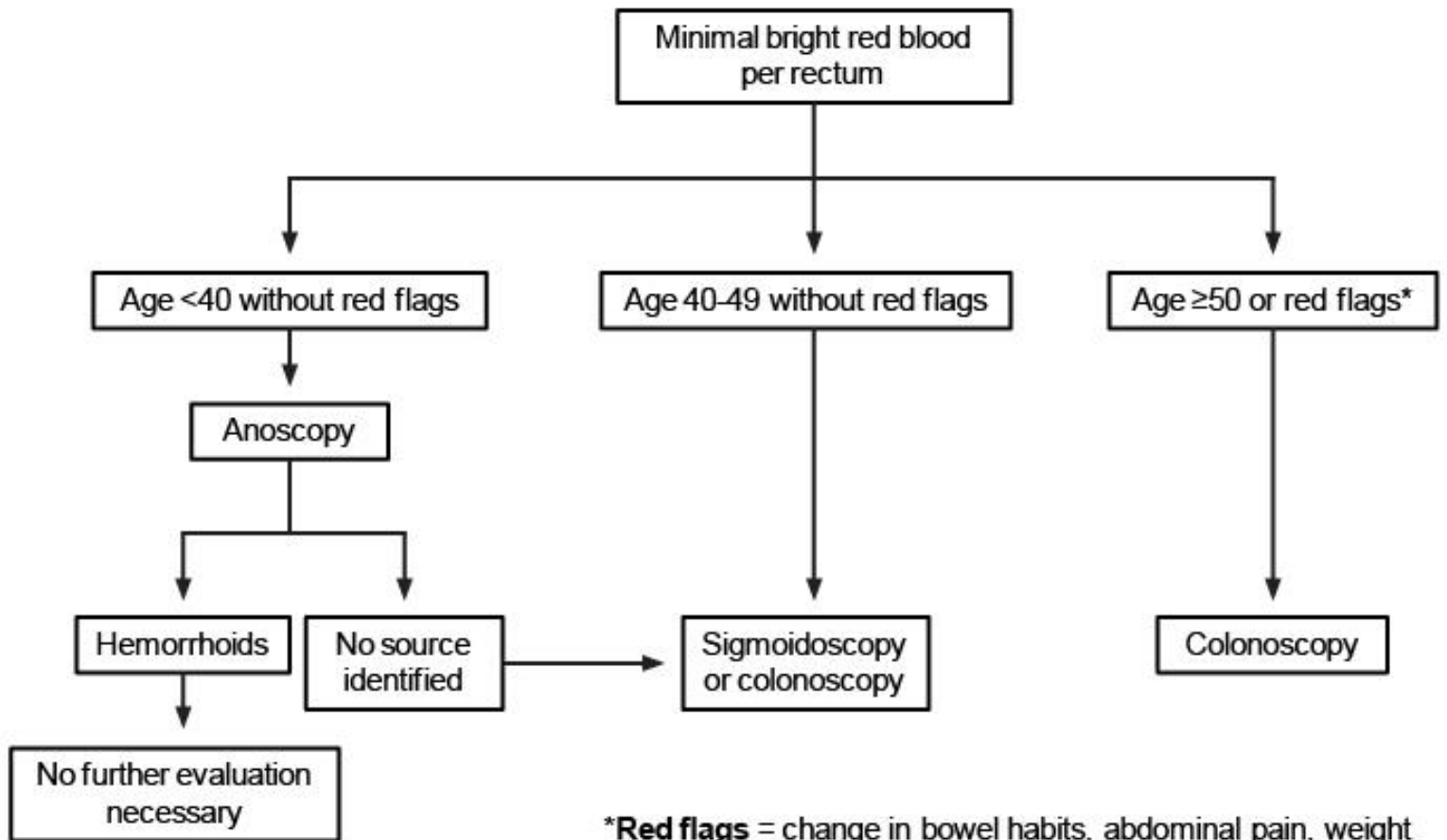
AMA = antimitochondrial antibody; **ERCP** = endoscopic retrograde cholangiopancreatogram; **GGT** = gamma-glutamyltransferase; **RUQ** = right upper quadrant.

Approach to hyperbilirubinemia in adults



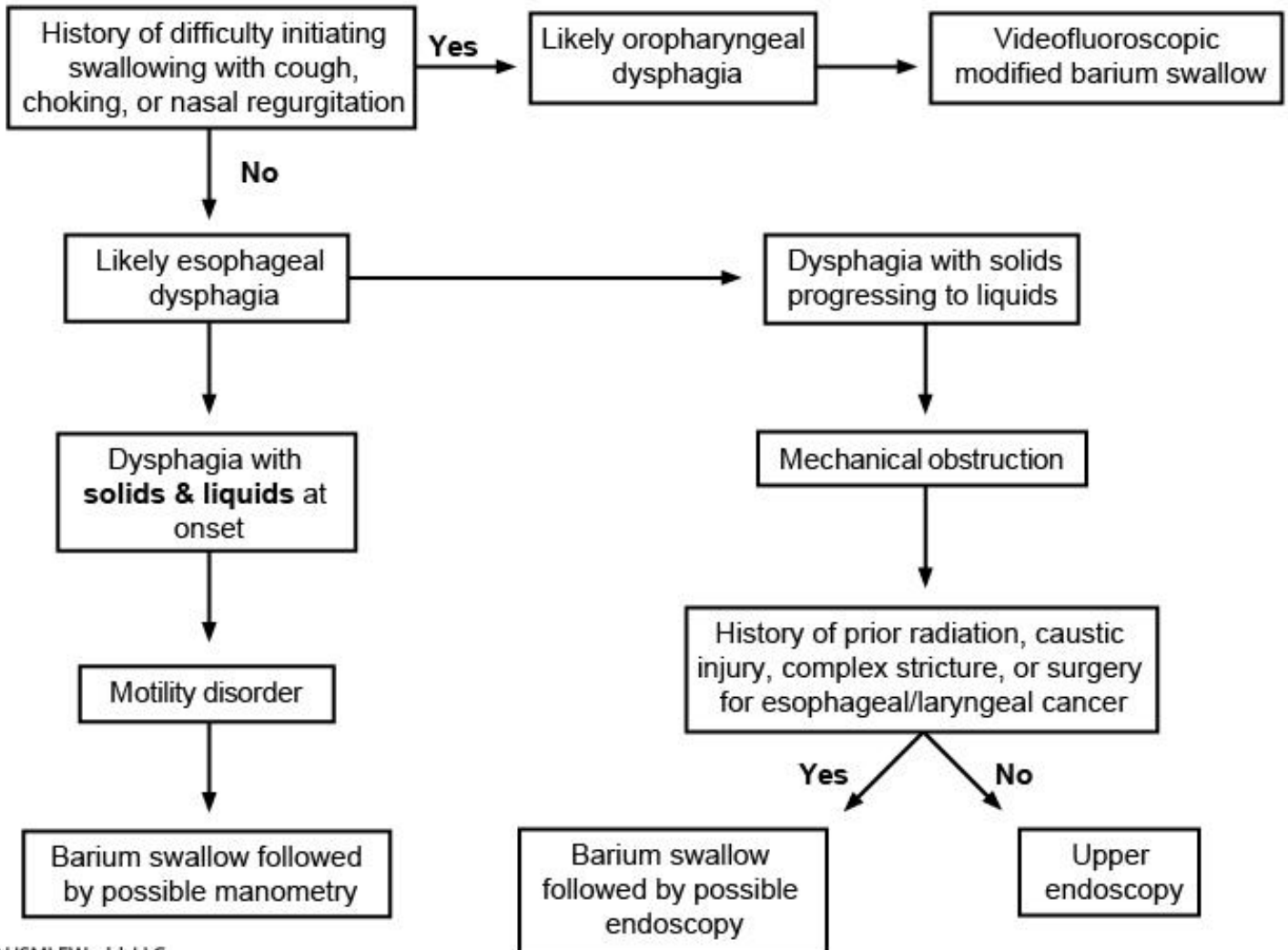
ALT = alanine transaminase; AST = aspartate transaminase.

Evaluation of minimal bright red blood per rectum

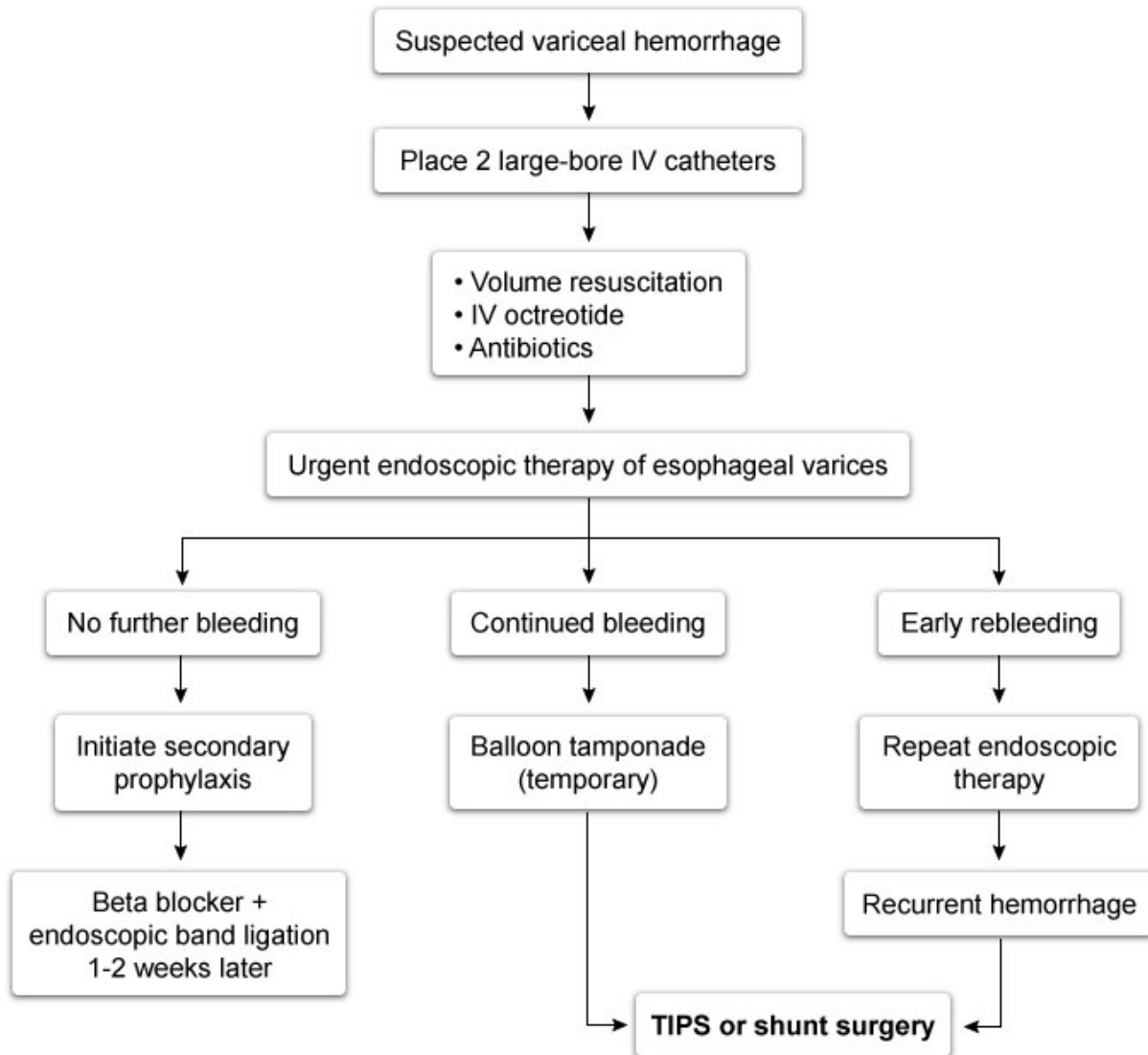


***Red flags** = change in bowel habits, abdominal pain, weight loss, iron deficiency anemia, family history of colon cancer.

Evaluation of dysphagia



Variceal hemorrhage bleed algorithm

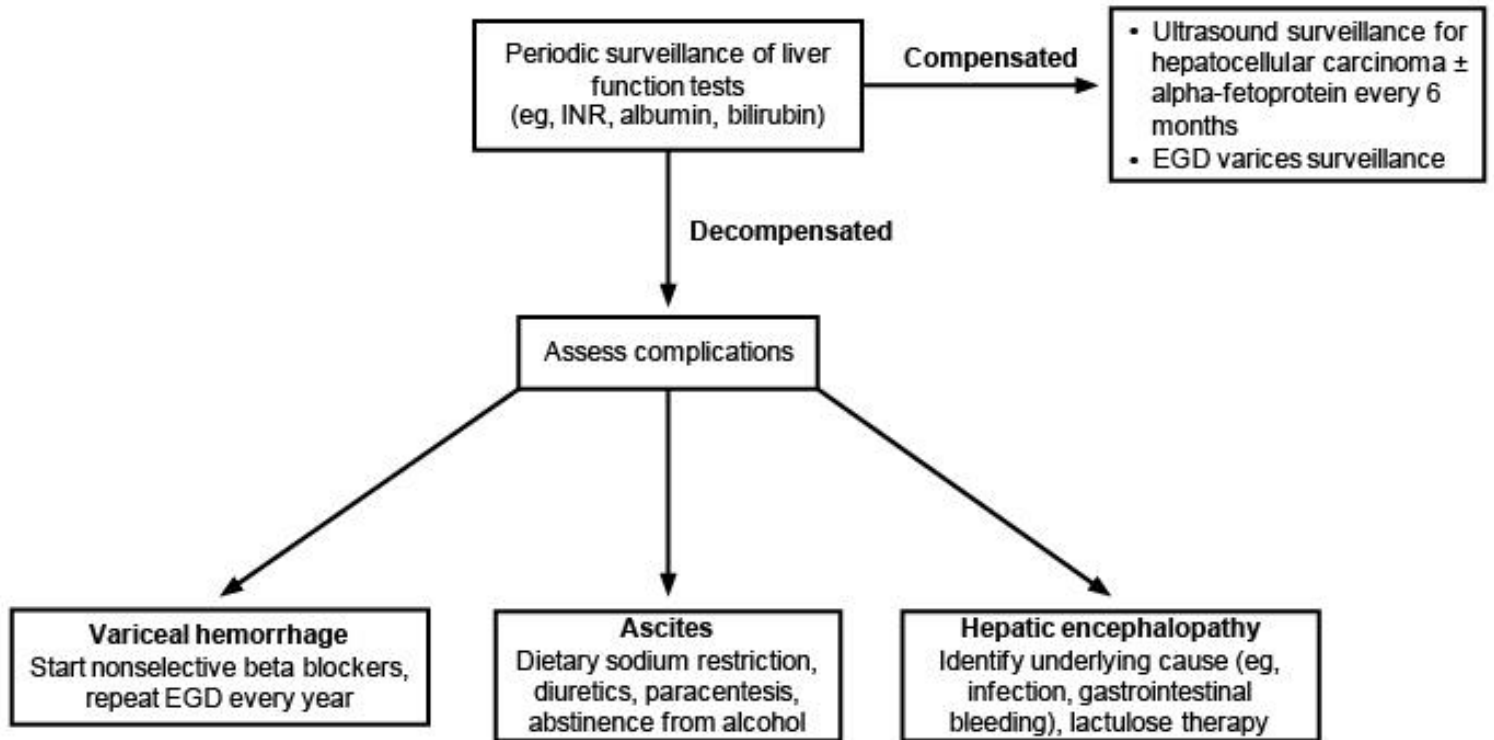


IV = intravenous; TIPS = transjugular intrahepatic portosystemic shunt.

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Urgent **endoscopy** (within 12 hours) can diagnose and treat (eg, endoscopic band ligation, sclerotherapy) active bleeding. Patients with uncontrollable bleeding require temporary balloon tamponade (eg, Sengstaken-Blakemore, Minnesota, Linton-Nachlas tubes) as a short-term measure until more definitive therapy, including transjugular intrahepatic portosystemic shunt (TIPS) or shunt surgery (**Choice E**). Patients without further bleeding after endoscopy can be monitored and receive secondary prophylaxis (**beta blocker**) with repeat endo

Management of cirrhosis

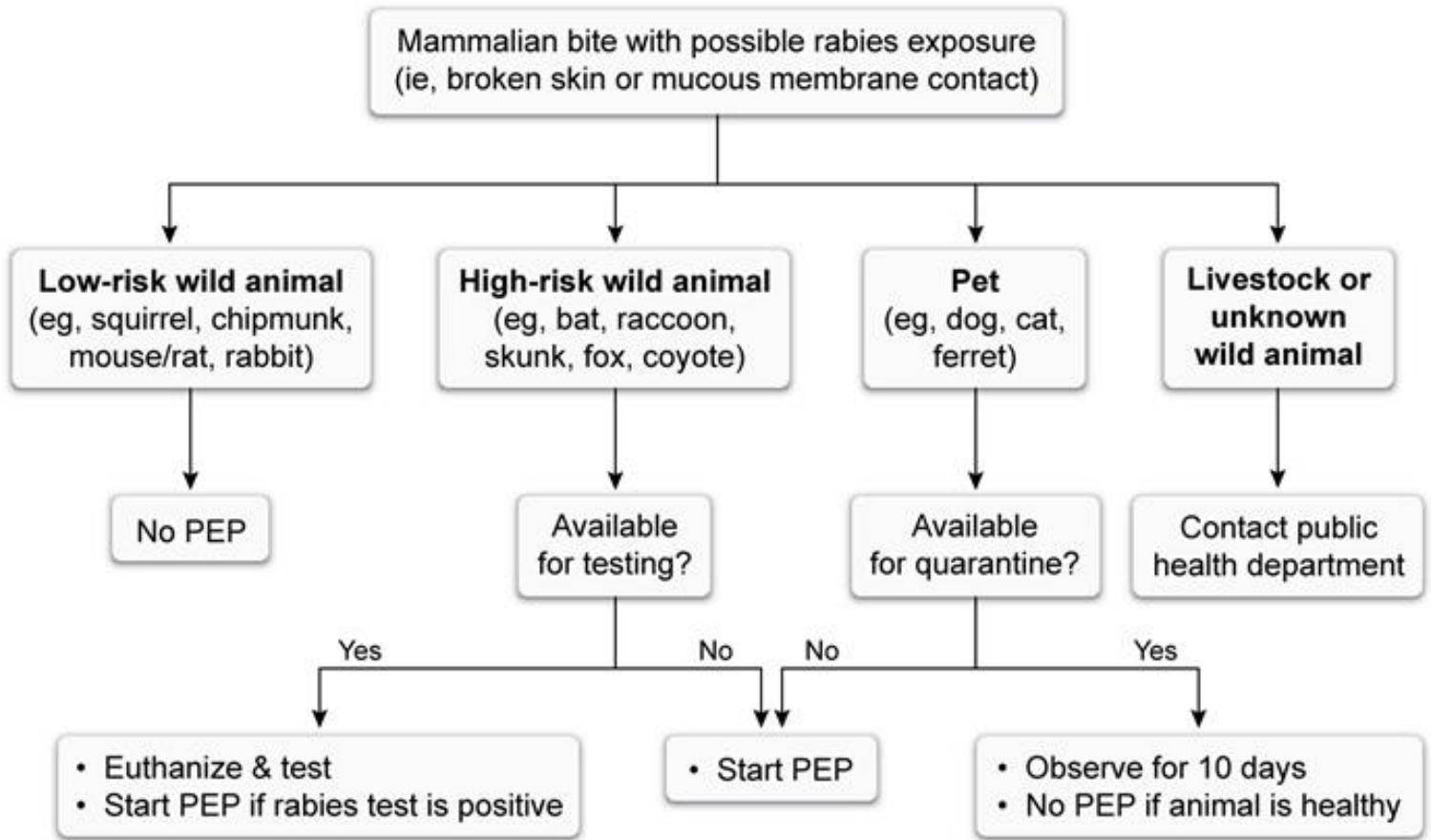


EGD = esophagogastroduodenoscopy.

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scopic band ligation 1-2 weeks later.

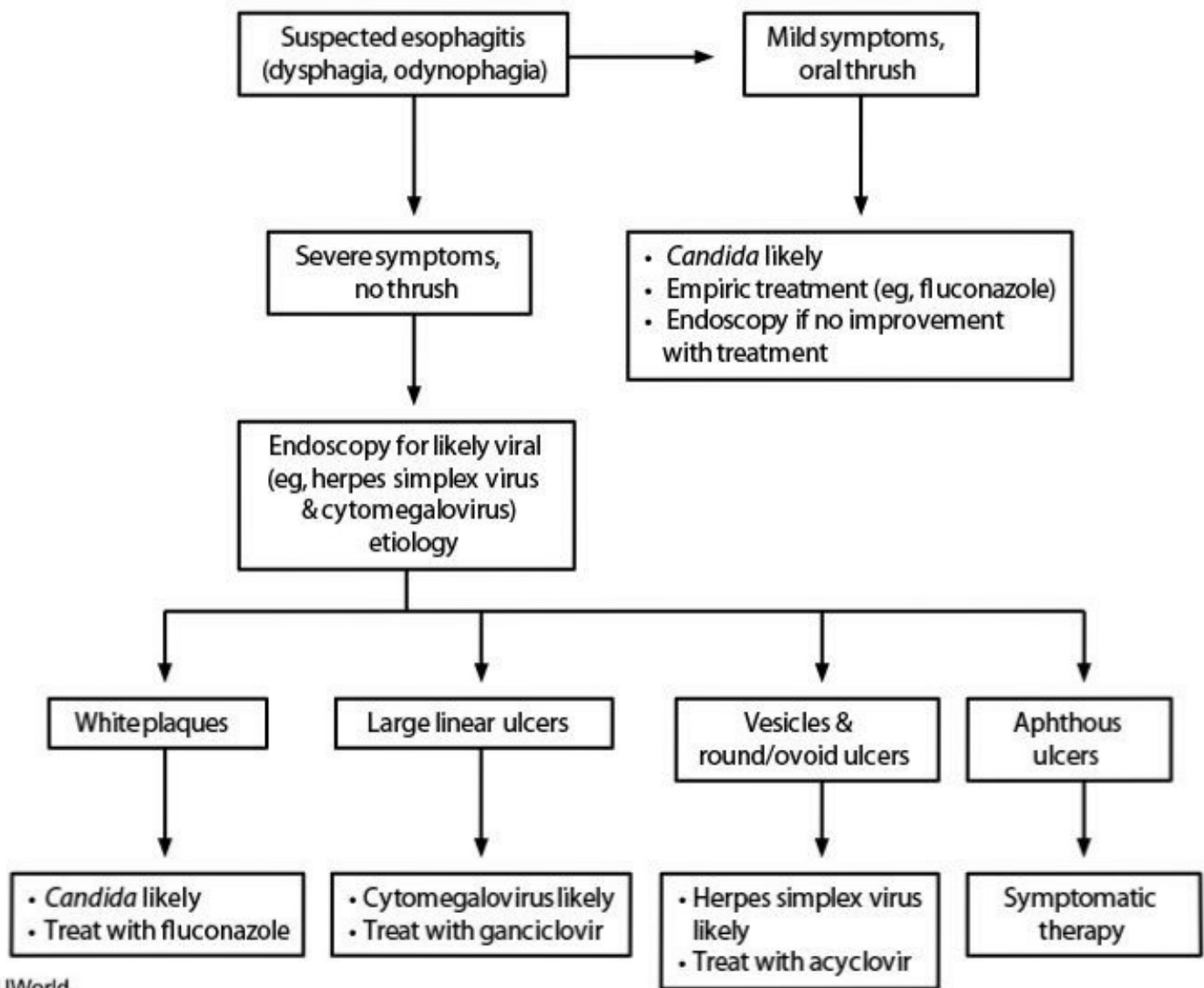
Rabies PEP



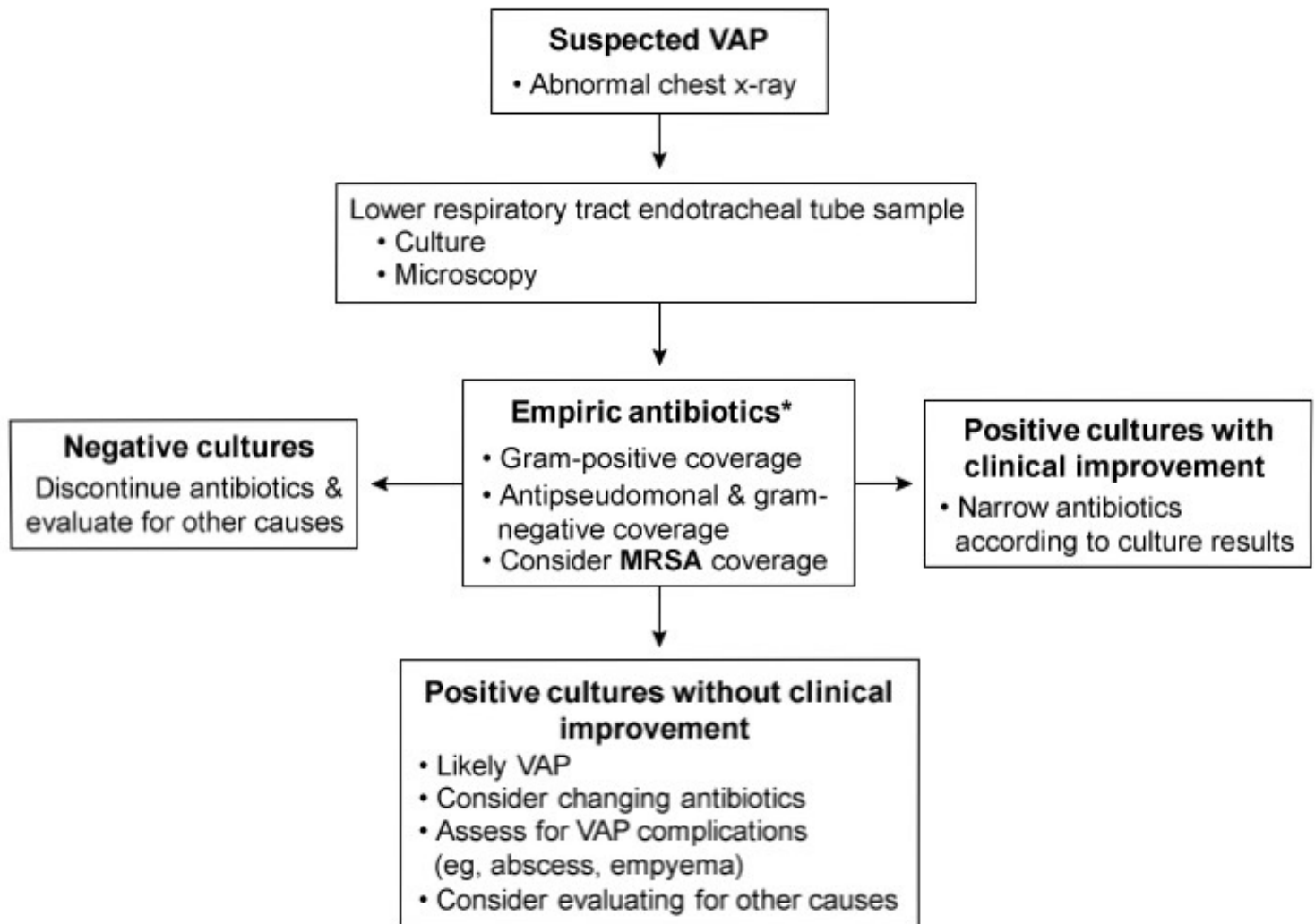
PEP = postexposure prophylaxis.

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Approach to odynophagia & dysphagia in patients with HIV



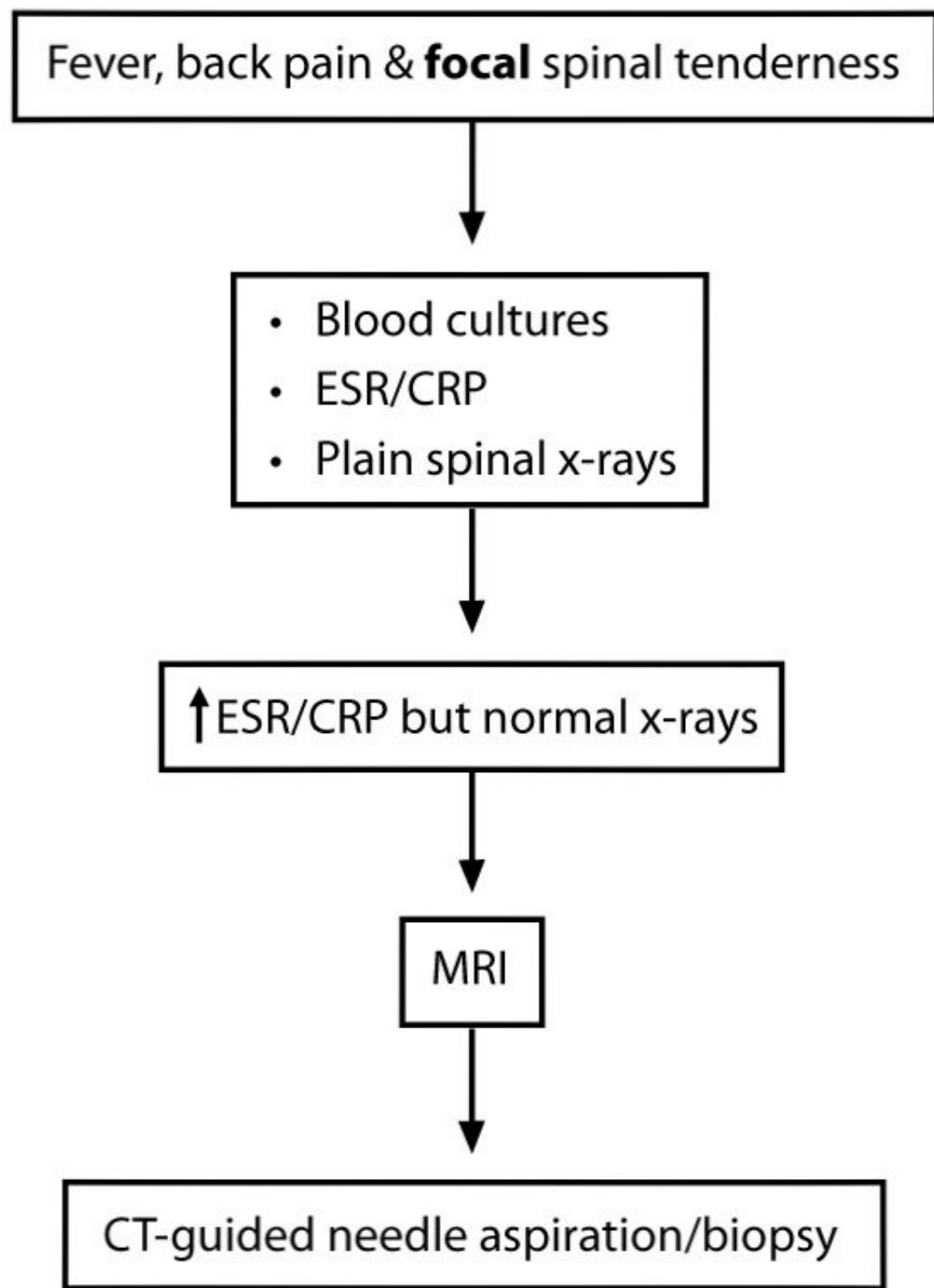
Evaluation of suspected ventilator-associated pneumonia



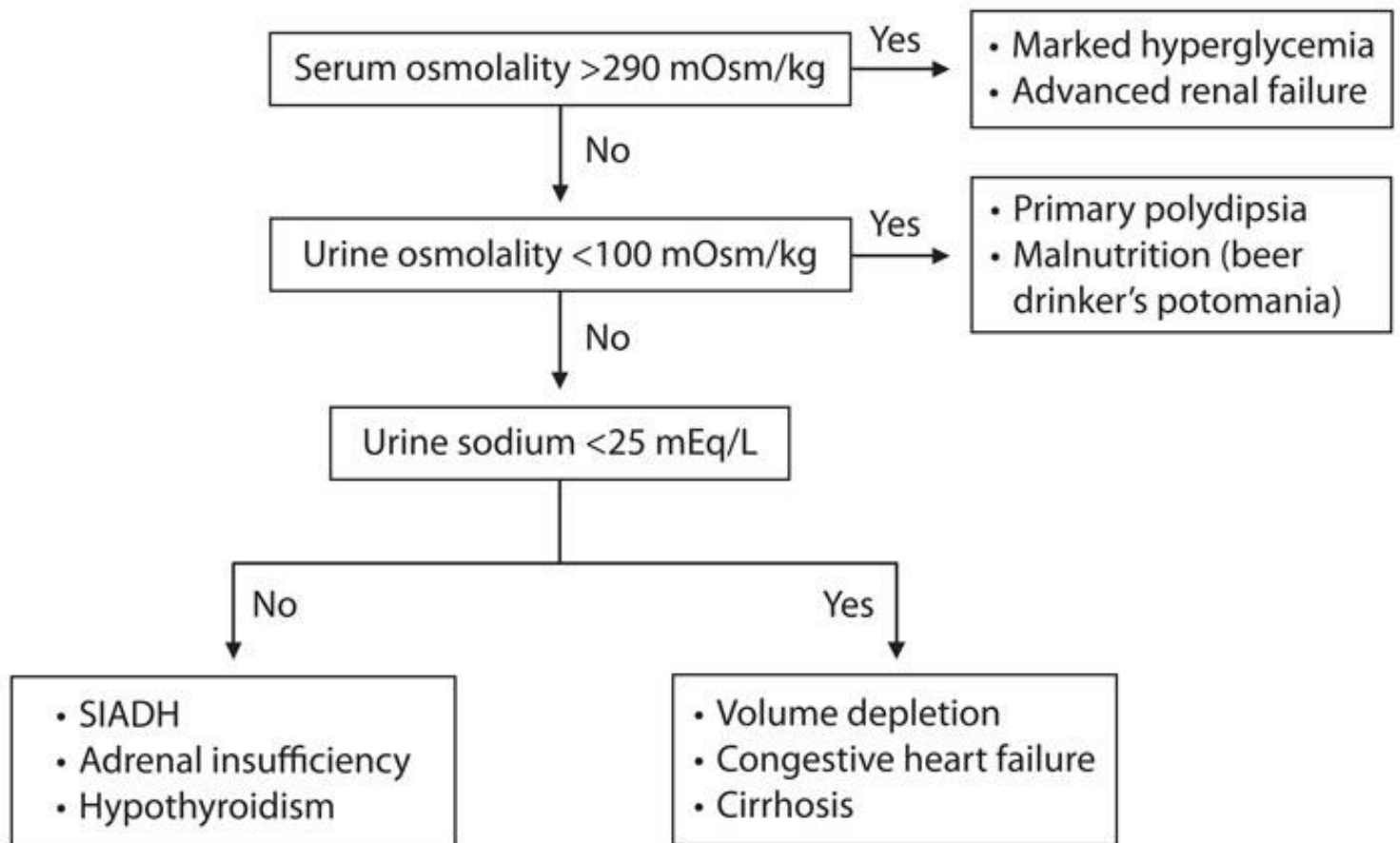
MRSA = methicillin-resistant *Staphylococcus aureus*; **VAP** = ventilator-associated pneumonia.

*Empiric coverage depends on drug-resistance pattern at institution.

Evaluation of vertebral osteomyelitis

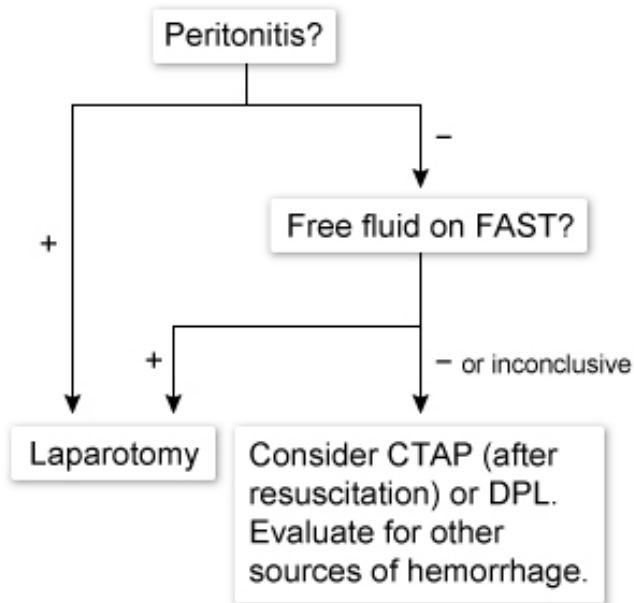


Evaluation of hyponatremia

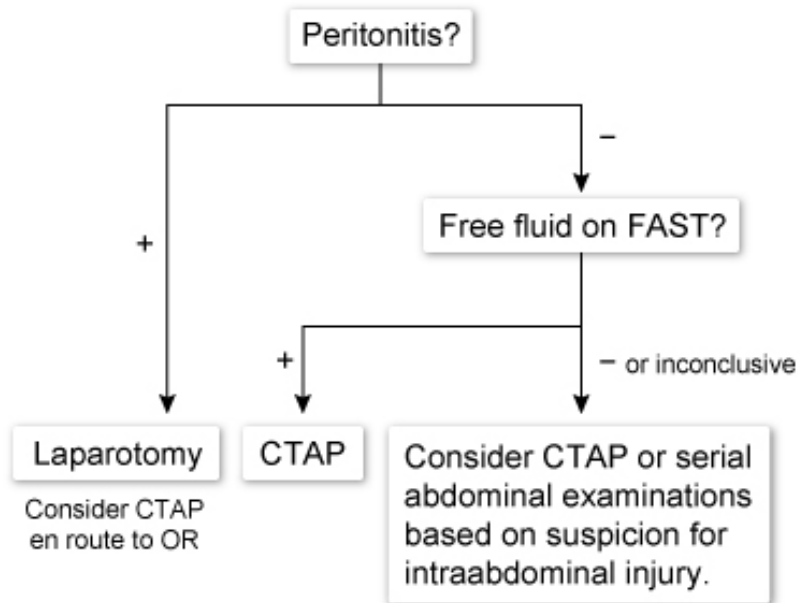


Blunt abdominal trauma

Hemodynamically unstable

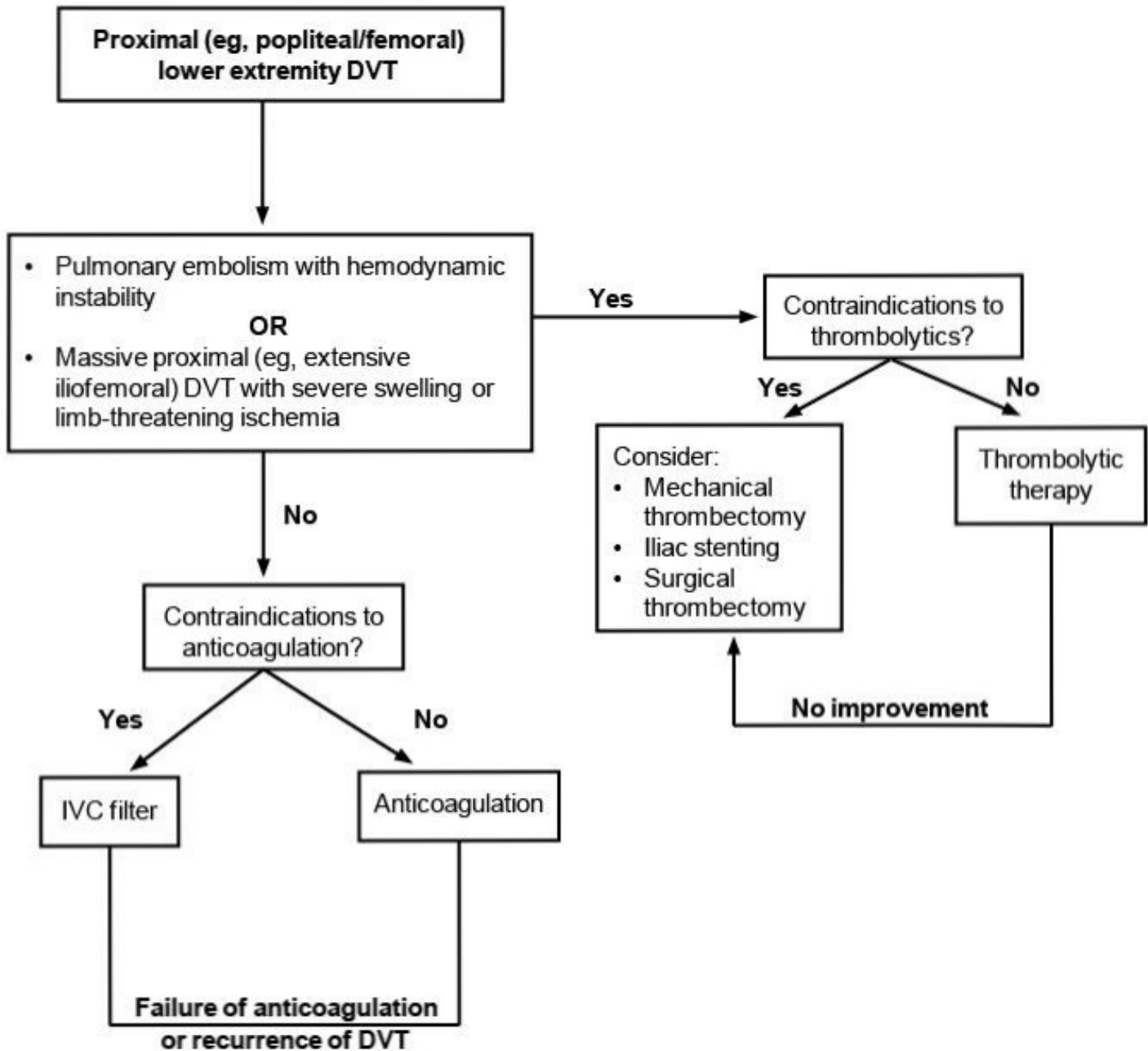


Hemodynamically stable



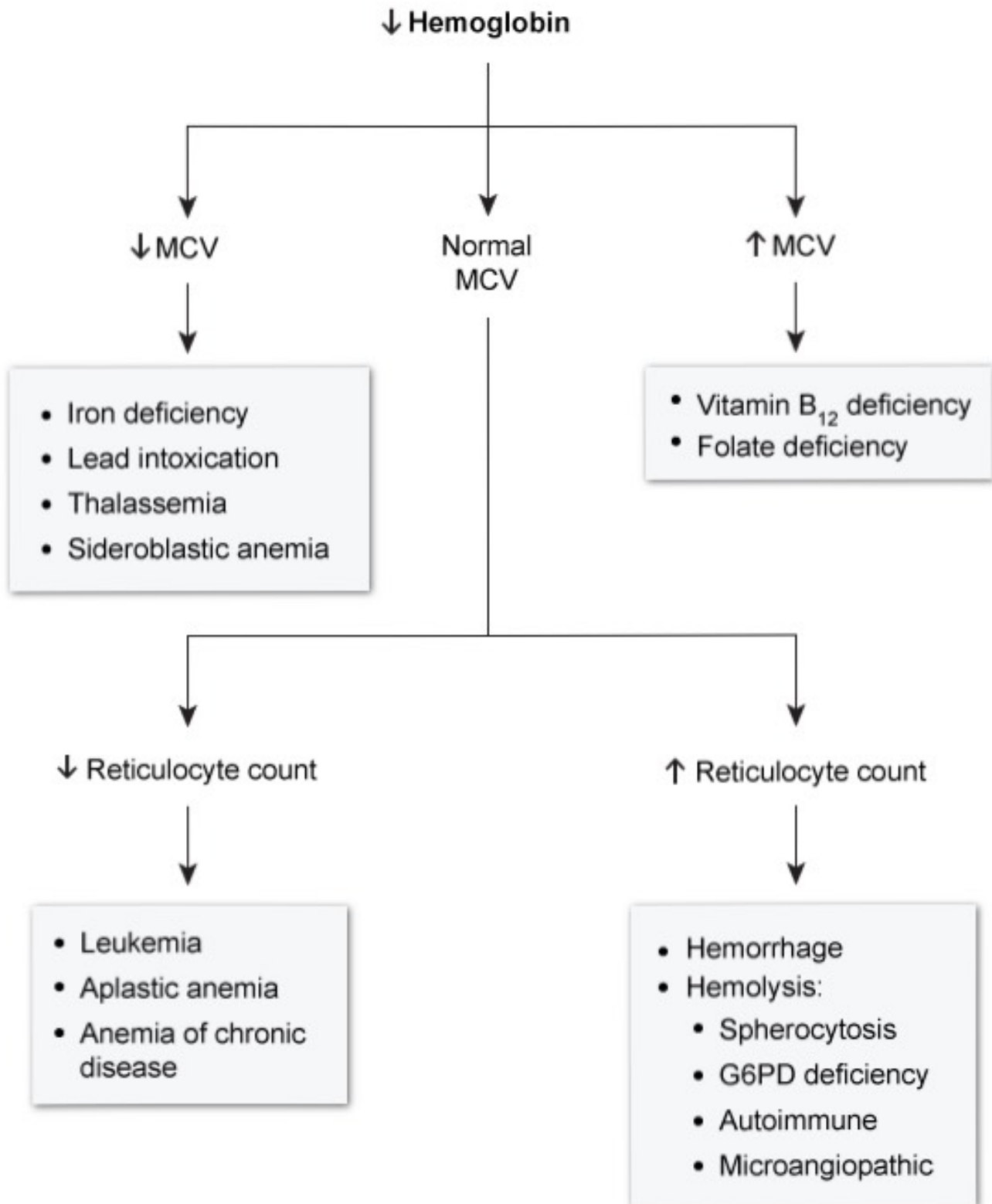
CTAP = CT scan of the abdomen & pelvis; **DPL** = diagnostic peritoneal lavage;
FAST = Focused Assessment with Sonography for Trauma; **OR** = operating room.

Treatment of deep venous thrombosis



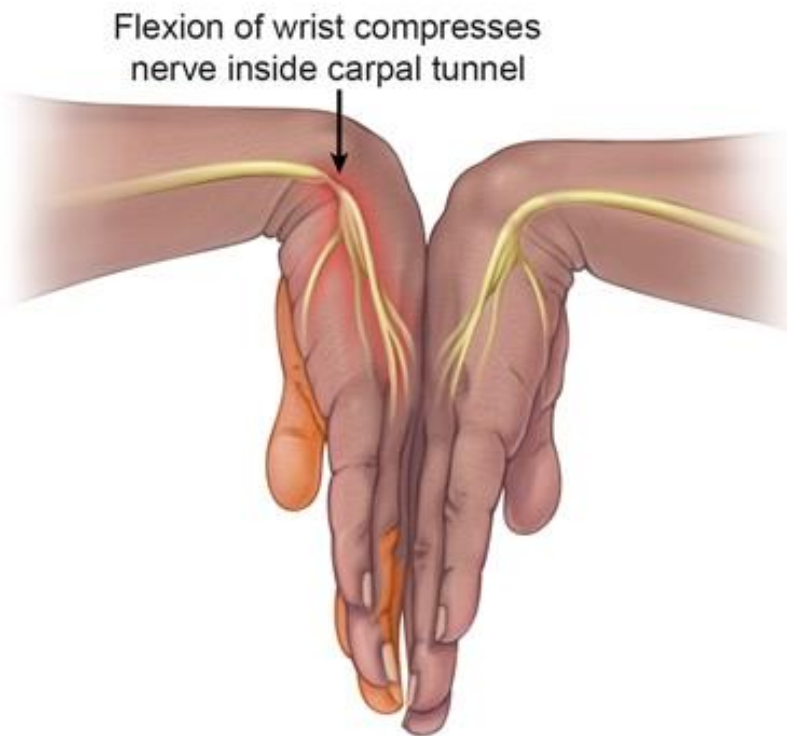
DVT = deep venous thrombosis; IVC = inferior vena cava.

Evaluation of anemia



Recommendations for lung cancer screening	
Recommended test	• Low-dose chest CT scan
Recommended interval	• Yearly
Age for screening	• 55-80
Eligibility	• Patient has ≥ 30-pack-year smoking history AND • Patient is a current smoker or quit smoking within the last 15 years
Termination of screening	• Age > 80 OR • Patient successfully quit smoking for ≥ 15 years OR • Patient has other medical conditions that significantly limit life expectancy or ability/willingness to undergo lung cancer surgery

Provocative tests for carpal tunnel syndrome

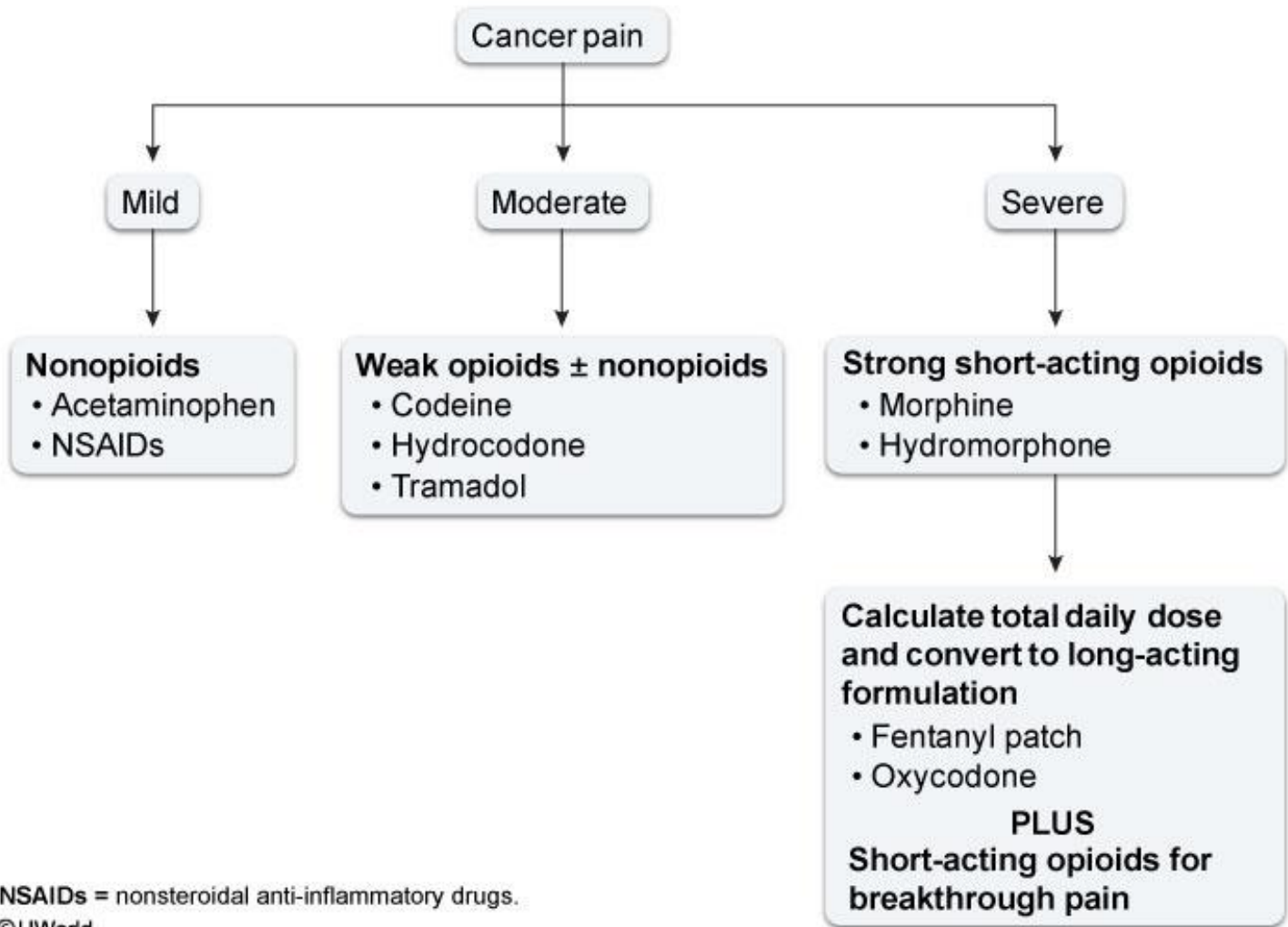


Phalen test



Tinel sign

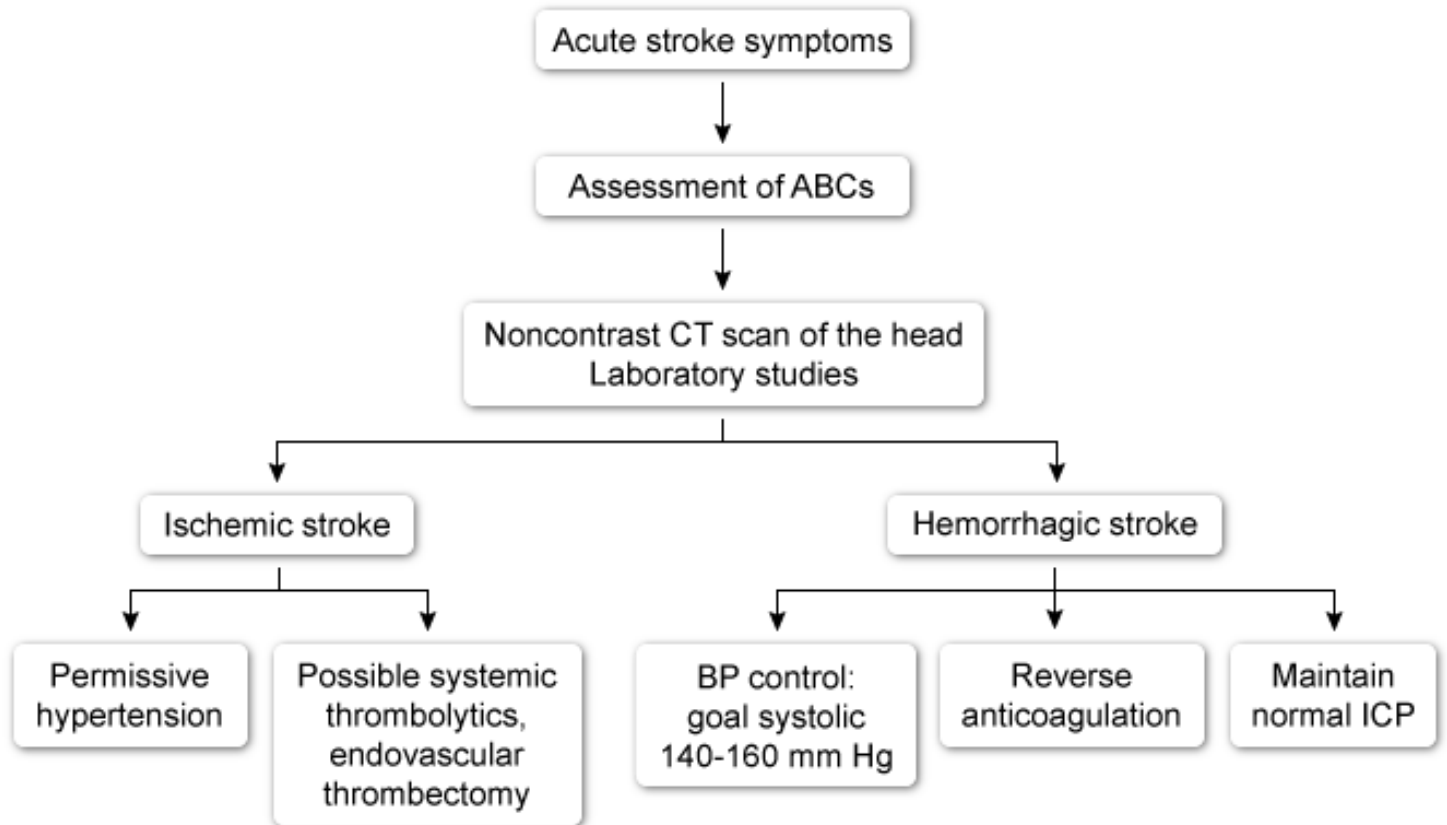
Cancer pain management



NSAIDs = nonsteroidal anti-inflammatory drugs.

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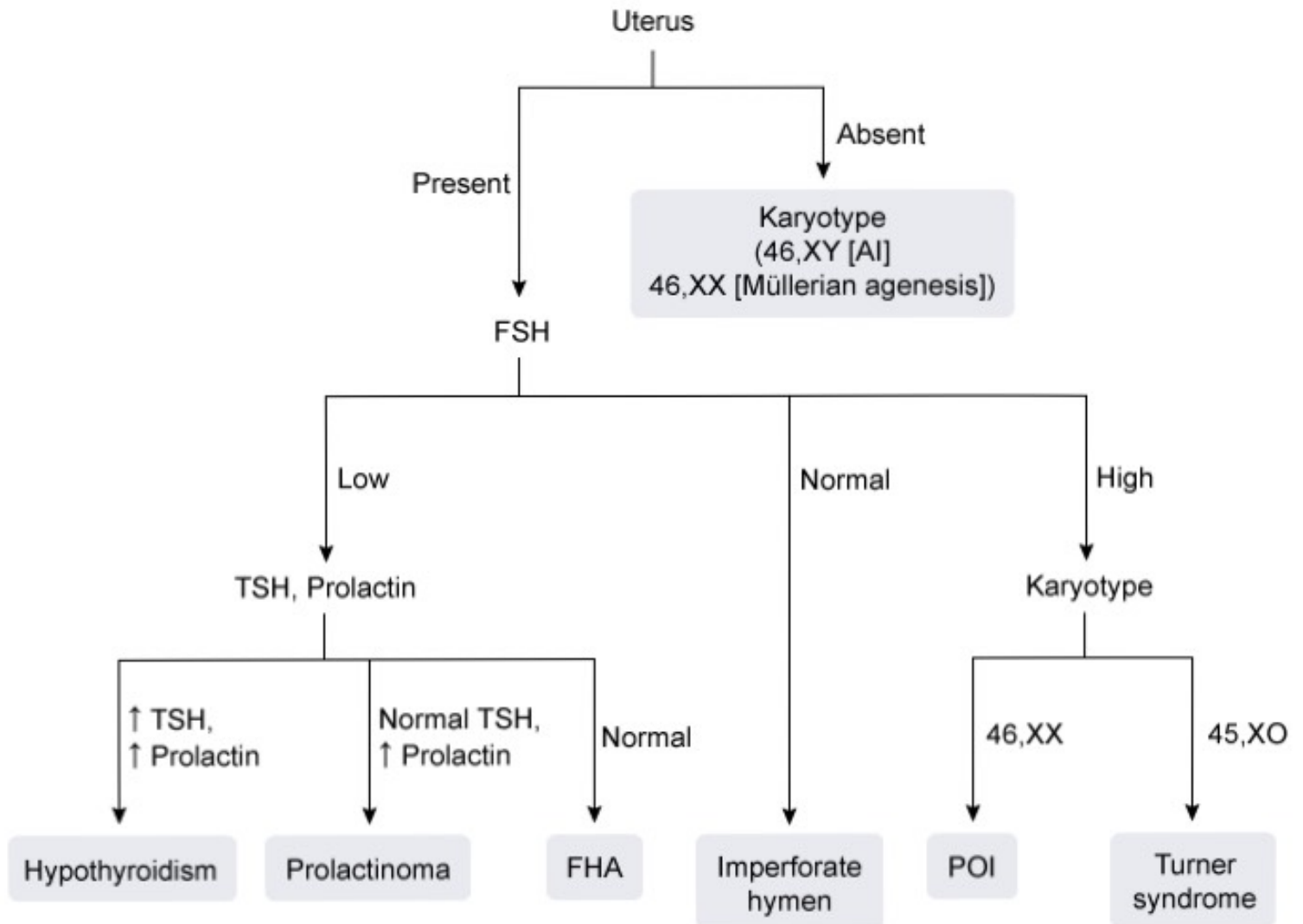
Initial management of stroke



BP = blood pressure; **ICP** = intracranial pressure.

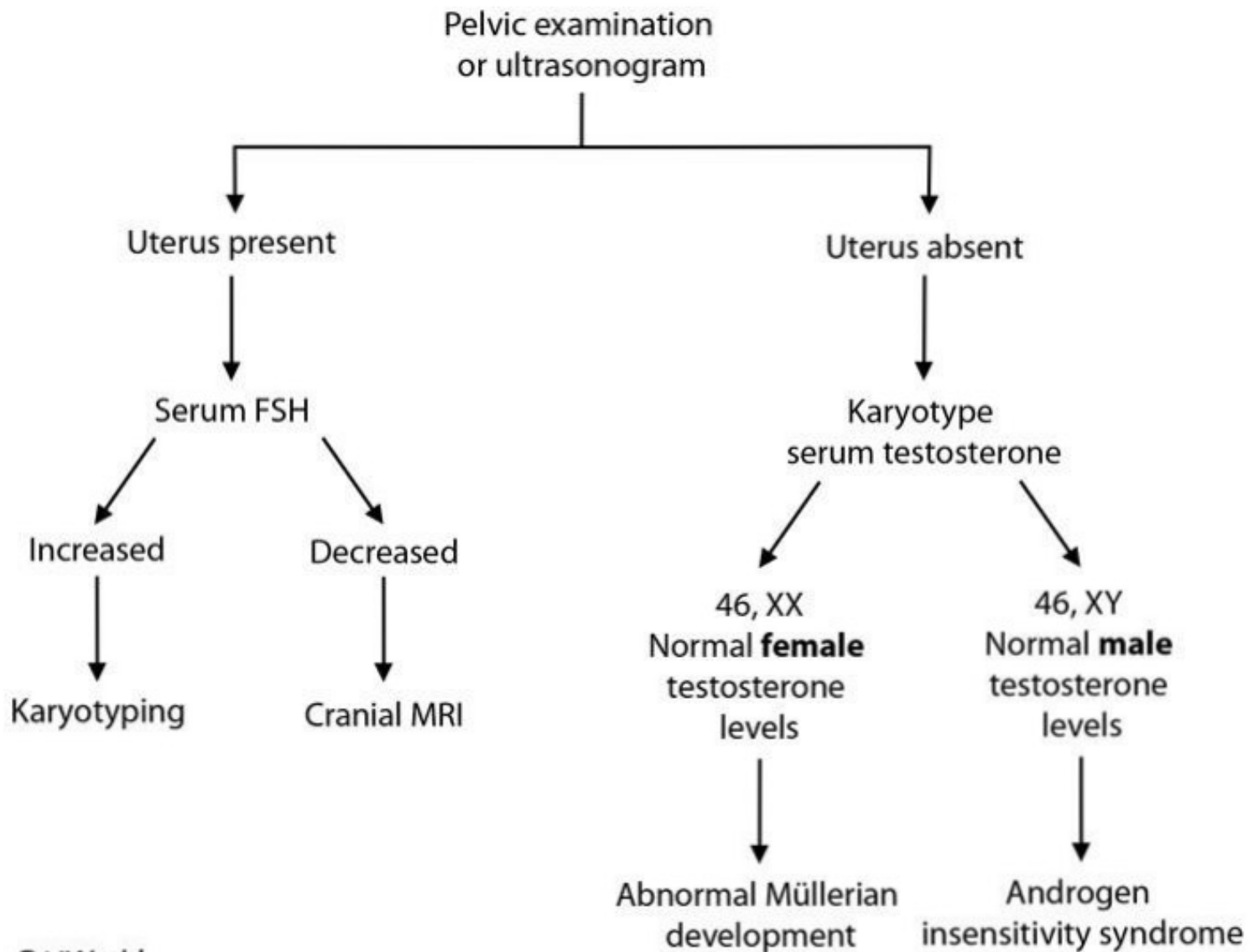
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Primary amenorrhea evaluation



AI = androgen insensitivity; **FHA** = functional hypothalamic amenorrhea; **POI** = primary ovarian insufficiency.

Evaluation of primary amenorrhea



Management of drowning

Prehospital care

- Administer rescue breaths first, then chest compressions for cardiac arrest
- Give supplemental oxygen as needed
- Remove wet clothing to improve hypothermia
- Transport to emergency department

Emergency department care

Symptomatic

Maintain oxygenation & ventilation

- Continuous pulse oximetry
- Supplemental oxygen, NIV, or intubation
- Bronchodilators

Evaluate

- Chest x-ray
- ECG
- ABG, CBC, electrolytes
- Drug screen

Asymptomatic

Observe for (≥ 8 hours)

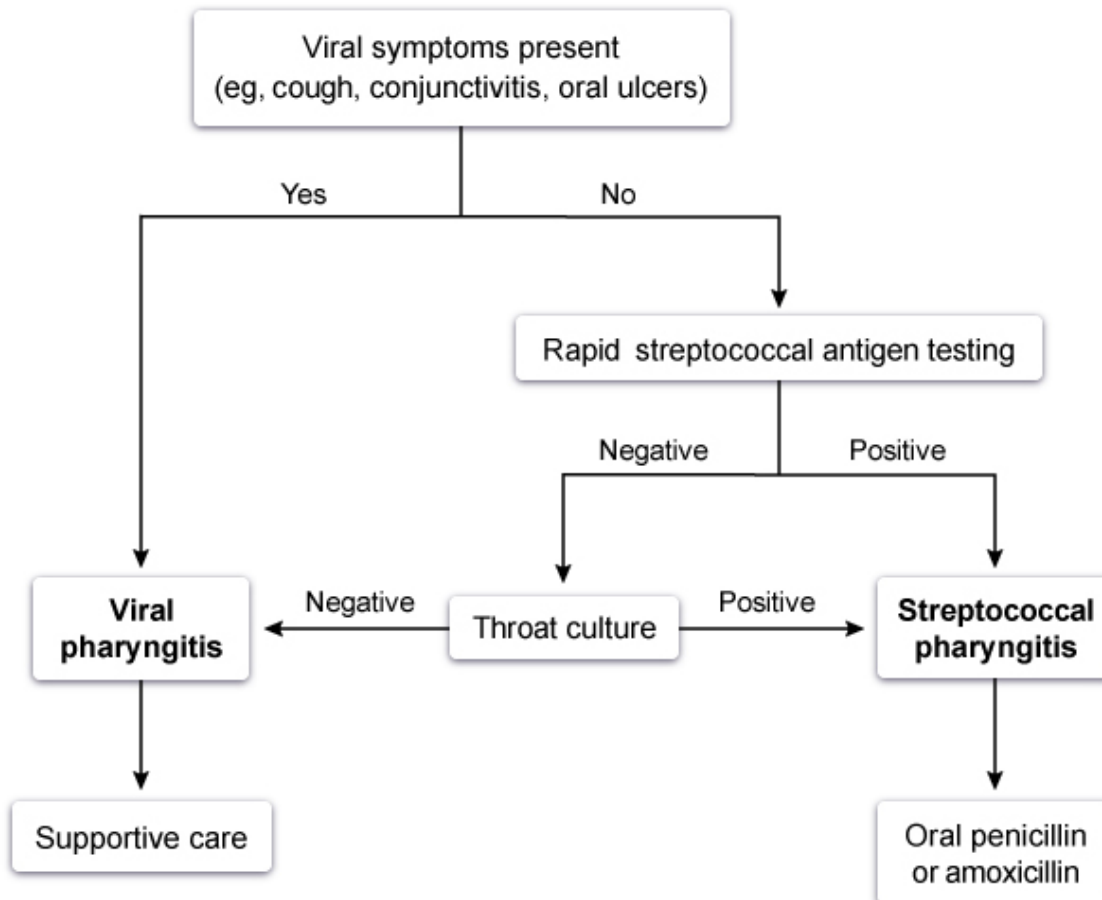
- Continuous pulse oximetry
- Frequent examinations

Evaluate

- Chest x-ray at the end of observation
- +/- ABG, CBC, electrolytes
- Drug screen

Drowning injuries	
Complications	• Acute respiratory distress syndrome • Cerebral edema • Arrhythmia
Poor prognostic indicators	• Submersion time >5 min • Delay in initiation of cardiopulmonary resuscitation • Prolonged resuscitative efforts • Age >14 • Arterial blood pH <7.1

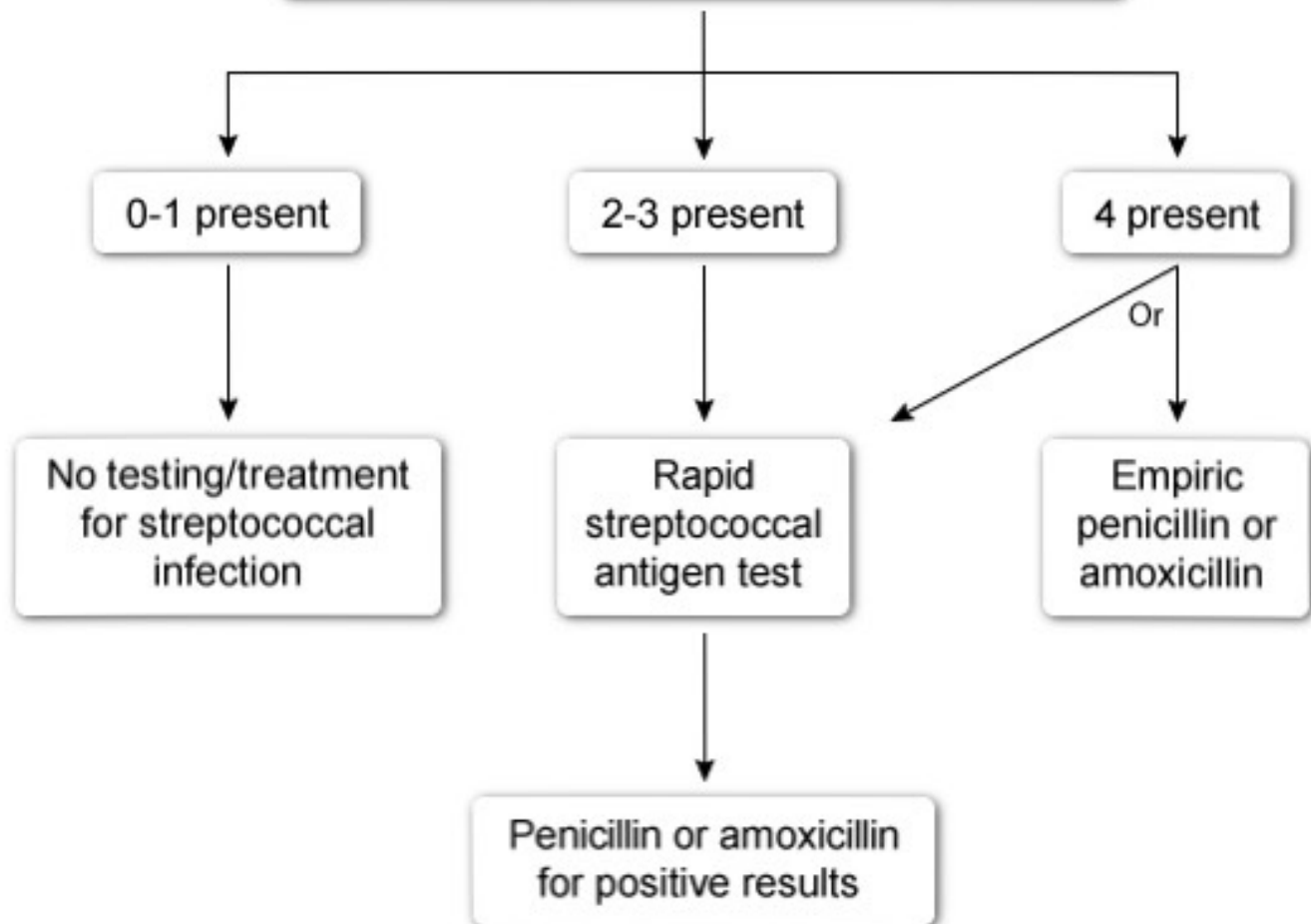
Evaluation of acute pharyngitis in children



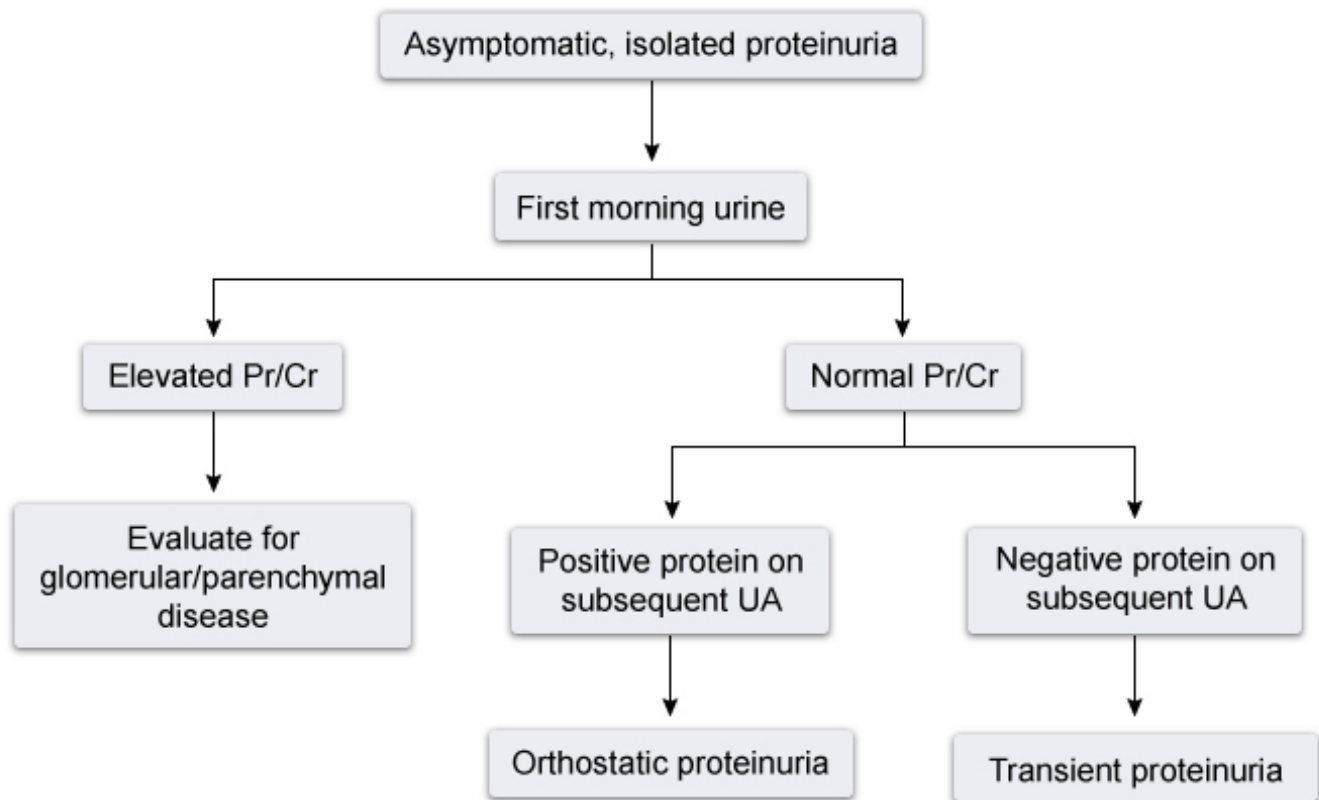
Evaluation & management of pharyngitis

Centor criteria

- Fever by history
- Tender anterior cervical lymphadenopathy
- Tonsillar exudates
- Absence of cough



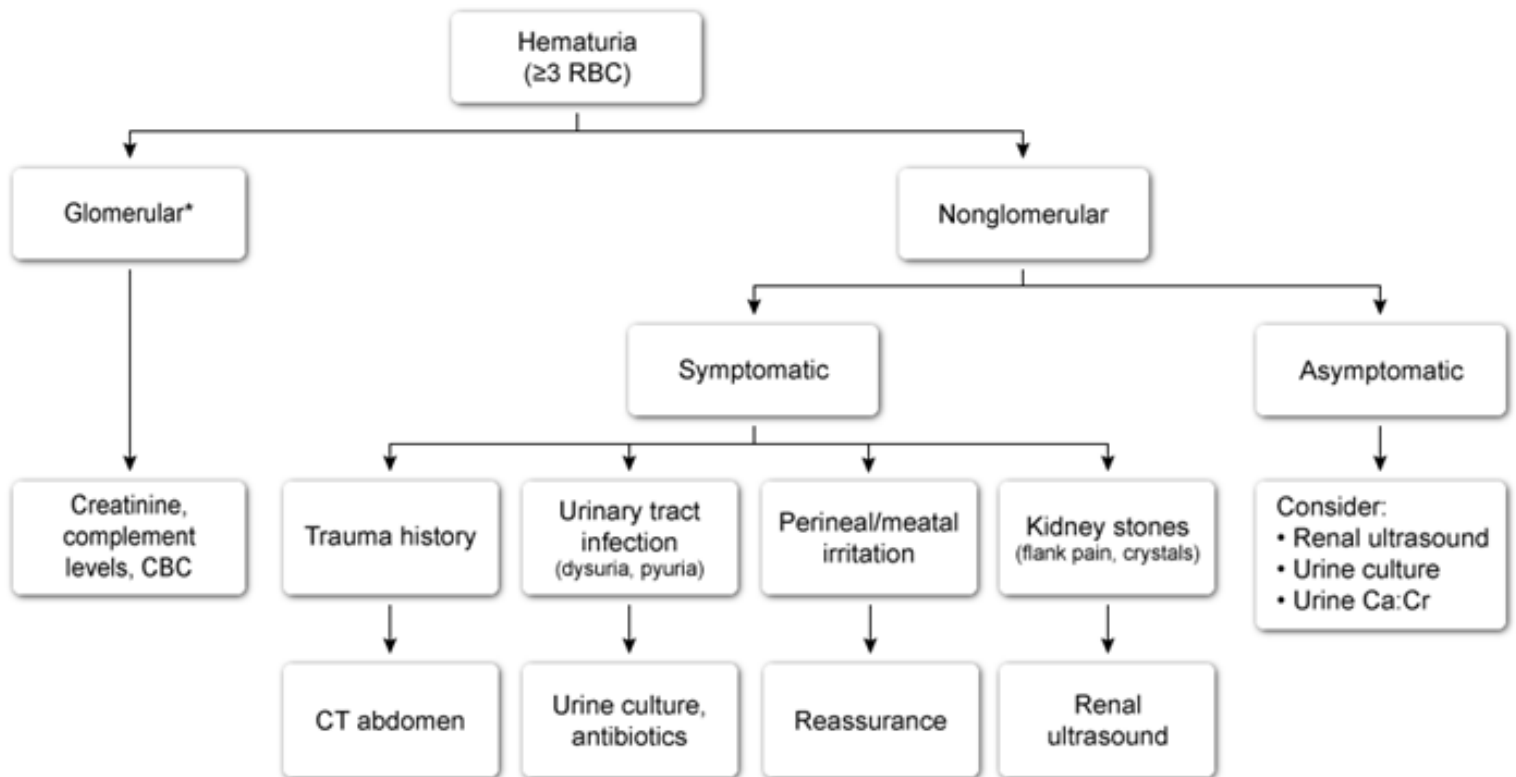
Diagnosis of proteinuria



Pr/Cr = protein/creatinine ratio; **UA** = urinalysis.

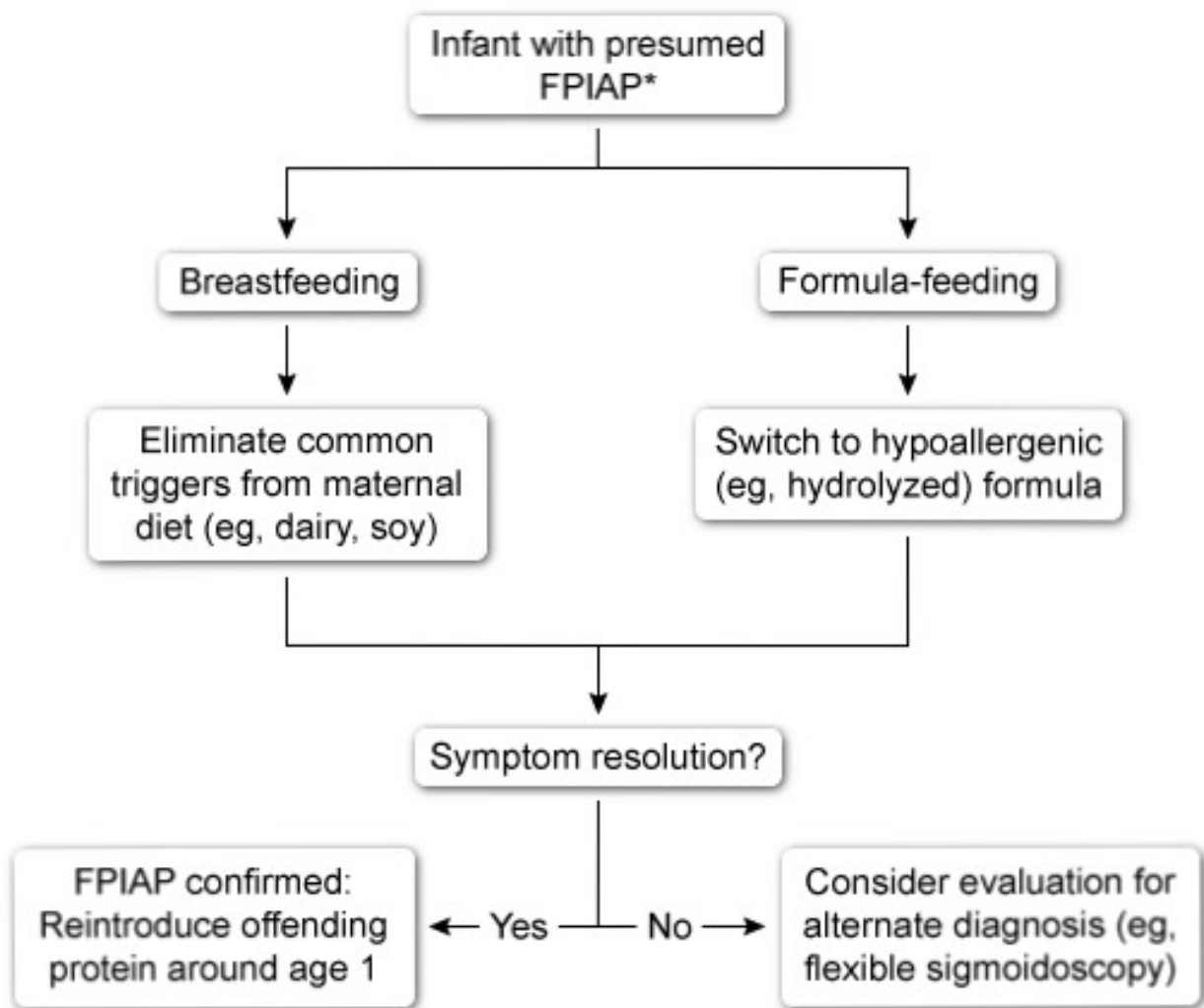
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Evaluation of hematuria in children



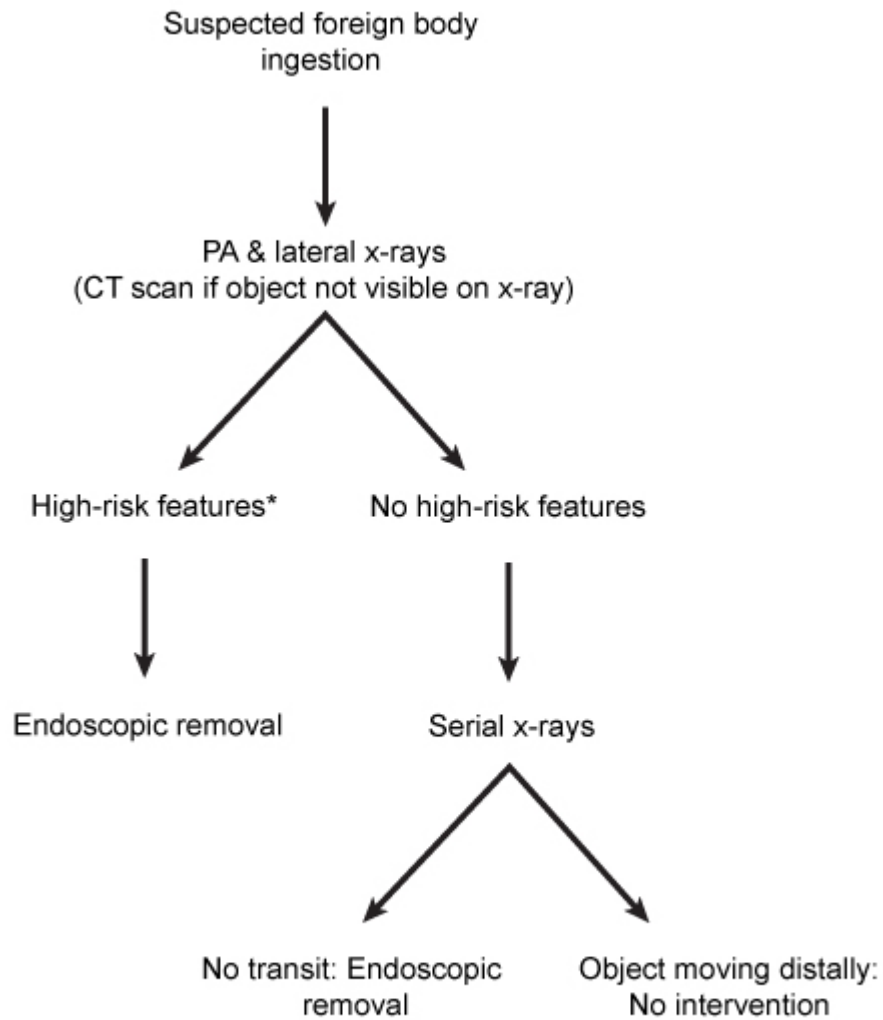
*Findings of glomerular disease include brown urine, edema, hypertension, proteinuria, and RBC casts
RBC = red blood cell, CBC = complete blood count; Ca:Cr = calcium to creatinine ratio

Management of food protein-induced allergic proctocolitis (FPIAP)



*Well-appearing infant age <6 months with blood-streaked stools and nonfocal examination

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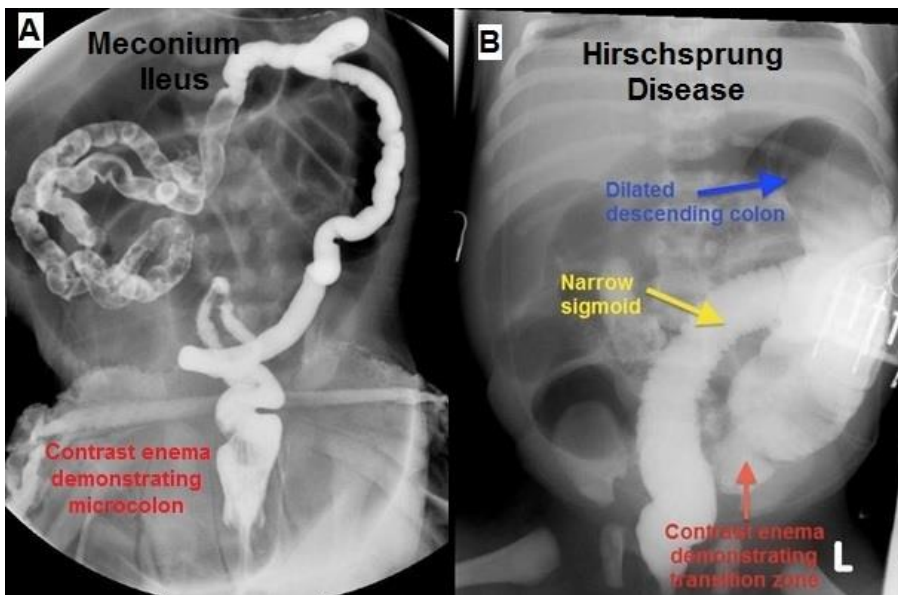
PA = posteroanterior.

*Patient has respiratory or obstructive symptoms;
object is a button battery, magnet, or sharp item.

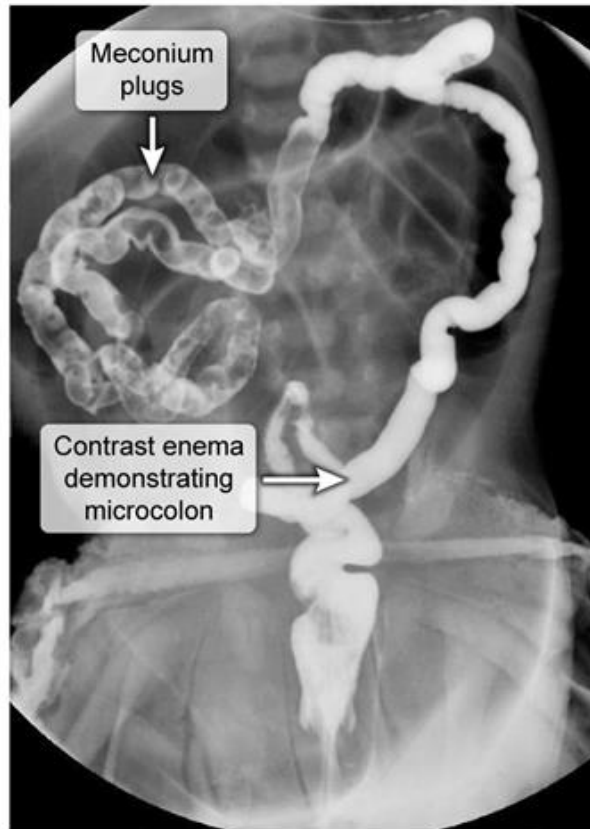
Ingestion of multiple magnets



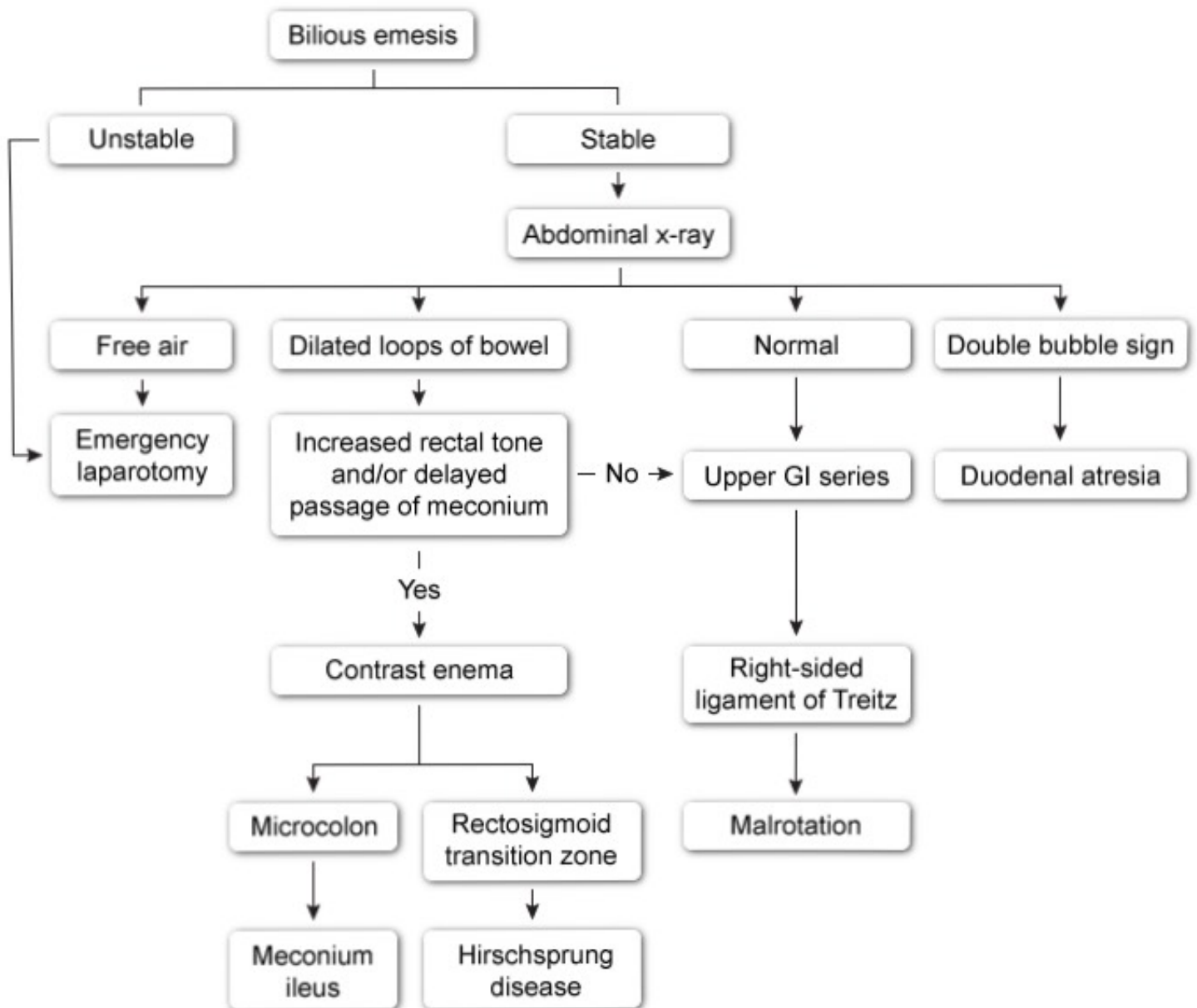
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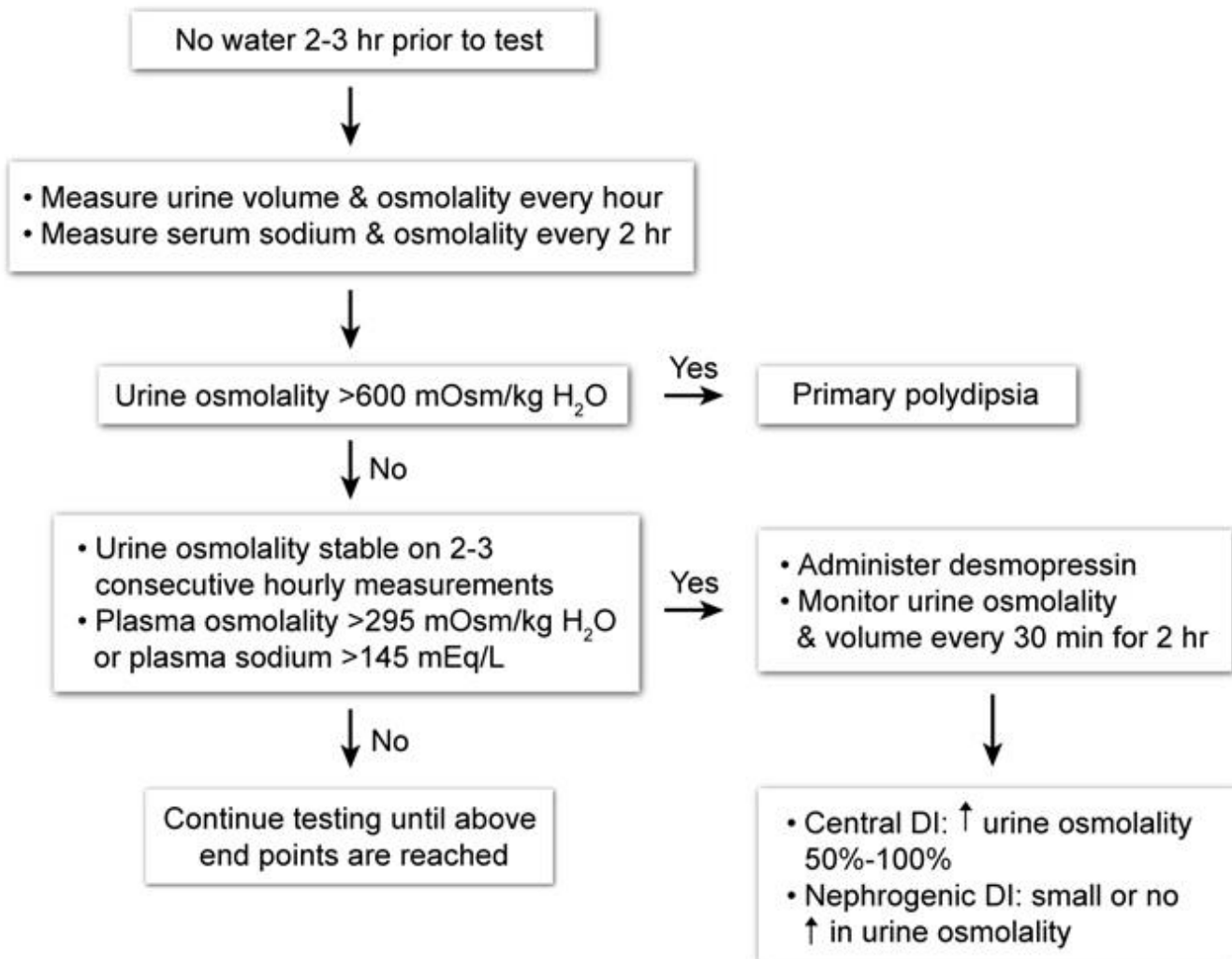
Meconium ileus - microcolon

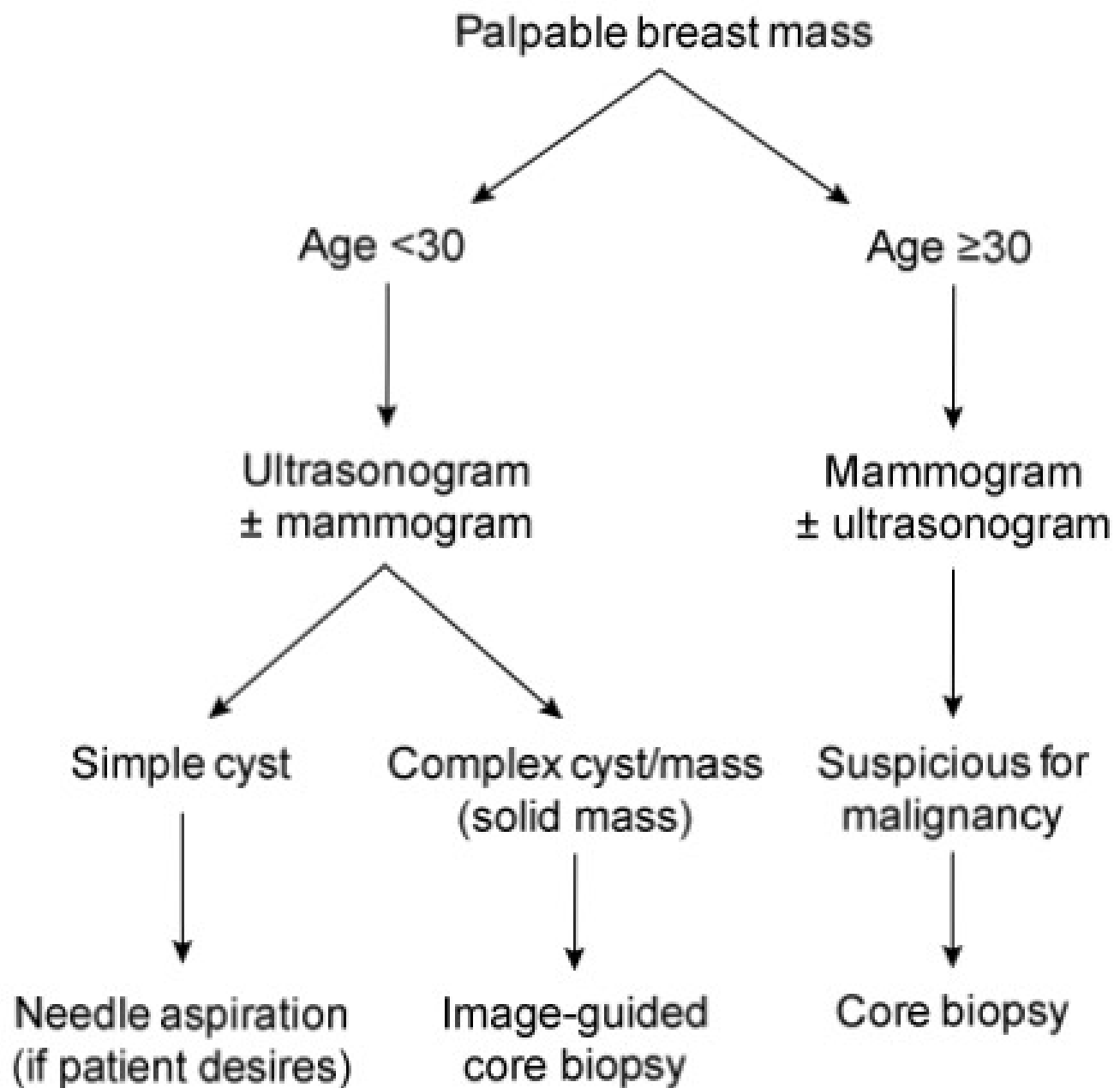


Evaluation of bilious emesis in the neonate

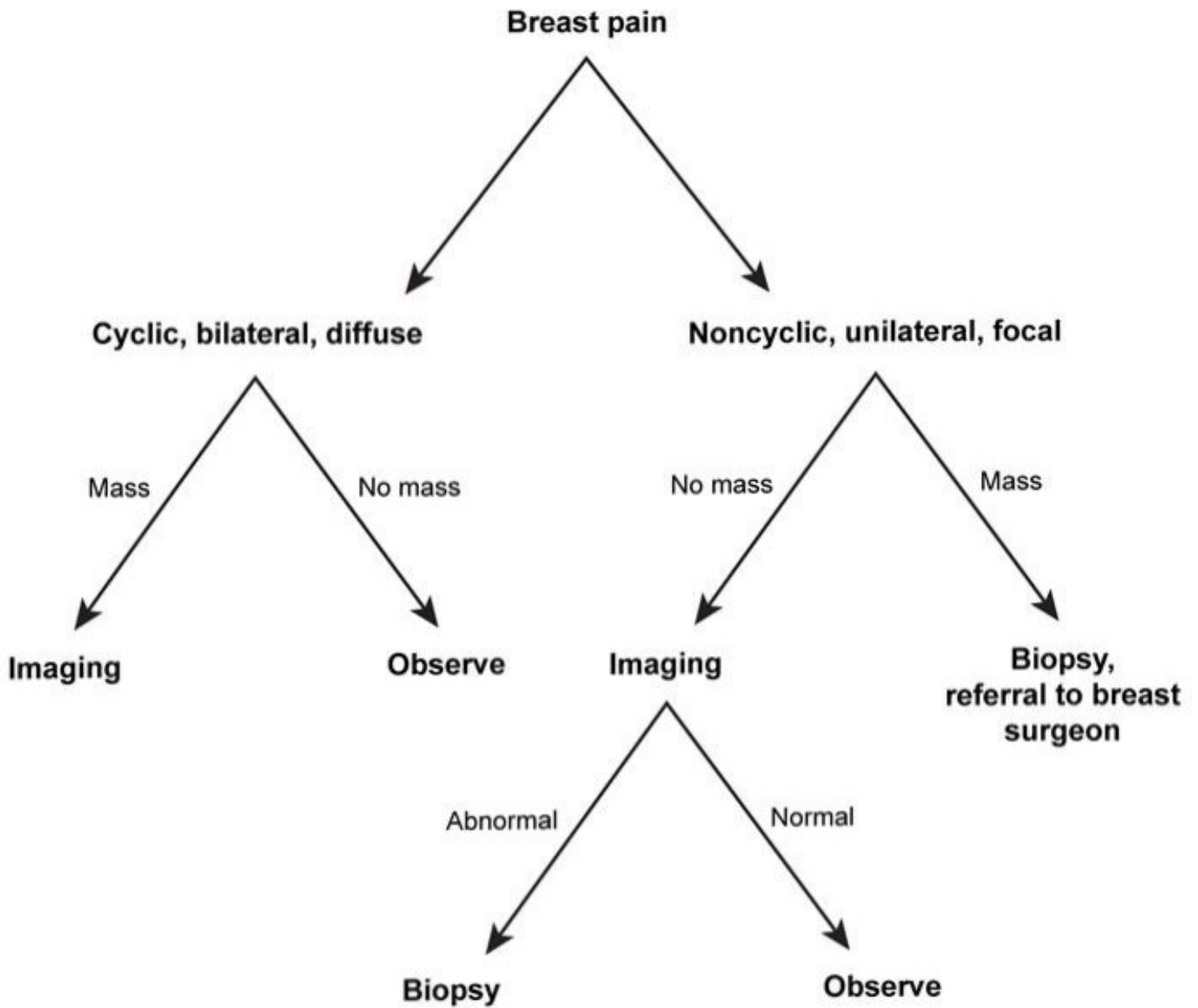


Water deprivation test

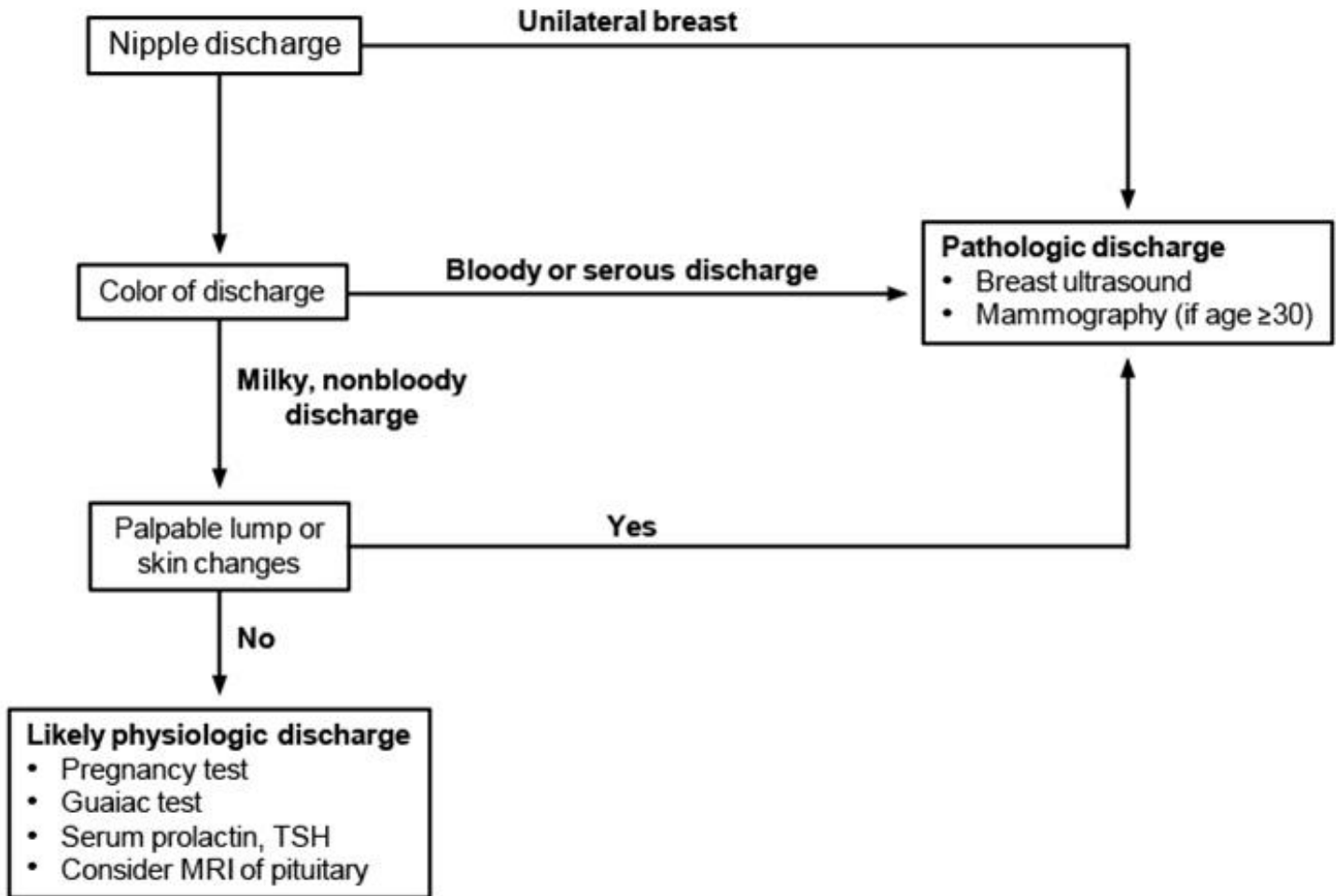




Management of breast pain

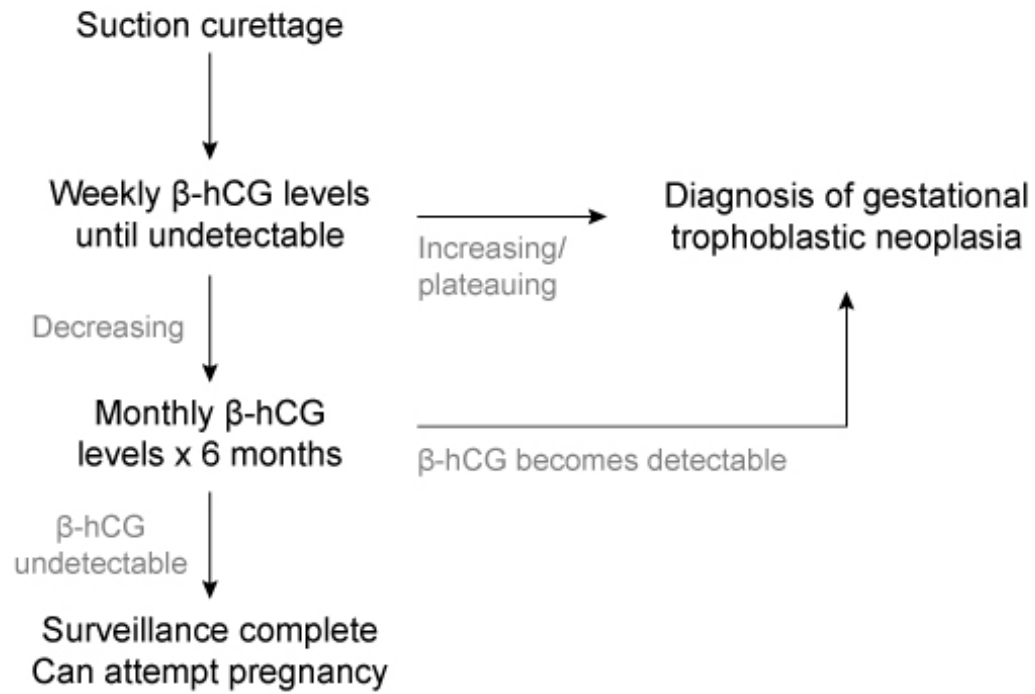


Evaluation of nipple discharge

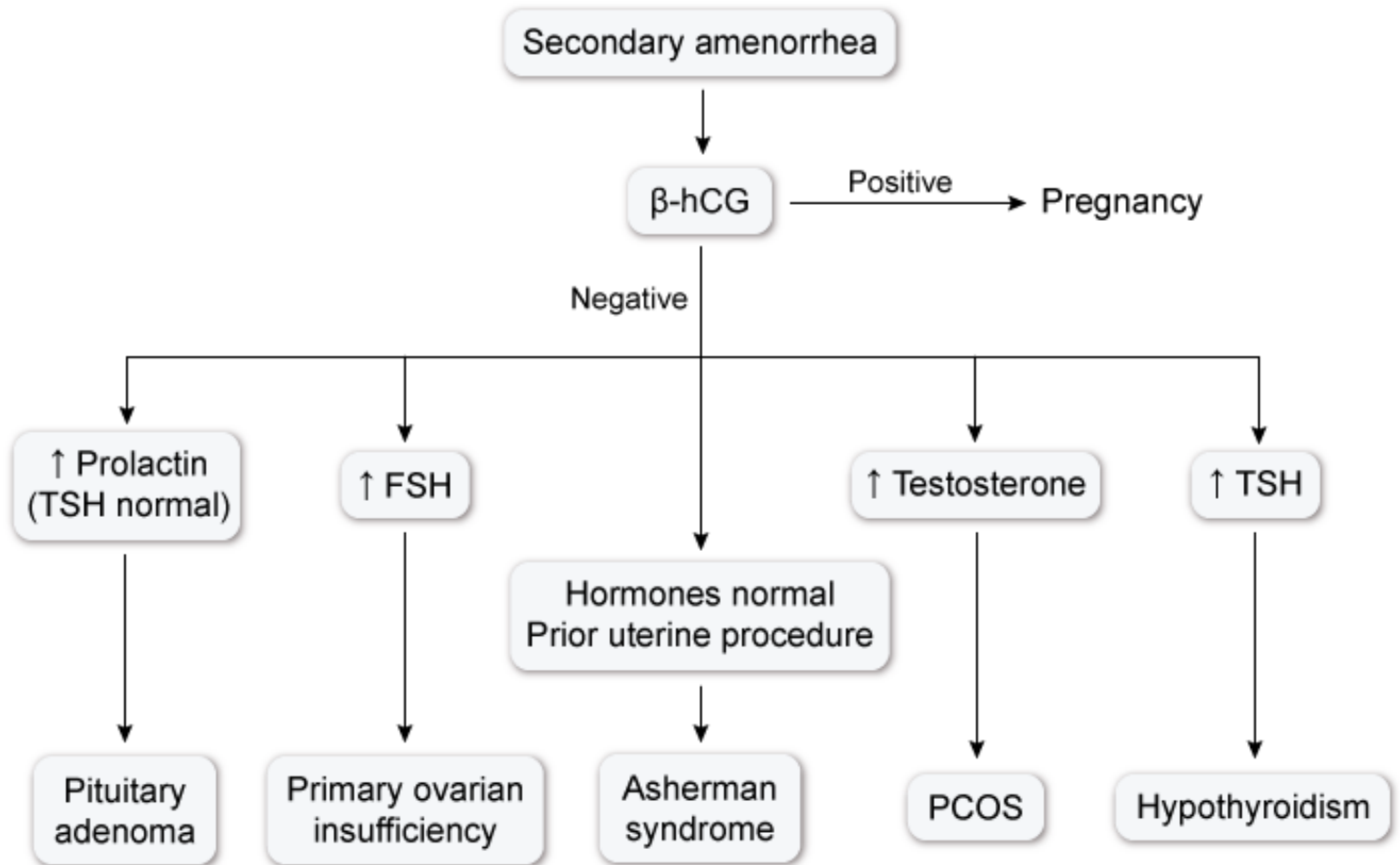


TSH = Thyroid-stimulating hormone.

Management of hydatidiform mole



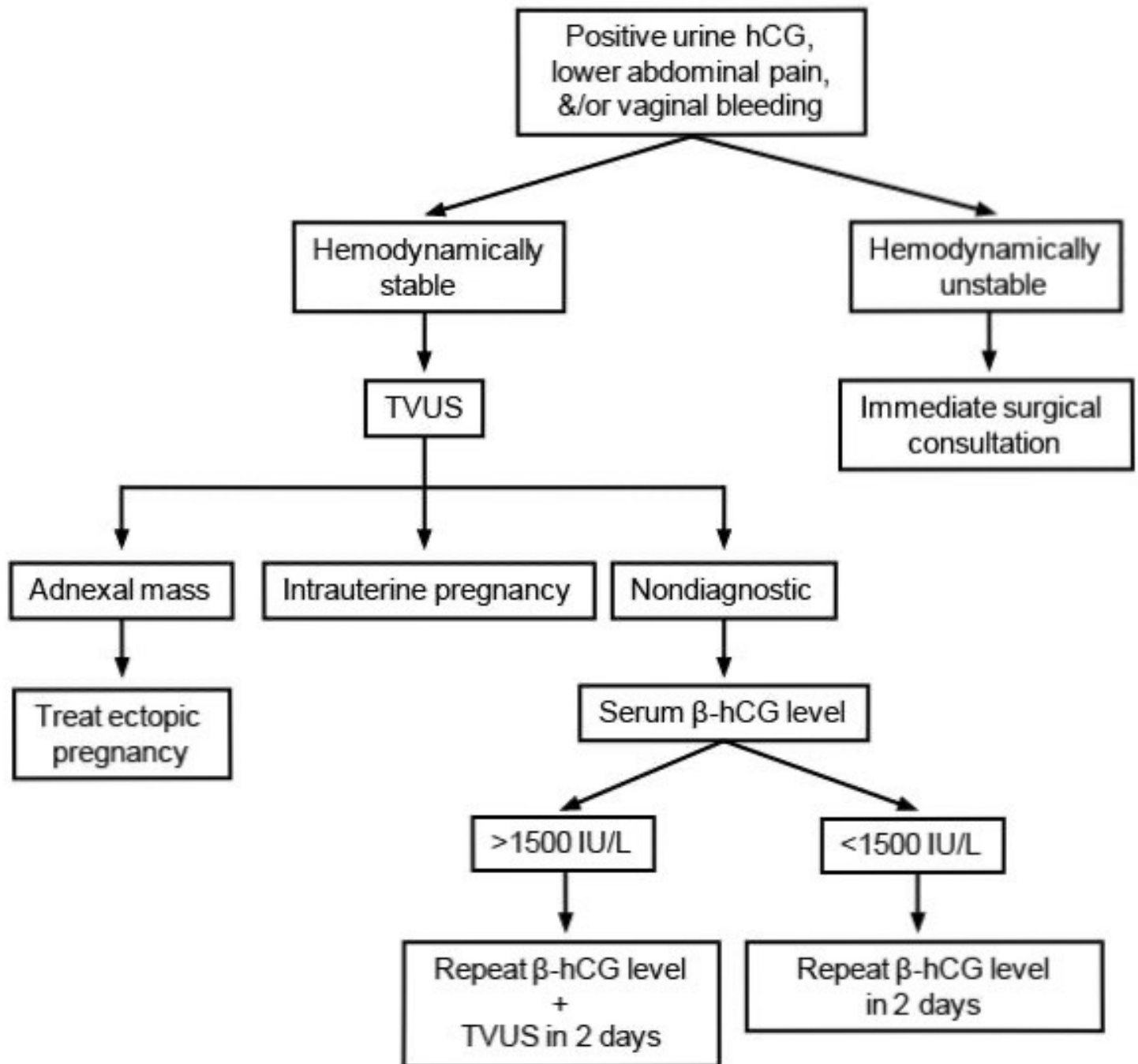
Secondary amenorrhea evaluation



PCOS = polycystic ovary syndrome.

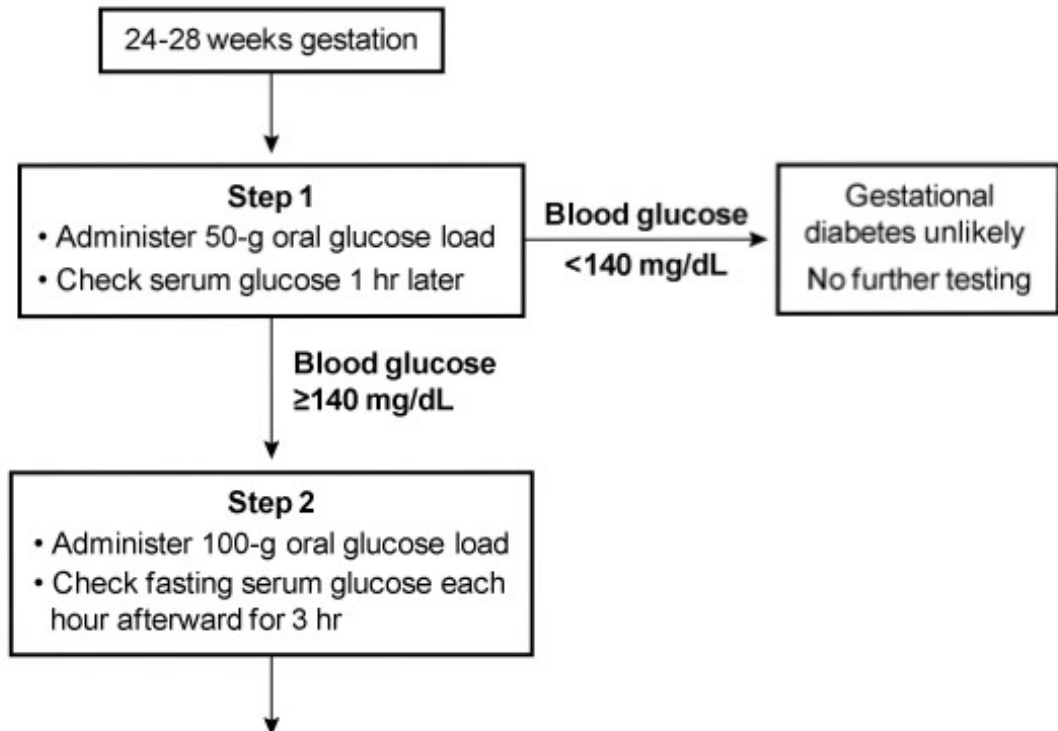
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Management of suspected ectopic pregnancy



TVUS = transvaginal ultrasound.

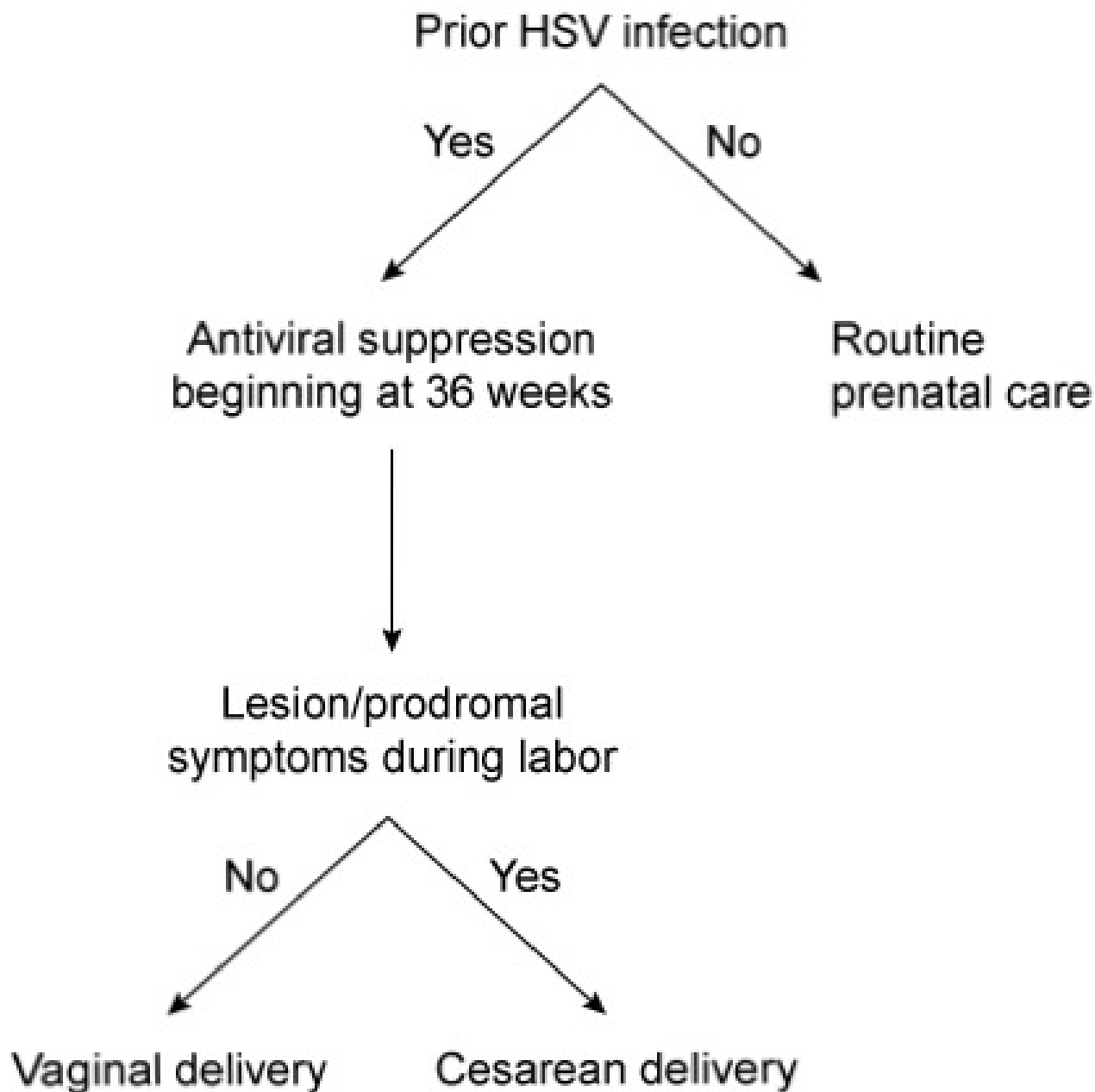
2-step approach for screening & diagnosing gestational diabetes mellitus



Diagnosis of gestational diabetes mellitus (≥2 abnormal values)		
Blood glucose level	Carpenter & Coustan	NDDG
Fasting	≥95 mg/dL (5.3 mmol/L)	≥105 mg/dL (5.3 mmol/L)
1 hr	≥180 mg/dL (10 mmol/L)	≥190 mg/dL (10.6 mmol/L)
2 hr	≥155 mg/dL (8.6 mmol/L)	≥165 mg/dL (9.2 mmol/L)
3 hr	≥140 mg/dL (7.8 mmol/L)	≥145 mg/dL (8 mmol/L)

NDDG = National Diabetes Data Group criteria.

Pregnancy management in patients with HSV



HSV = herpes simplex virus.

Management of CIN 3

CIN 3



Not currently pregnant



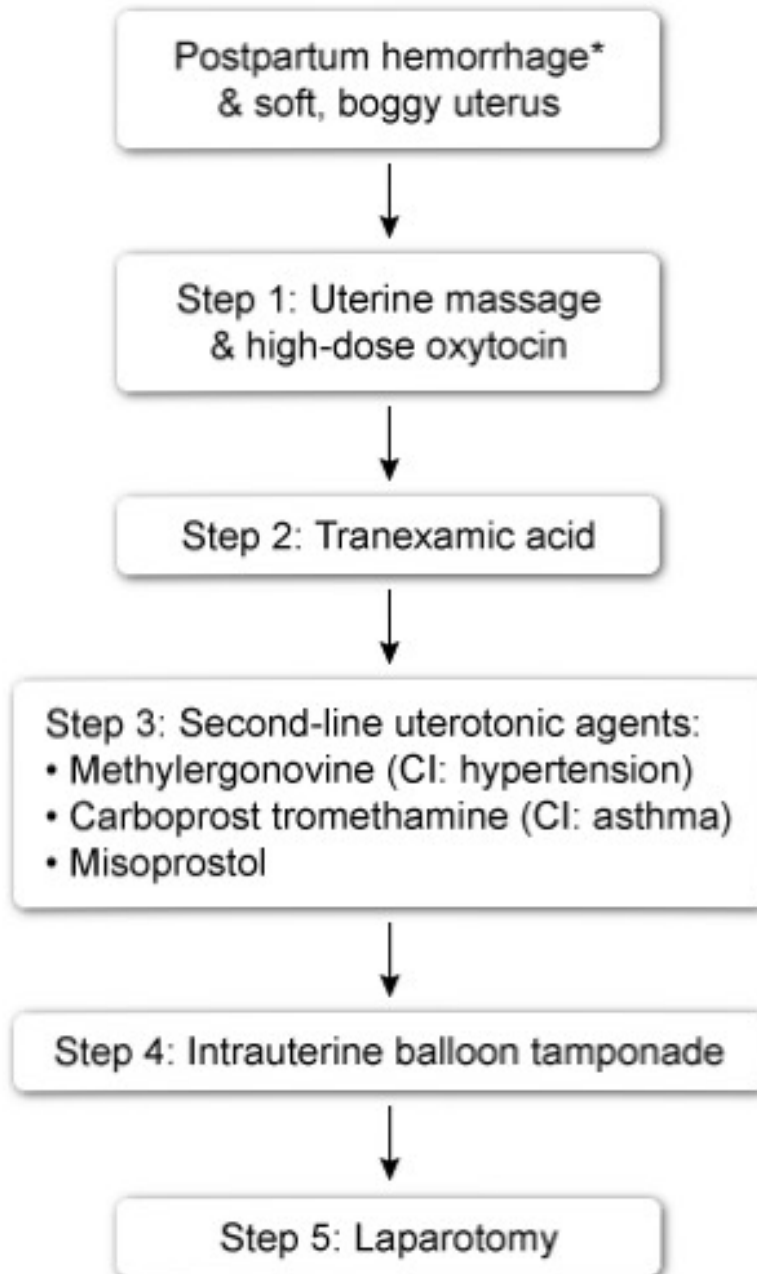
- LEEP
- Cold knife conization
- Cryoablation



Pap testing with HPV cotesting
1 and 2 years postprocedure

CIN 3 = cervical intraepithelial neoplasia 3;
LEEP = loop electrosurgical excision procedure;
HPV = human papillomavirus.

Management of postpartum hemorrhage due to uterine atony

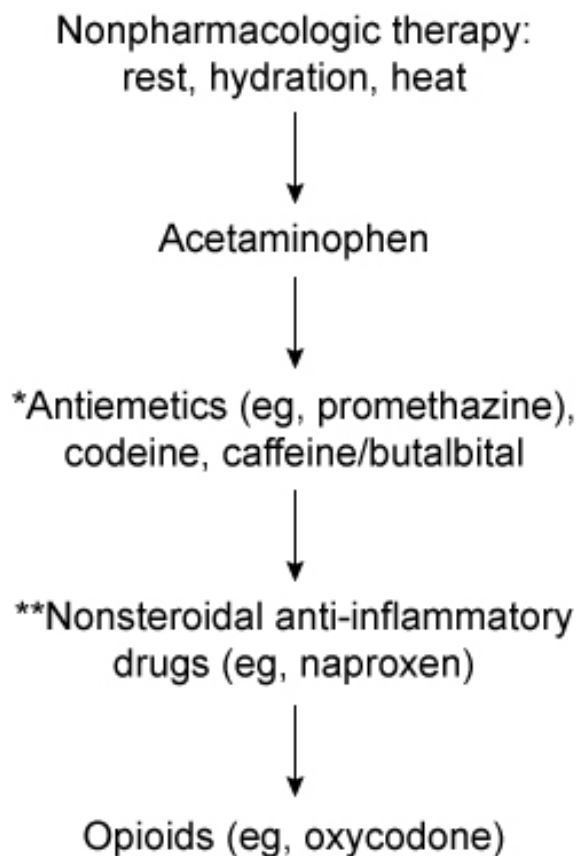


*Estimated blood loss $\geq 1,000$ mL or bleeding + hypovolemia.

CI = contraindication.

Visual assessment of melanoma	
ABCDE criteria (≥1-2 is suspicious)	• Asymmetry • Border irregularity • Color variation (within lesion or compared to other lesions) • Diameter ≥6 mm • Evolving appearance over time
7-point checklist (≥1 major or ≥3 minor is suspicious)	• Major criteria: change in size, shape, or color • Minor criteria: size ≥7 mm, local inflammation, crusting/bleeding, sensory symptoms
Ugly duckling sign	• One lesion is significantly different from others on the patient

Management of migraines in pregnancy

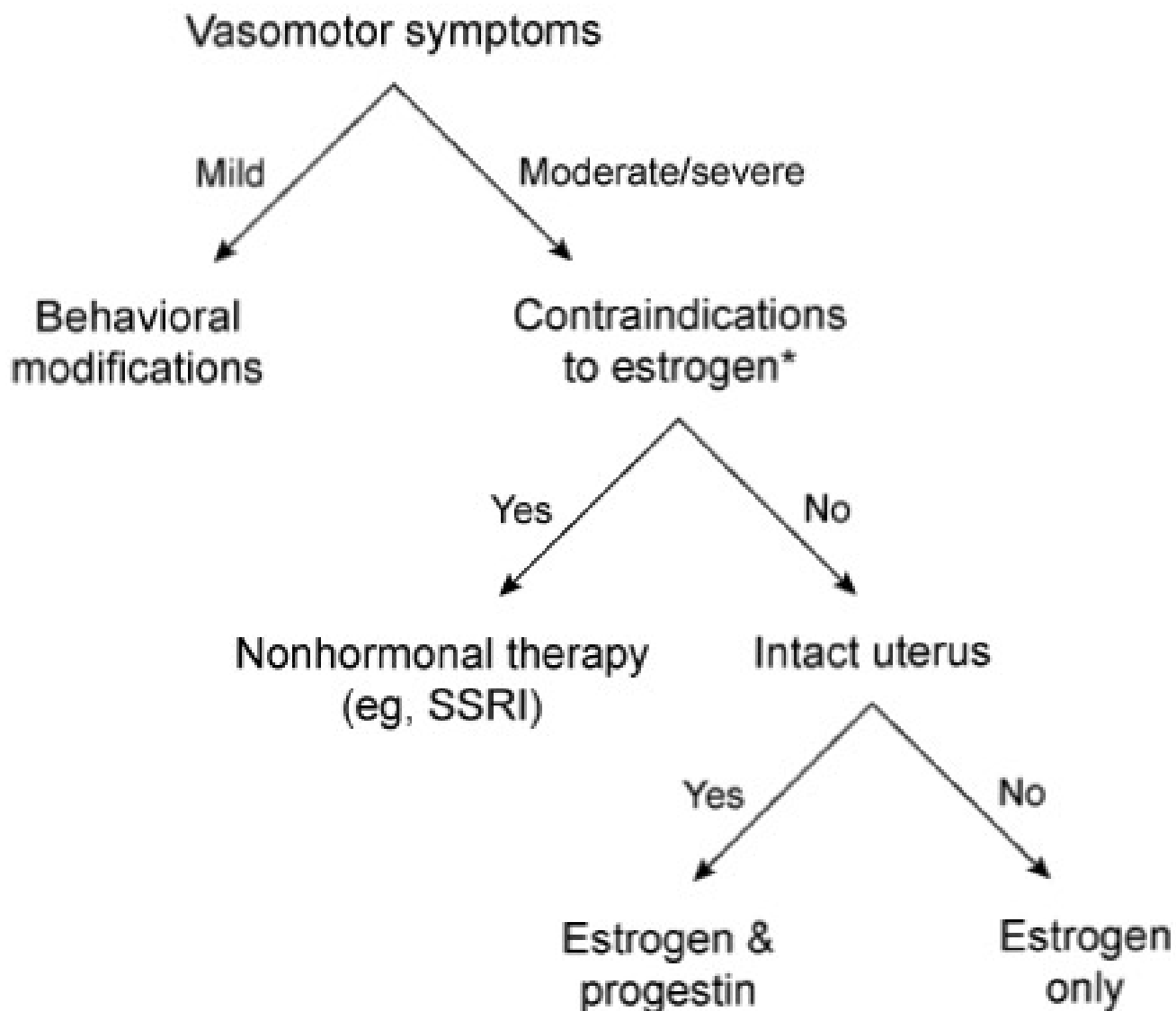


*Can be used in conjunction with acetaminophen.

**2nd trimester only.

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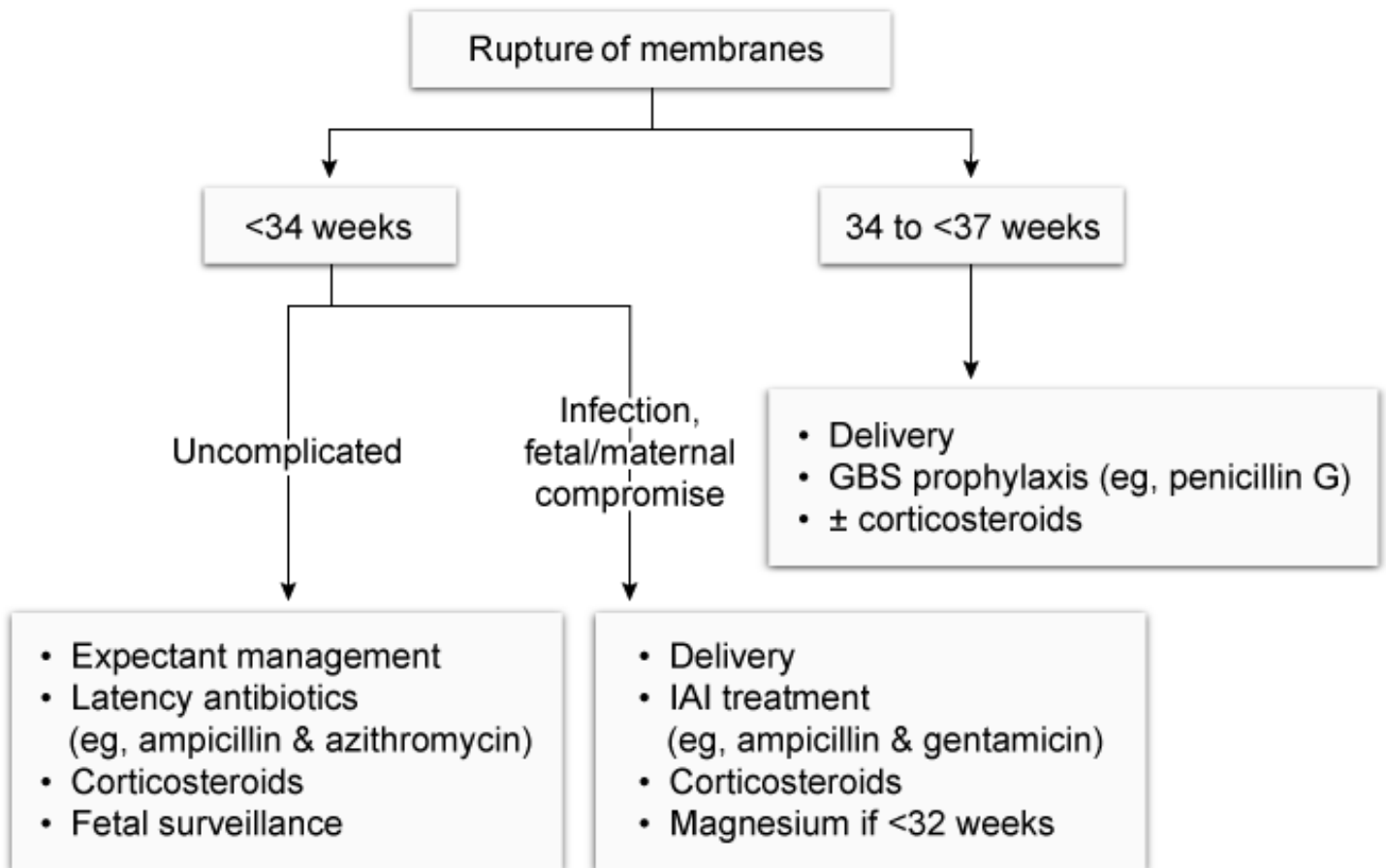
Treatment of menopause



*Contraindications to estrogen: Breast cancer, coronary heart disease, endometrial cancer, liver disease, thromboembolism.

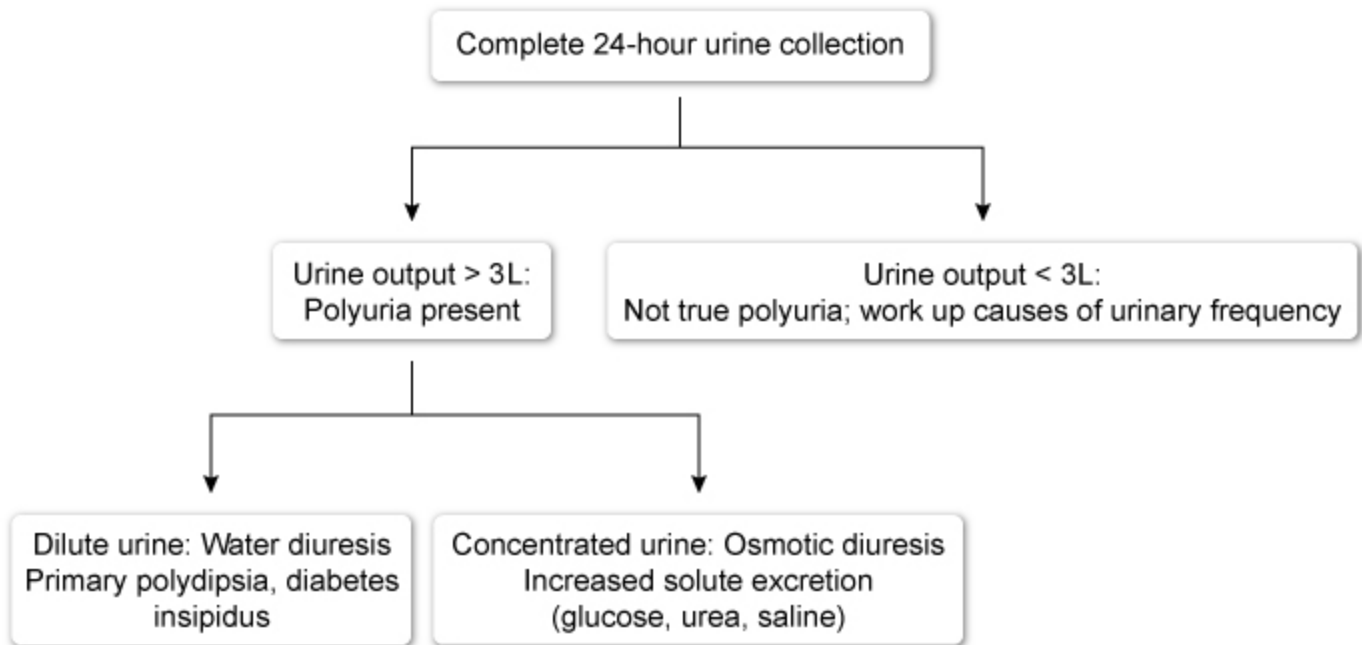
SSRI = selective serotonin reuptake inhibitor.

Management of preterm prelabor ROM



GBS = group B streptococcal; **IAI** = intra-amniotic infection; **ROM** = rupture of membranes.

Evaluation of suspected polyuria

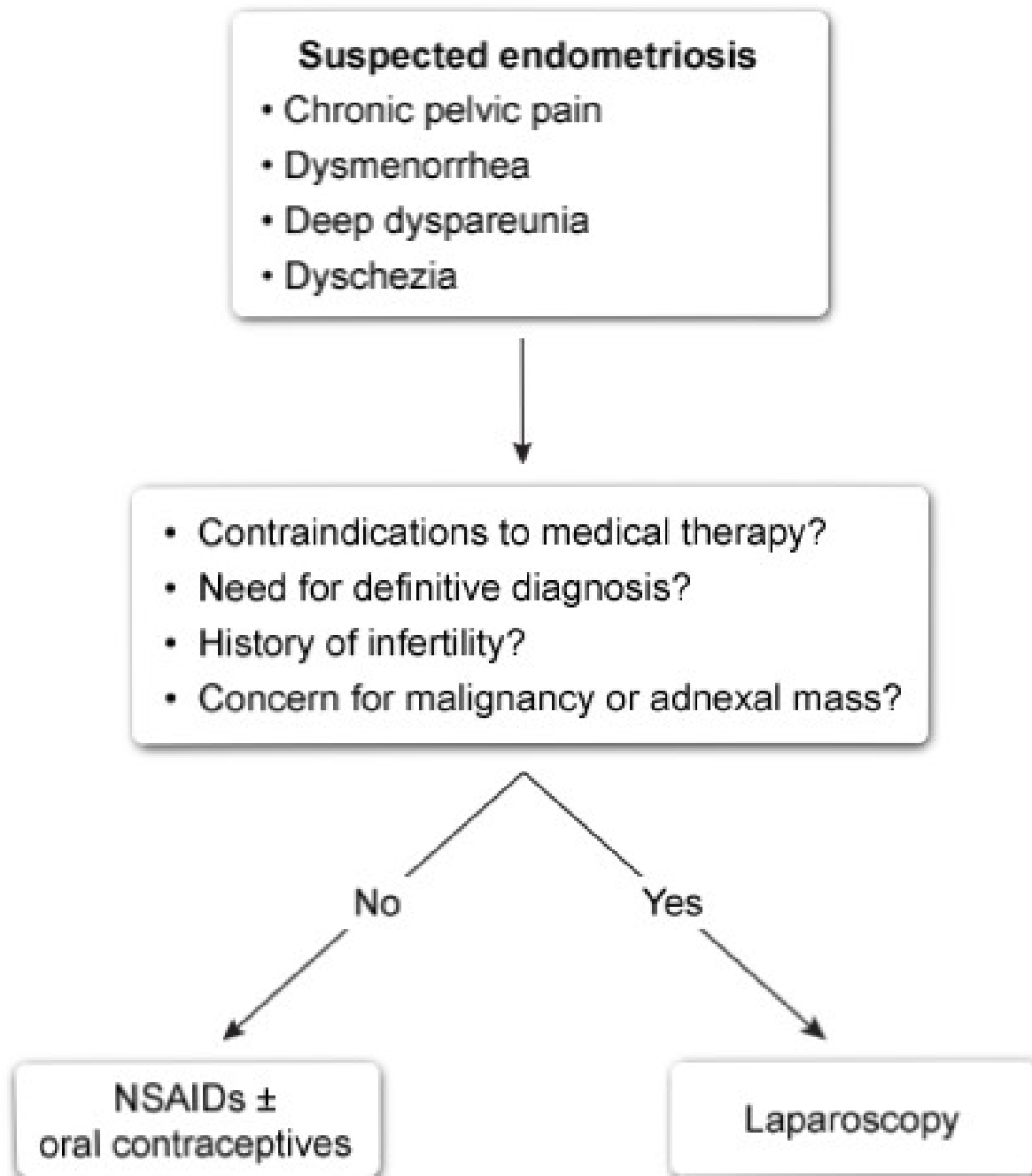


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can remain undiagnosed because the body often compensates for water losses with a demand for increased fluid intake (eg, increased thirst). However, during **pregnancy**, placentally produced enzymes (eg, vasopressinase) increase ADH breakdown and can worsen symptoms, thereby **unmasking undiagnosed DI**. Some women develop transient DI of pregnancy due to an exaggerated response to vasopressinase. This phenomenon may resolve with delivery but recur in subsequent pregnancies.

In this patient, a more complete workup (eg, challenge with desmopressin, an ADH analogue) is required to confirm the precise etiology of DI (ie, central DI due to insufficient ADH levels vs nephrogenic DI due to renal resistance to ADH).

Management of endometriosis



NSAIDs = nonsteroidal anti-inflammatory drugs.

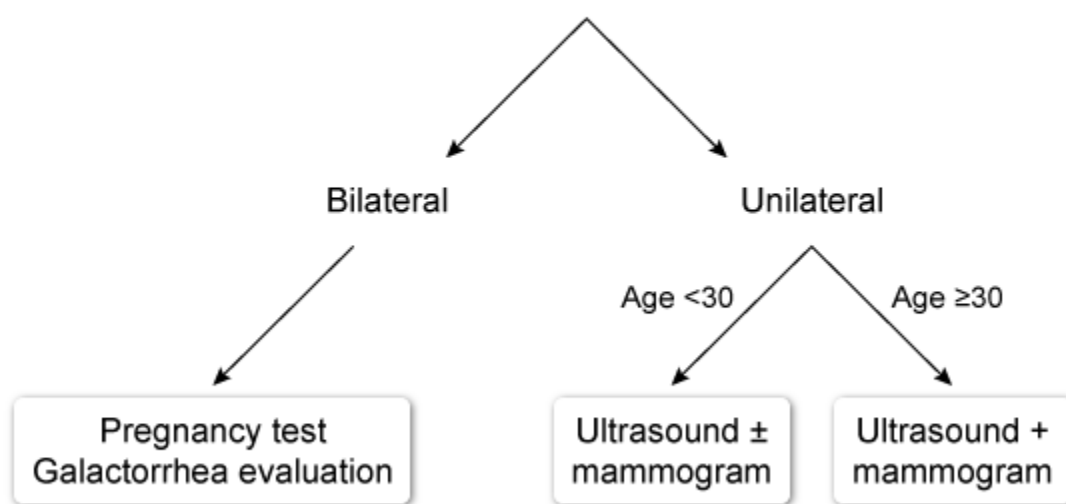
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Normal postpartum lochia		
	Expected duration	Description
Lochia rubra	• Birth to 3-4 days postpartum	• Dark or bright red (blood); odor similar to that of menstrual blood; occasional small clots; quantity decreasing each day
Lochia serosa	• 4th postpartum day to 10th or 14th postpartum day	• Serosanguineous (pink); brownish (old blood); quantity gradually decreasing in amount
Lochia alba	• 11th postpartum day to 6 weeks postpartum	• White/yellow; creamy; light quantity

Lochia may increase in quantity after breastfeeding (suckling releases oxytocin & causes uterus to contract) & 7-14 days postpartum, when scabbing on the placental site sloughs off (heavier bleeding for <2 hr); lochia may also feel increased after lying down & then standing (due to blood pooling in vagina).

Breast discharge evaluation

Nipple discharge*

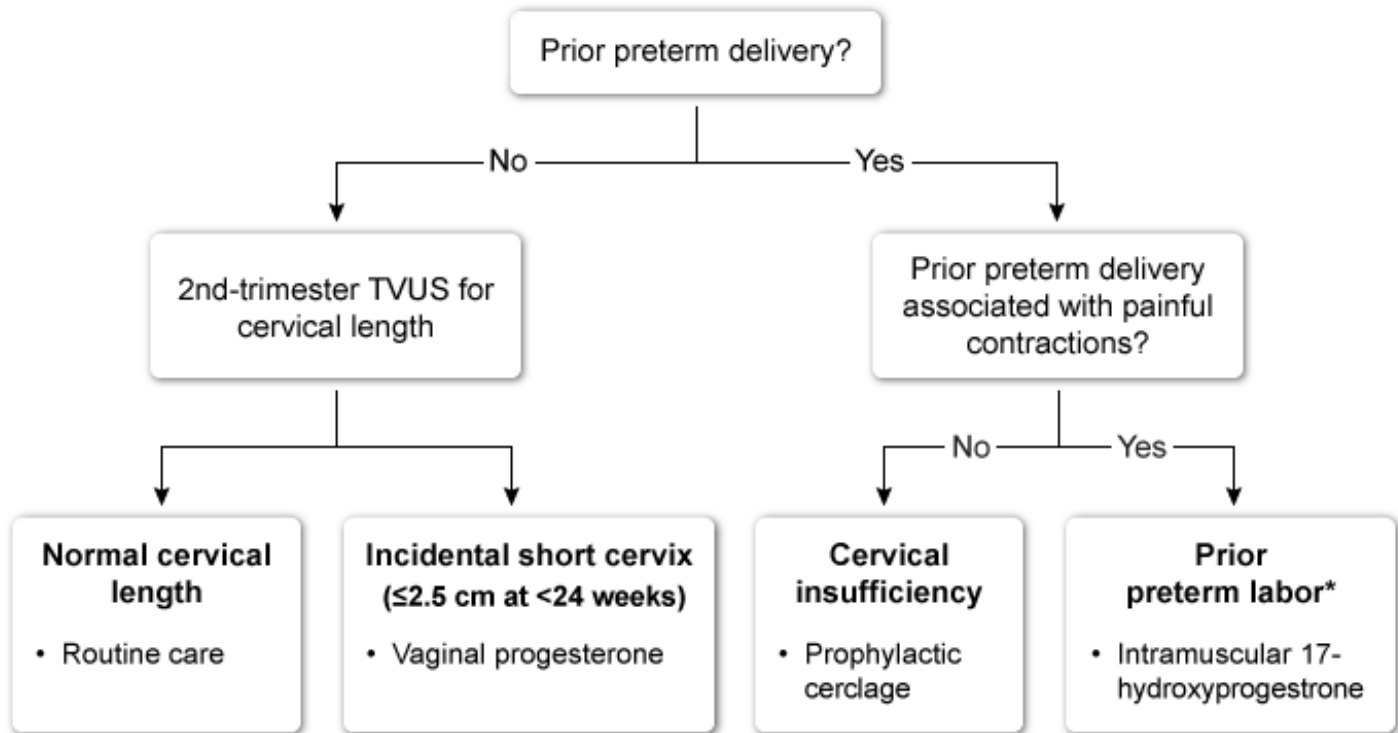


*Nonbloody discharge & normal breast examination (eg, no masses, lymphadenopathy, or skin changes).

©UWorld s patient, imaging is

Therefore, this patient with pathologic nipple discharge requires further evaluation with age-based imaging. In women **age ≥30**, as in thi with both ultrasound and mammography. Mammography is typically performed first because it has high specificity (94%-100%) for identification of suspicious breast tissue. Even if **mammography is normal**, an **ultrasound is required** because this modality increases the detection of small, noncalcified, or intraductal lesions that can cause nipple discharge (**Choice E**). Age-based imaging for women age <30 is with a breast ultrasound (± mammography) due to the greater density of breast tissue and higher false-positive mammography rates in this population

Preterm birth prevention

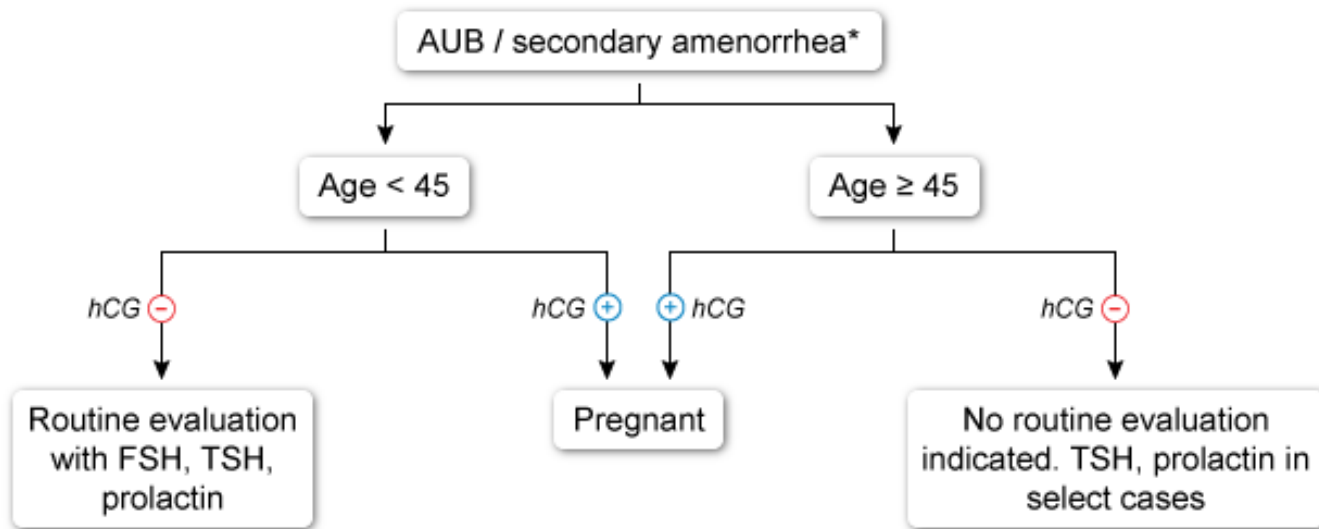


*Preterm labor = regular contractions causing cervical change at <37 weeks gestation with intact membranes.
TVUS = transvaginal ultrasound.

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This patient with a prior cold knife conization is at risk for **preterm delivery** (ie, delivery at <37 weeks gestation). Preterm delivery is a common obstetric complication and a leading cause of neonatal morbidity and mortality. The greatest risk factor for preterm delivery is prior spontaneous preterm delivery, but additional risk factors include tobacco use, multiple gestation, and **prior cervical surgery**. In particular, removal of part of the cervix by **cold knife conization** for cervical intraepithelial neoplasia can cause cervical shortening and scarring, which decreases cervical tensile strength and increases the risk for preterm labor.

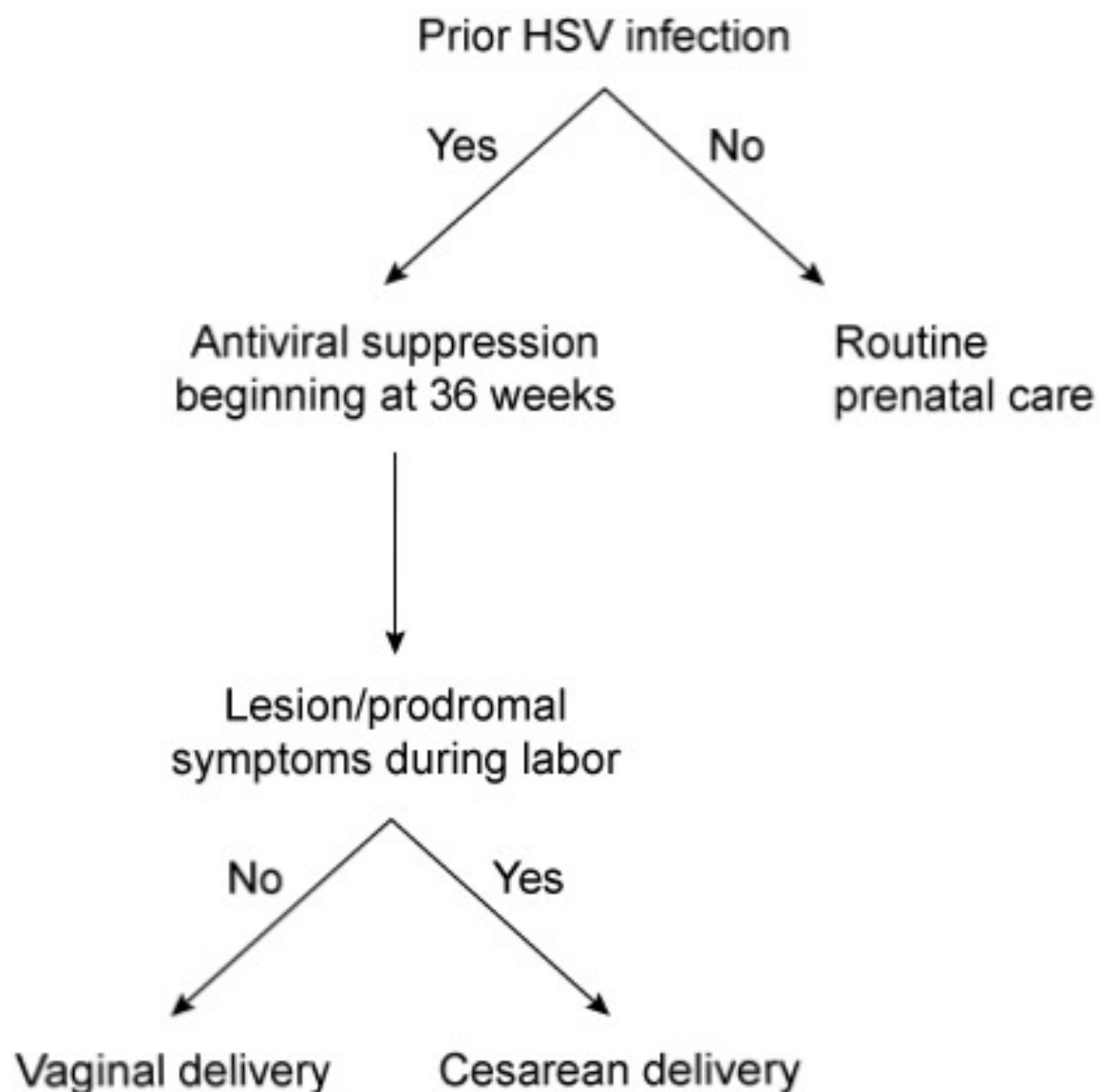
AUB & secondary amenorrhea evaluation



AUB = abnormal uterine bleeding.

***Secondary amenorrhea** = no menses >3 months with prior regular menses
OR no menses >6 months with prior irregular menses.

Pregnancy management in patients with HSV



HSV = herpes simplex virus.

Management of breast pain

